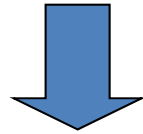


Chapter 7 Flip-Flops and Related Devices

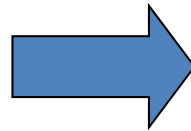
触发器以及相关设备

- **Logic circuits**

- **Combinational Circuits** (组合电路)
- **Sequential Circuits** (时序电路)



Basic Block



Latches
Flip-Flops

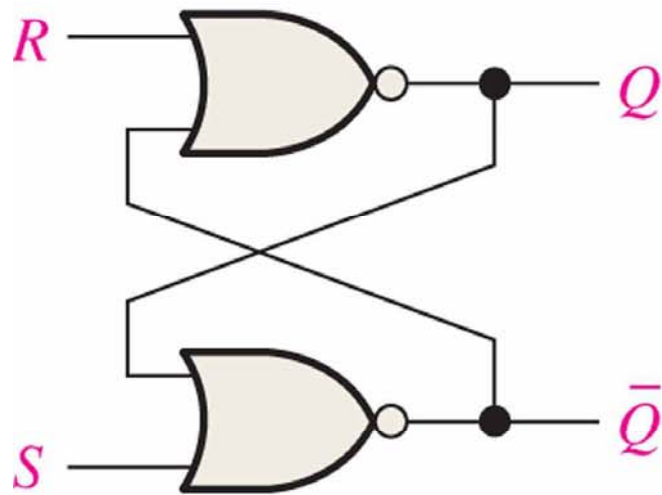
7.1 Latches (锁存器)

- The *latch* is a type of temporary device that has two stable states (*bistable*) and is normally placed in a category separate from that of *flip-flops*(触发器).
- The difference between latches and flip-flops is the method used for changing their states.

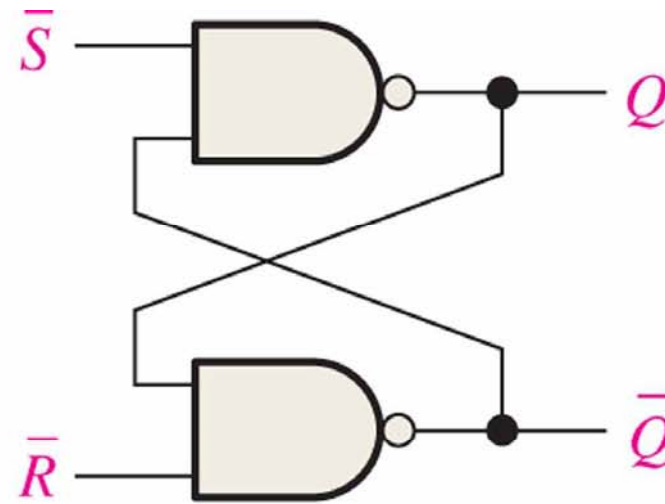
- *Latches*

- S-R (Set-Reset) Latch (SR锁存器)
- Gated S-R Latch (门控锁存器)
- Gated D (Data) Latch

7.1.1 S-R (Set-Reset) Latch

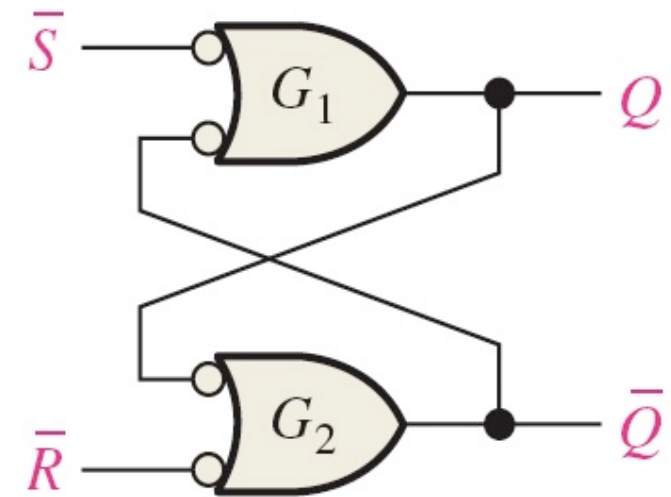
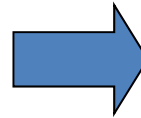
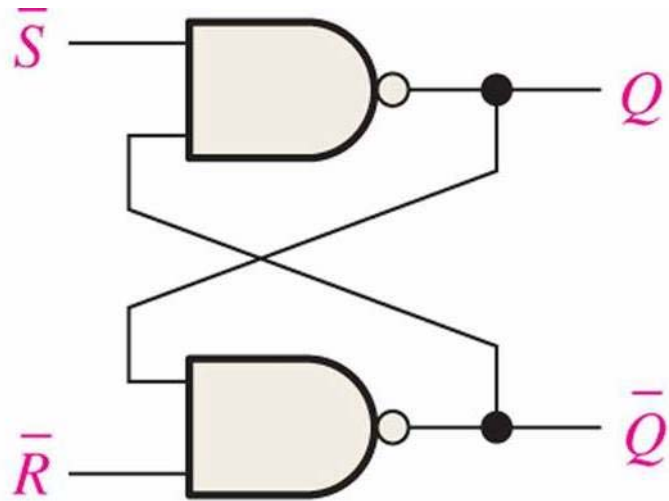


(a) Active-HIGH input S-R latch

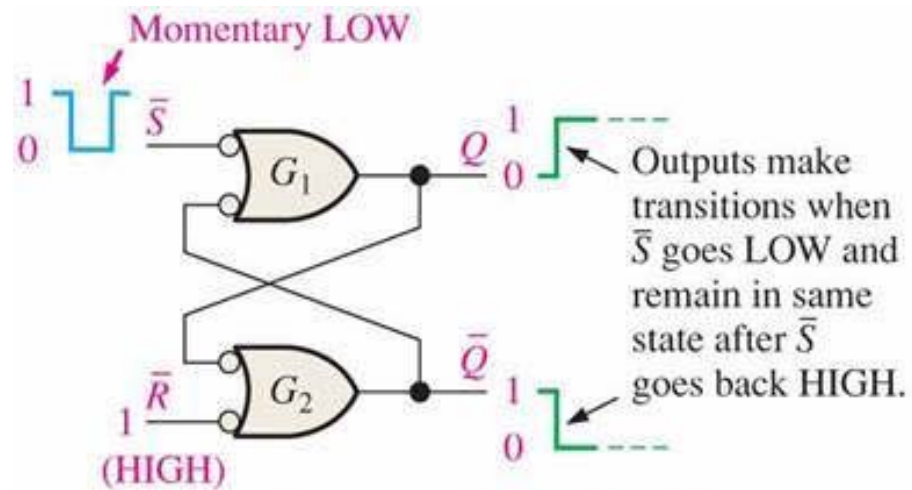


(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

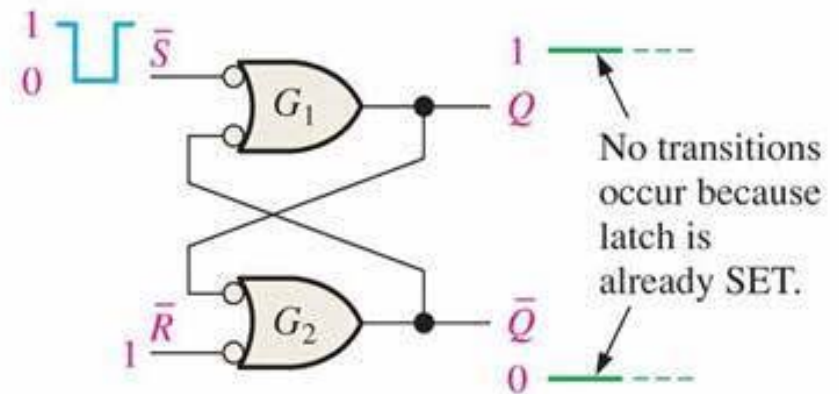
Active-LOW input S-R latch



(b) Active-LOW input $\bar{S}$ - $\bar{R}$ latch

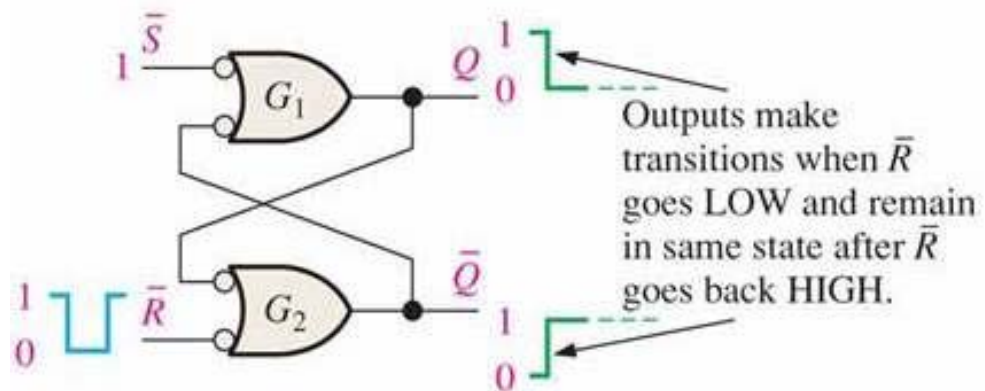


Latch starts out RESET ($Q = 0$).

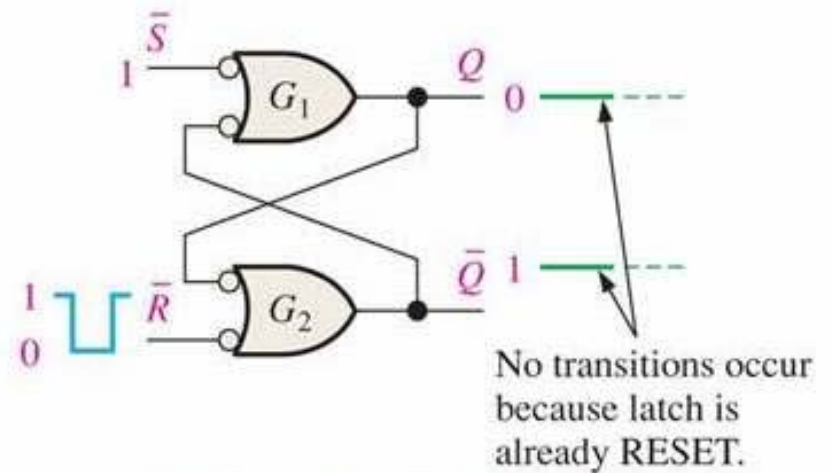


Latch starts out SET ($Q = 1$).

(a) Two possibilities for the SET operation

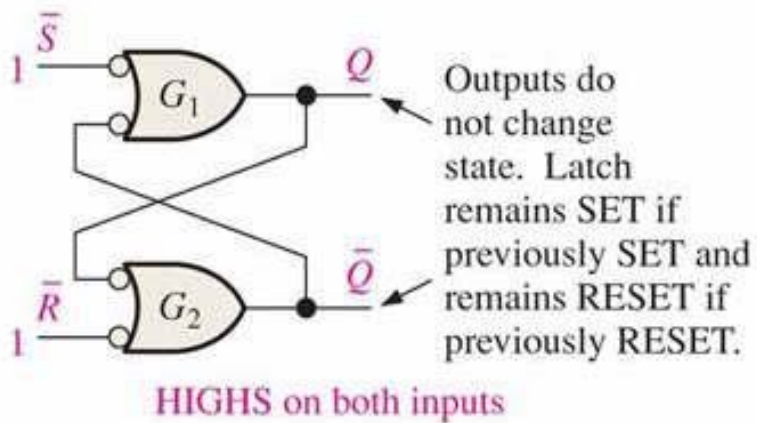


Latch starts out SET ($Q = 1$).

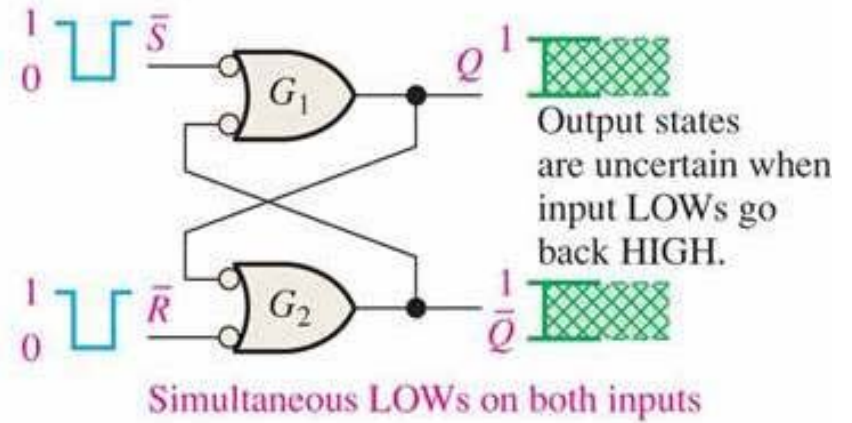


Latch starts out RESET ($Q = 0$).

(b) Two possibilities for the RESET operation



(c) No-change condition

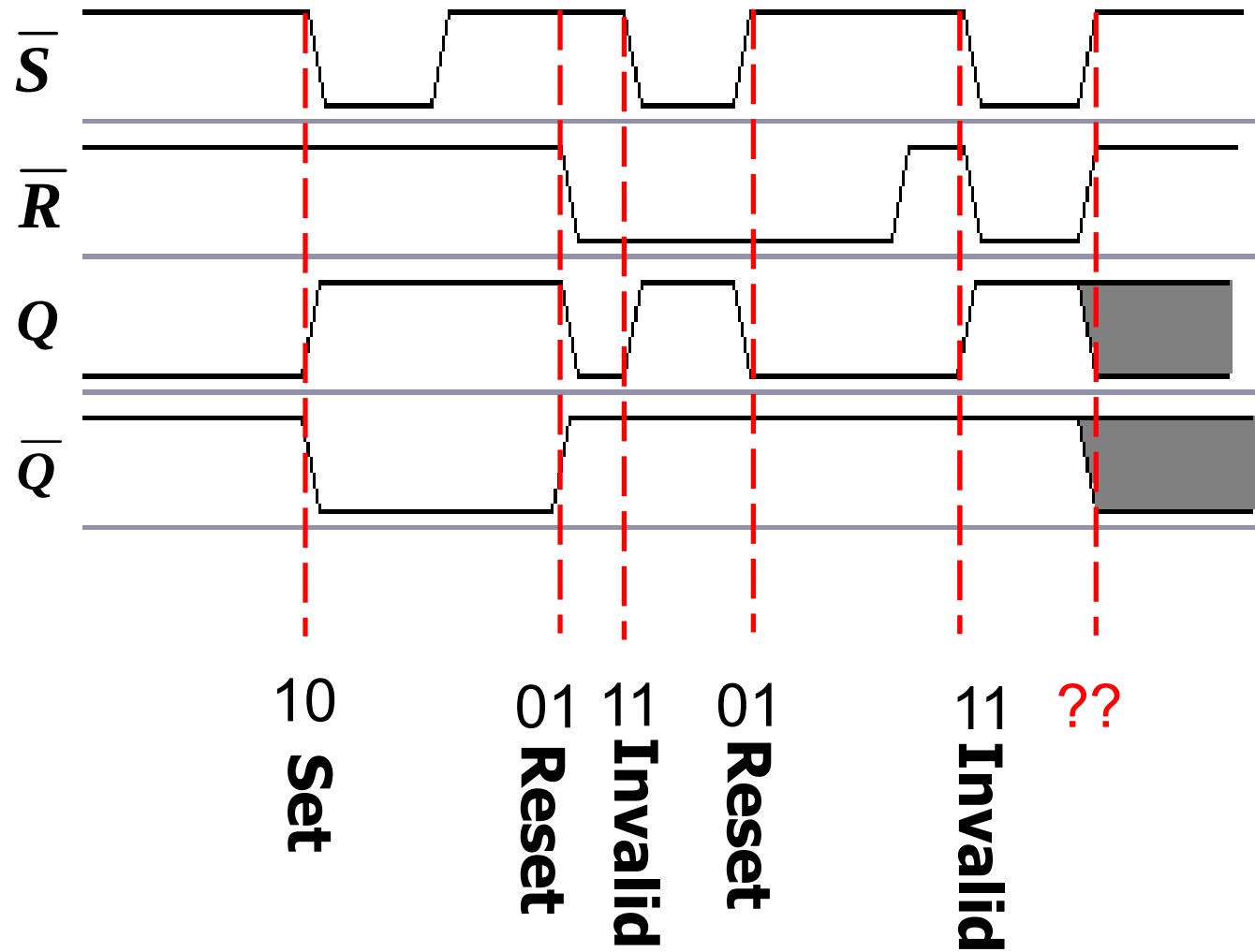


(d) Invalid condition

Truth Table (真値表)

Inputs		Outputs		Comments
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
1	1	NC	NC	No change. Latch remains in present state.
0	1	1	0	Latch SET.
1	0	0	1	Latch RESET.
0	0	1*	1*	Invalid condition

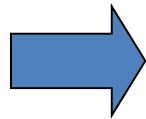
Timing diagram



Memory Function

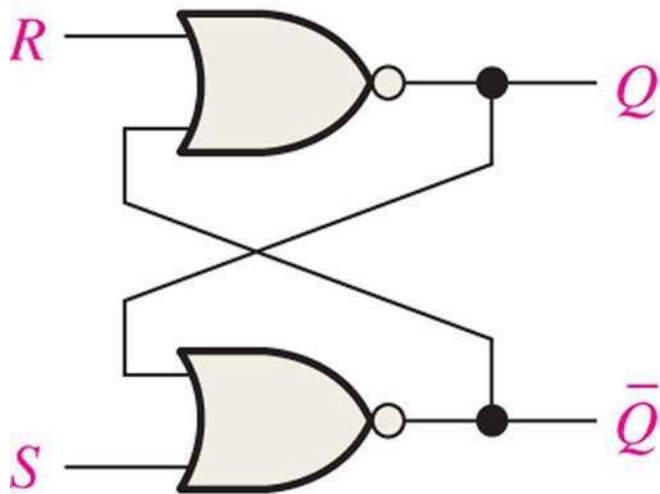
$$\begin{array}{l} S=1; \quad R=0; \left\{ \begin{array}{l} Q=0 \\ \bar{Q}=1 \end{array} \right. \quad \text{“0”,} \quad \text{remember “0”} \\ \\ S=0; \quad R=1; \left\{ \begin{array}{l} Q=1 \\ \bar{Q}=0 \end{array} \right. \quad \text{“1”,} \quad \text{remember “1”} \end{array}$$

Bi-stable



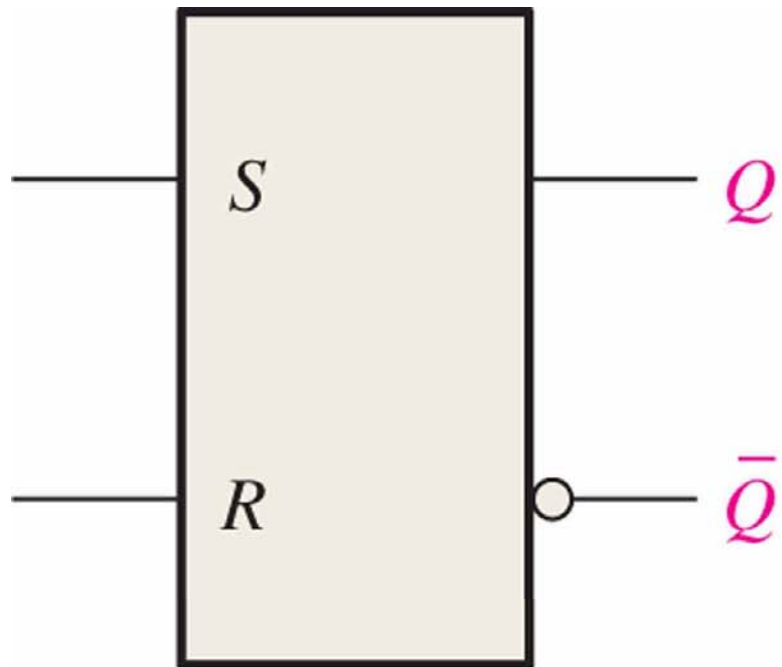
**Used to store 1 bit
binary number**

Active-HIGH input S-R latch

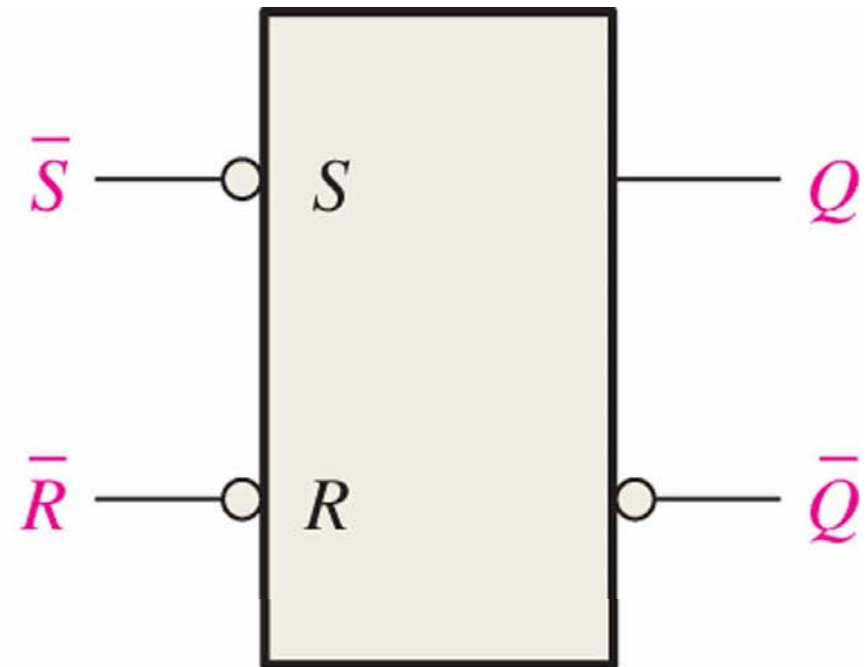


R	S	Q	$\bar{Q}$
0	0	Q_0	$\bar{Q}_0$
0	1	1	0
1	0	0	1
1	1	0^*	0^*

Logic Symbol

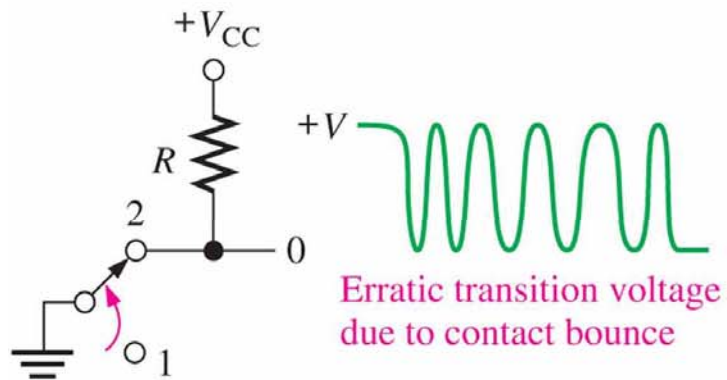


(a) Active-HIGH input
S-R latch

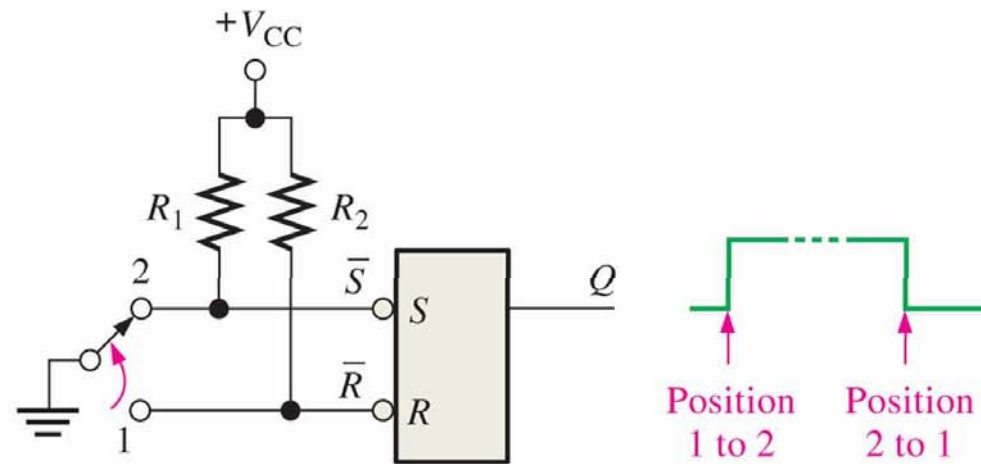


(b) Active-LOW input
 $\bar{S}$ - $\bar{R}$ latch

Application Example

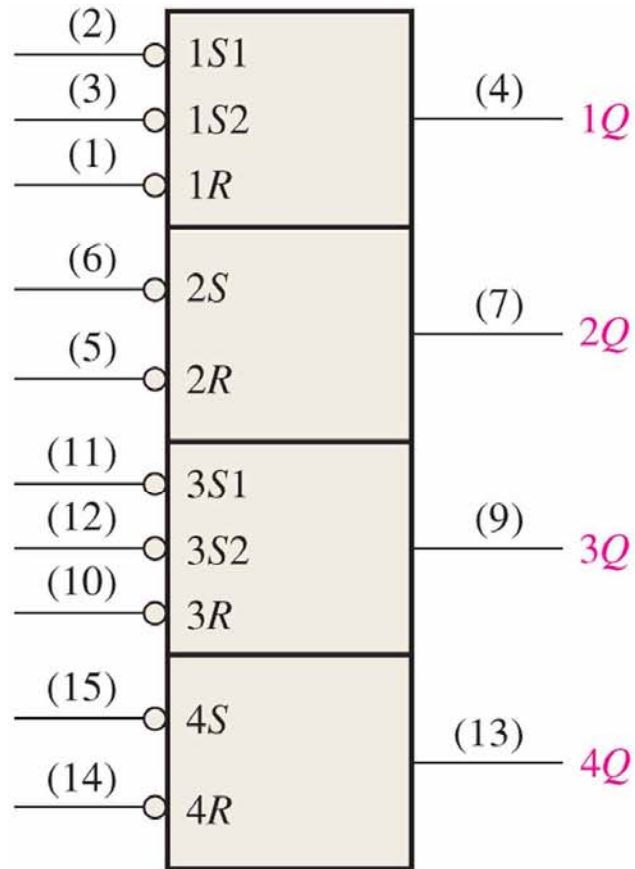


(a) Switch contact bounce

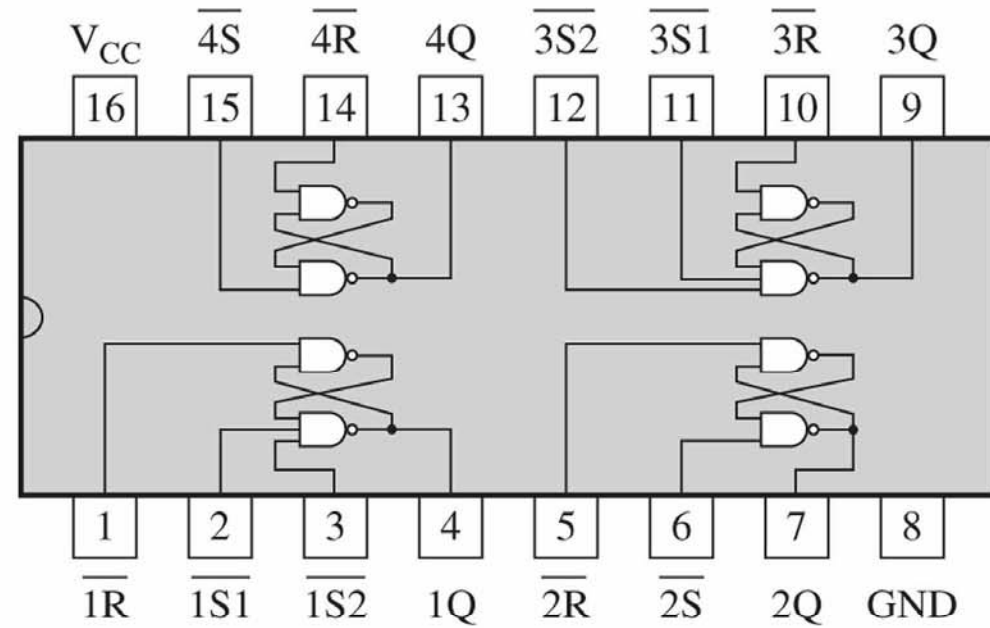


(b) Contact-bounce eliminator circuit

74LS279



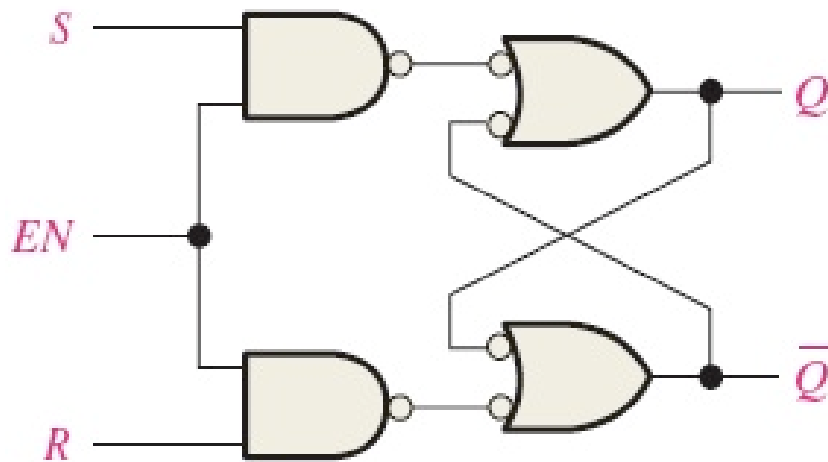
(a) Logic diagram



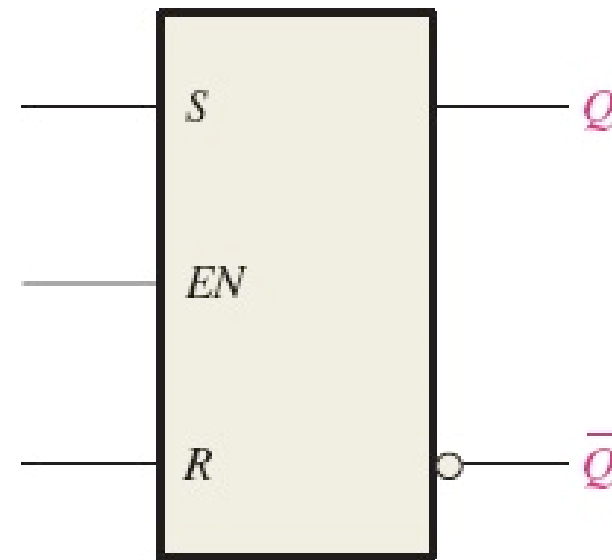
(b) Pin diagram

7.1.2 The Gated S-R Latch (门控S-R锁存器)

EN: control the time when the inputs S and R can control the output. (Level-triggered)

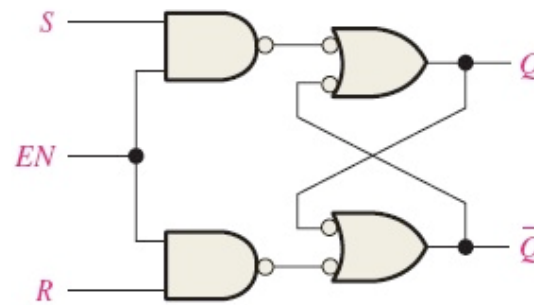
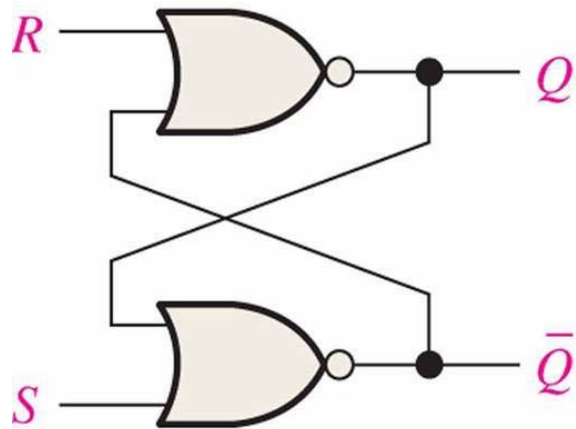


(a) Logic diagram

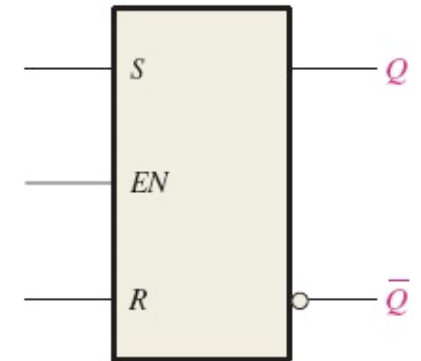


(b) Logic symbol

Comparison



(a) Logic diagram

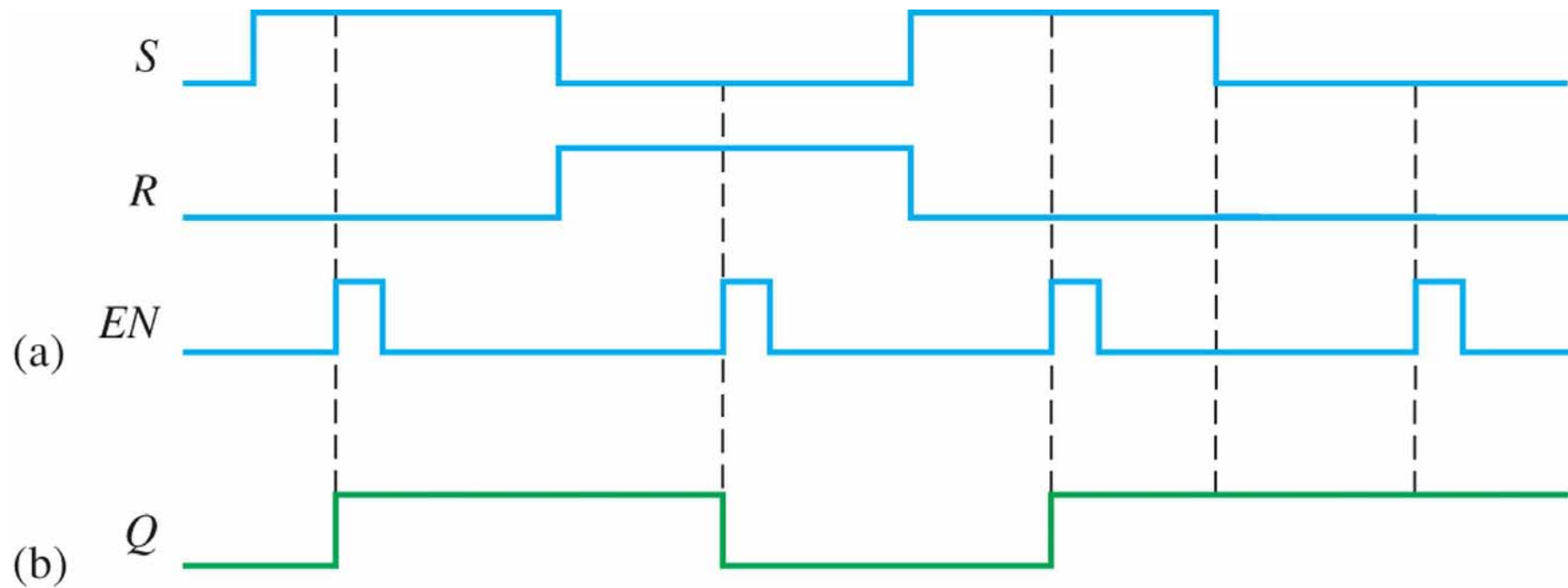


(b) Logic symbol

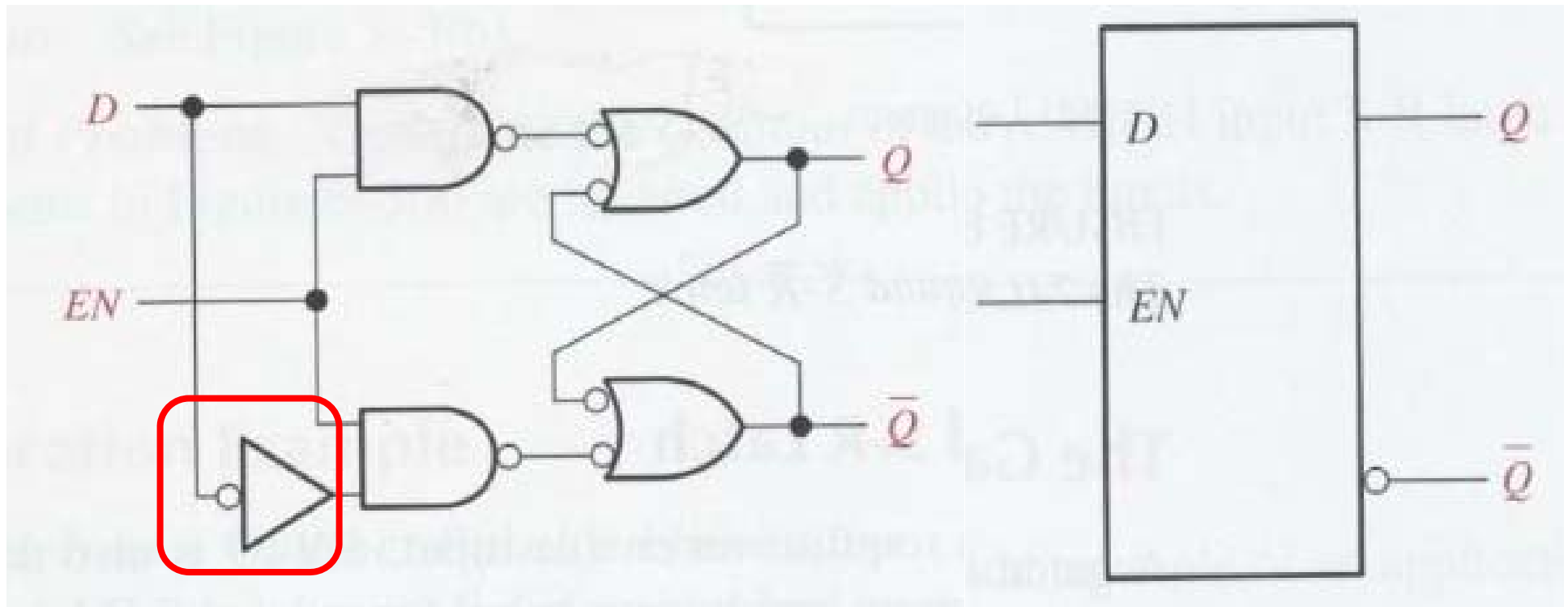
R	S	Q	$\bar{Q}$
0	0	Q_0	$\bar{Q}_0$
0	1	1	0
1	0	0	1
1	1	1^*	1^*

E	R	S	Q	$\bar{Q}$
0	X	X	Q_0	$\bar{Q}_0$
1	0	0	Q_0	$\bar{Q}_0$
1	0	1	1	0
1	1	0	0	1
1	1	1	1^*	1^*

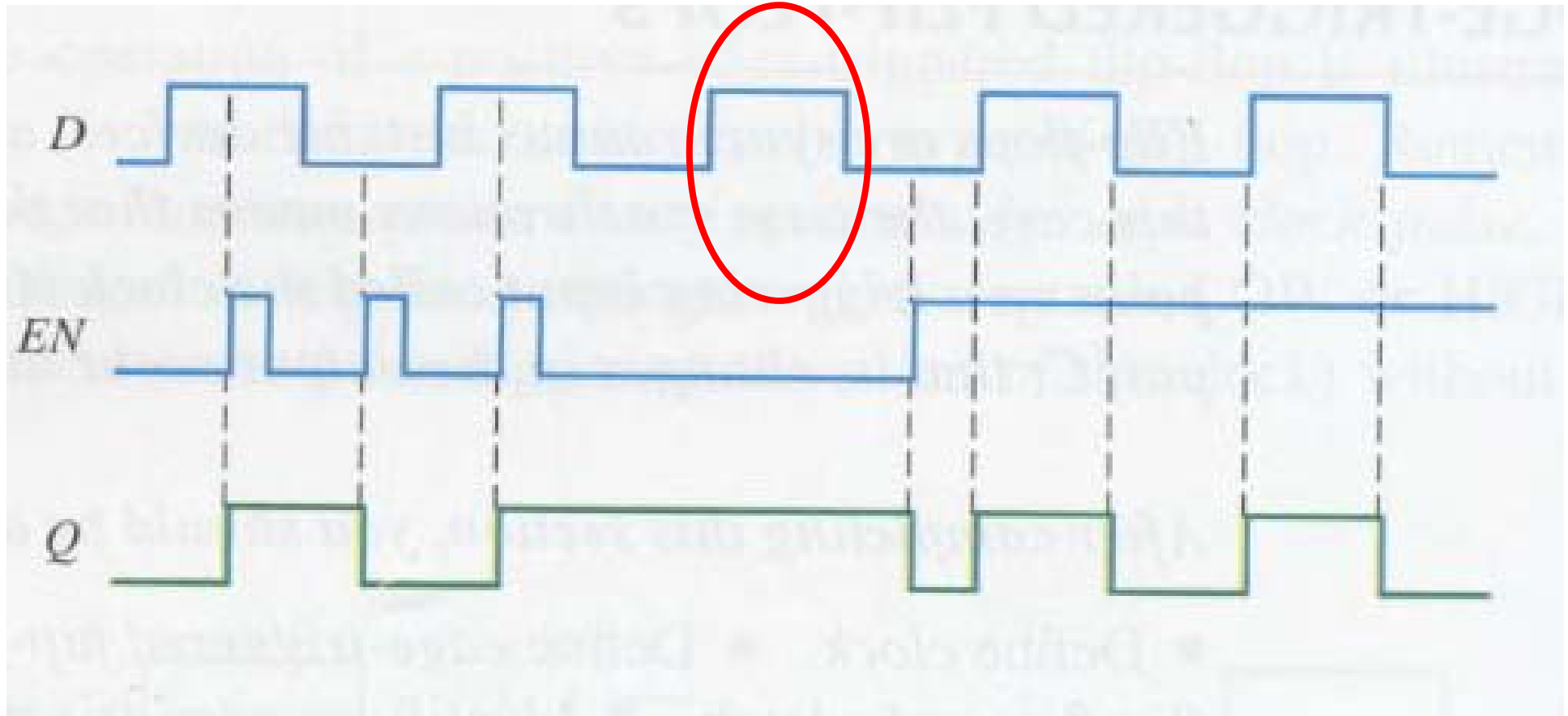
Gated S-R Latch Waveform



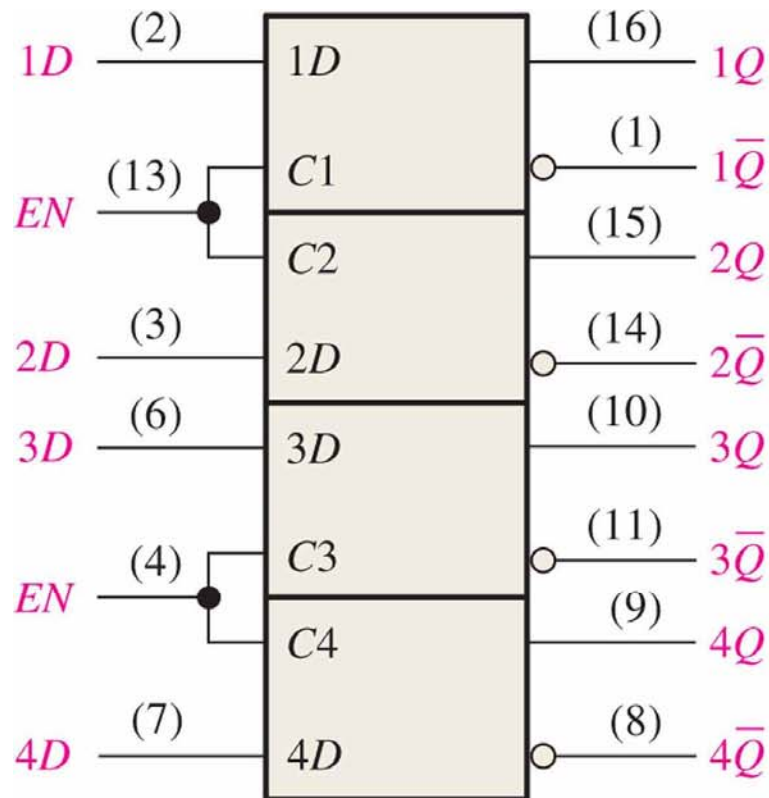
7.1.3 The Gated D Latch



Gated D Latch Waveform



74LS75



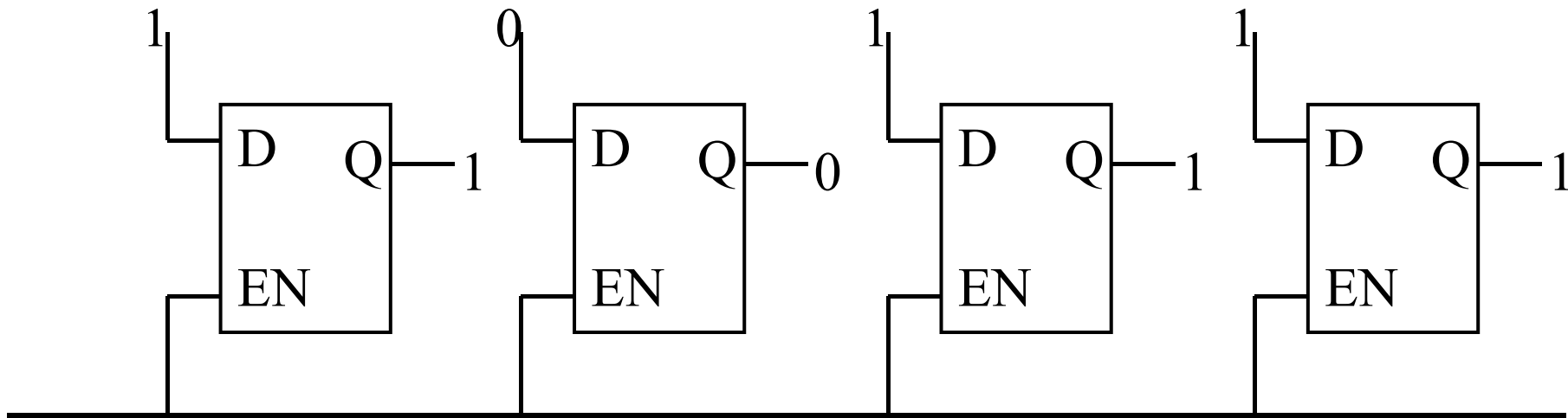
(a) Logic symbol

INPUTS		OUTPUTS		COMMENTS
D	EN	Q	$\bar{Q}$	
0	1	0	1	RESET
1	1	1	0	SET
X	0	Q_0	$\bar{Q}_0$	No change

Note: Q_0 is the prior output level before the indicated input conditions were established.

(b) Truth table (each latch)

Application example: Latches for temporary data storage



7.2 Edge-triggered Flip-Flops

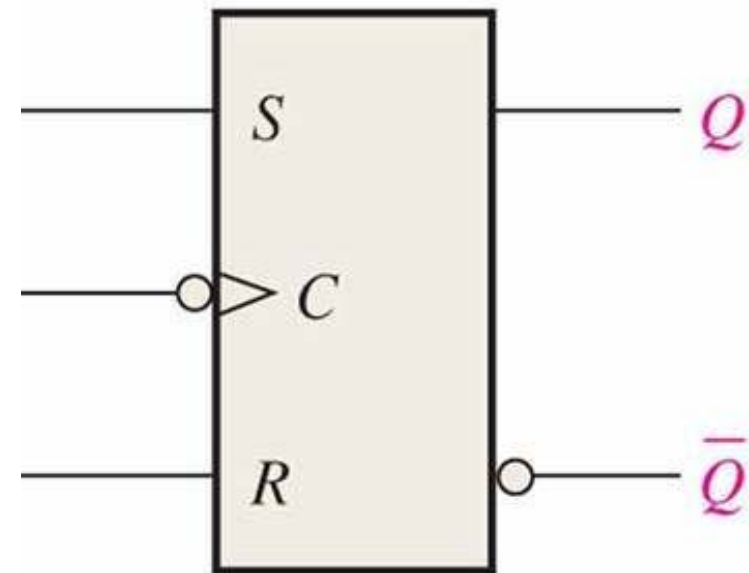
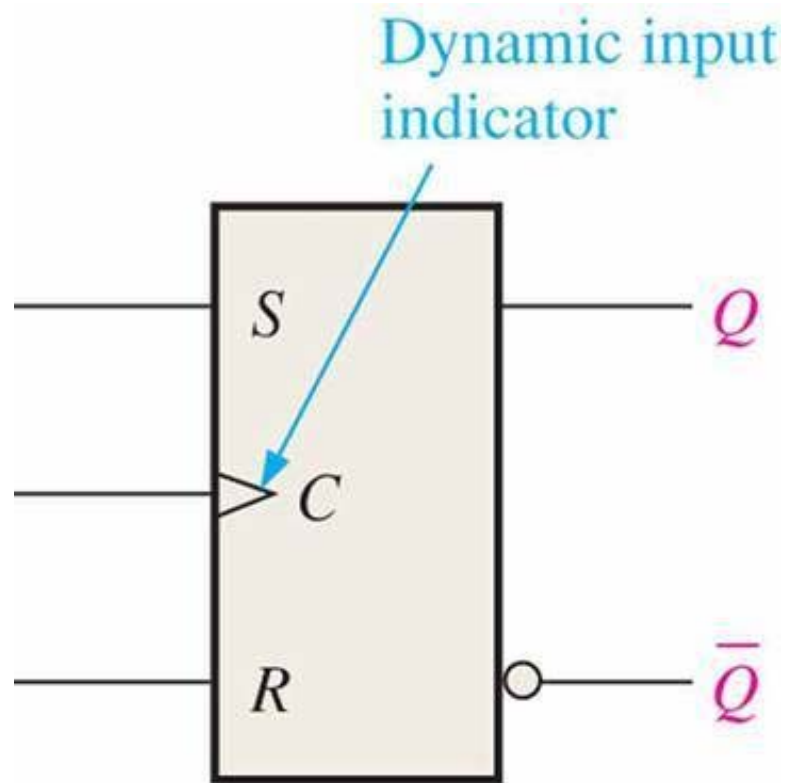
边沿触发的触发器

- Flip-flops are *synchronous bi-stable* devices. The term *synchronous* means that the output changes state only at a specified point on a *triggering input* called the *clock* (CLK) is designated as a control input C. i.e., changes in the output occur in synchronous with the clock.
- An edge-triggered flip-flop changes state either at the *positive edge* (rising edge) or at the *negative edge* (falling edge) of the clock pulse and is sensitive to its input only at this transition of the clock.

7.2.1 The edge-triggered S-R FF

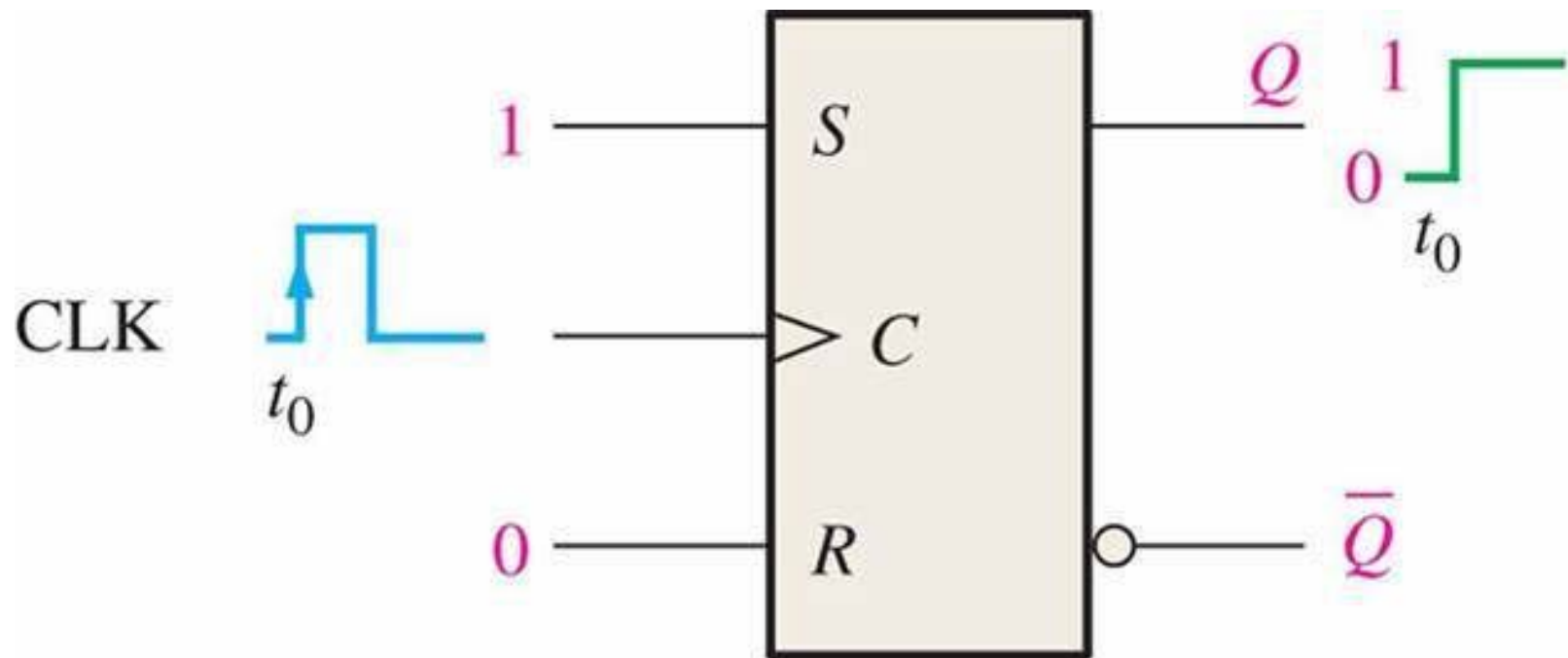
- The S and R inputs of the S-R flip-flop are called synchronous inputs because data on these inputs are transferred to the flip-flop's output only on the *triggering edge of the clock pulse*.
- When S is HIGH and R is LOW on the triggering edge of the clock pulse, Q is HIGH and the flip-flop is **SET**. When S is LOW, R is HIGH, Q is LOW and flip-flop is **RESET**. When S and R are both LOW, the output does not change from its prior state. When S and R are both HIGH, it is an invalid condition.

The edge-triggered S-R FF

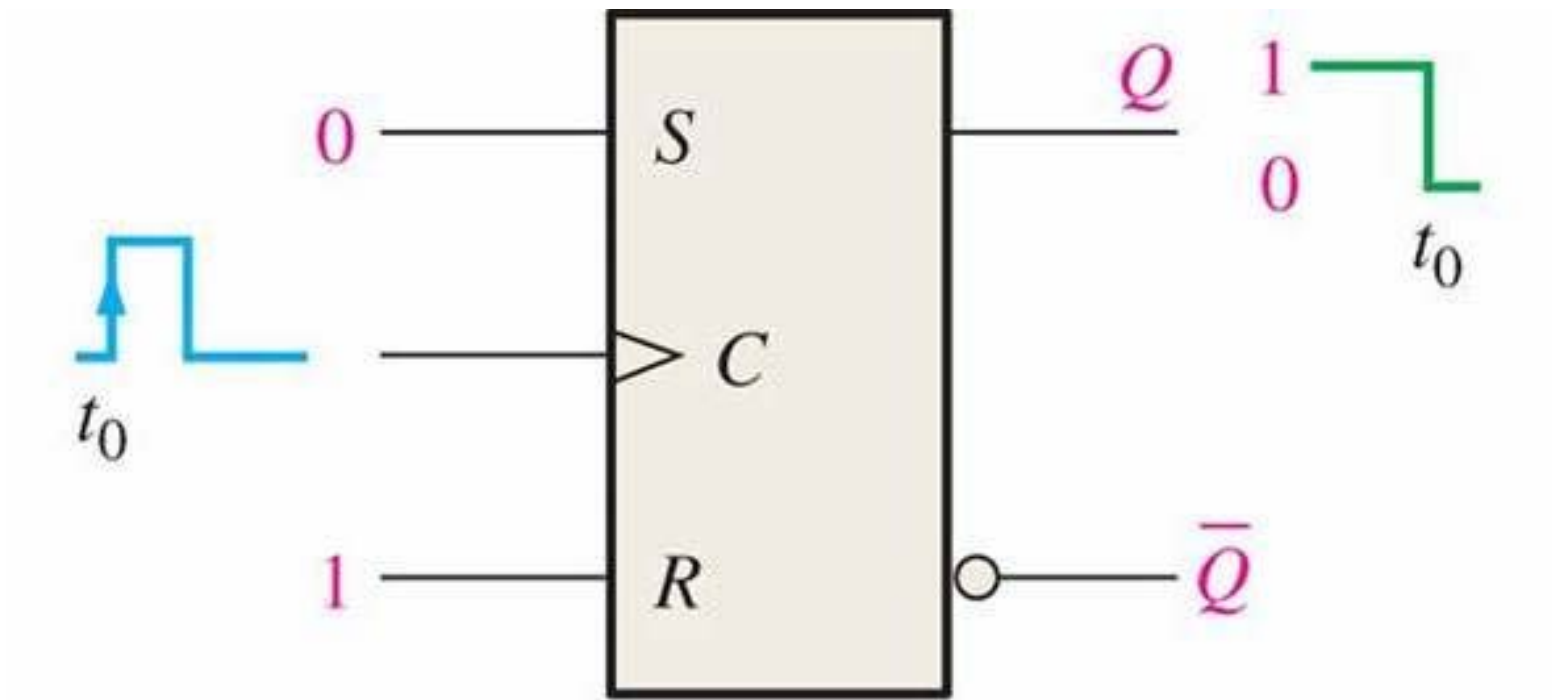


Truth Table of S-R FF

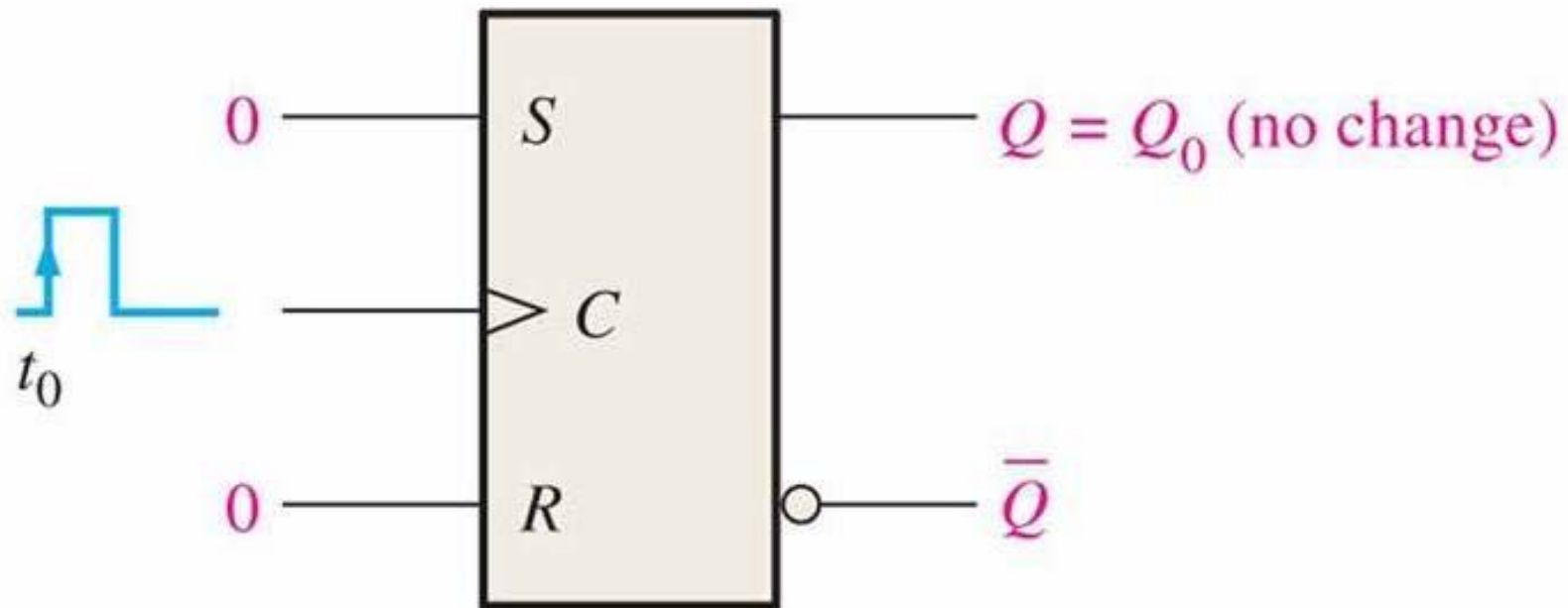
Inputs			Outputs		Comments
S	R	CLK	Q	$\bar{Q}$	
0	0	X	Q_0	$\bar{Q}_0$	No change
0	1	↑	0	1	RESET
1	0	↑	1	0	SET
1	1	↑	?	?	Invalid



(a) $S = 1, R = 0$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)

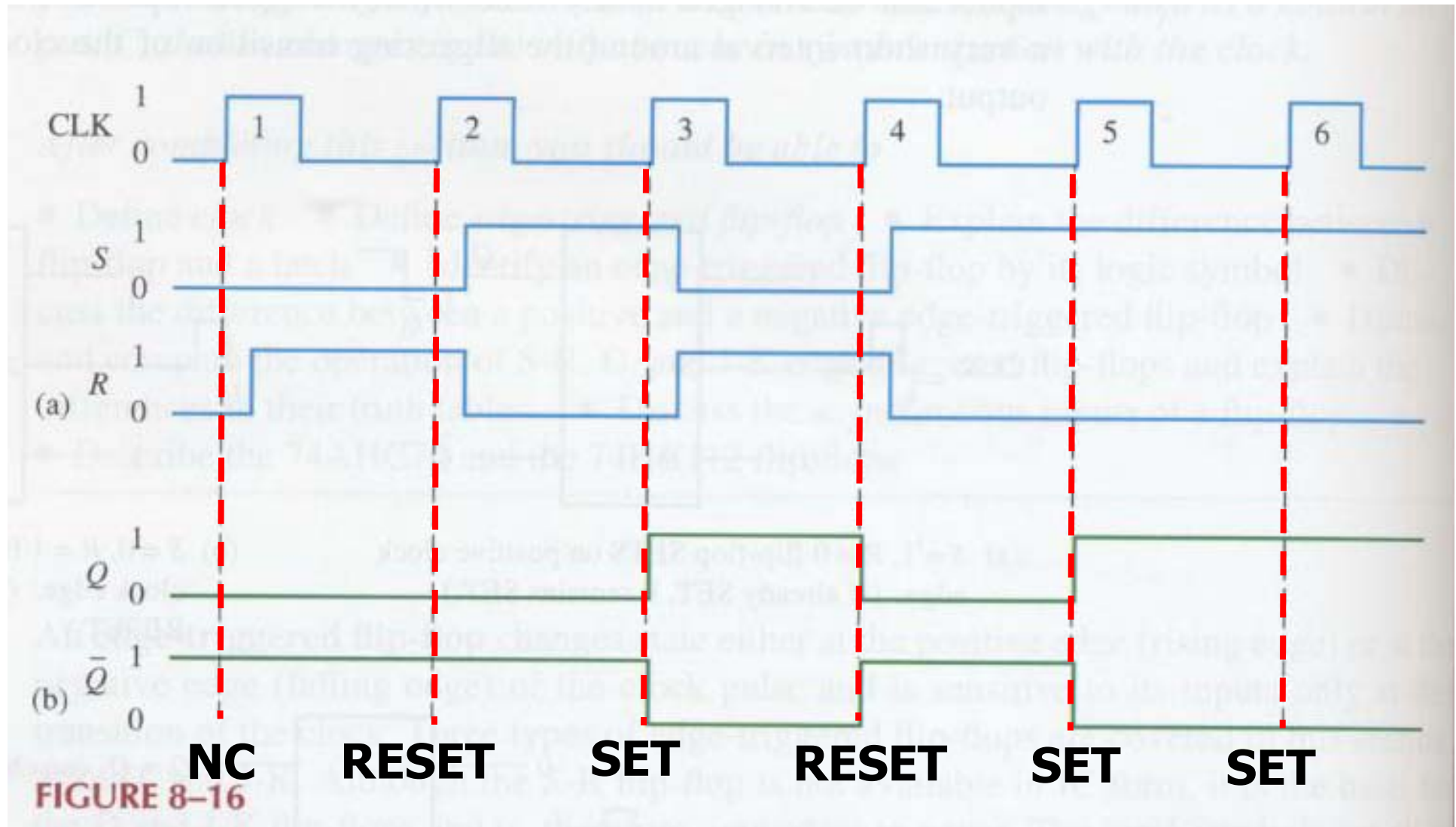


(b) $S = 0$, $R = 1$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)

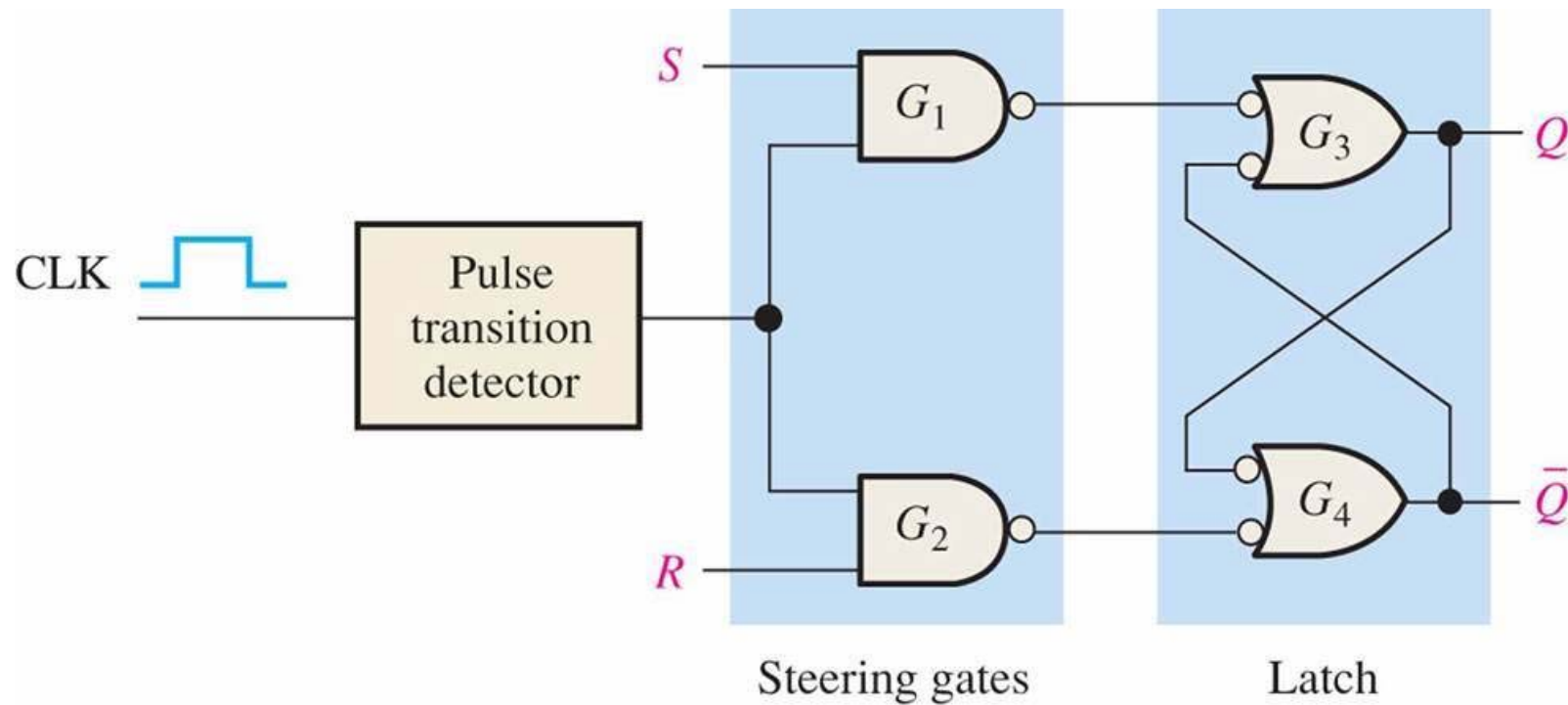


(c) $S = 0$, $R = 0$ flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

Example



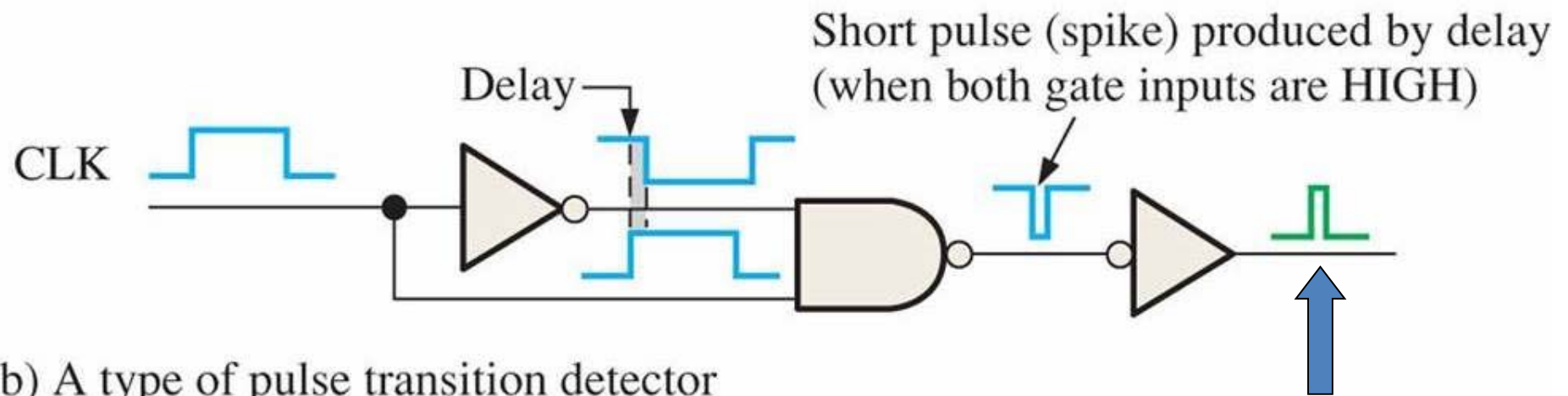
A method of Edge-triggering



(a) A simplified logic diagram for a positive edge-triggered S-R flip-flop

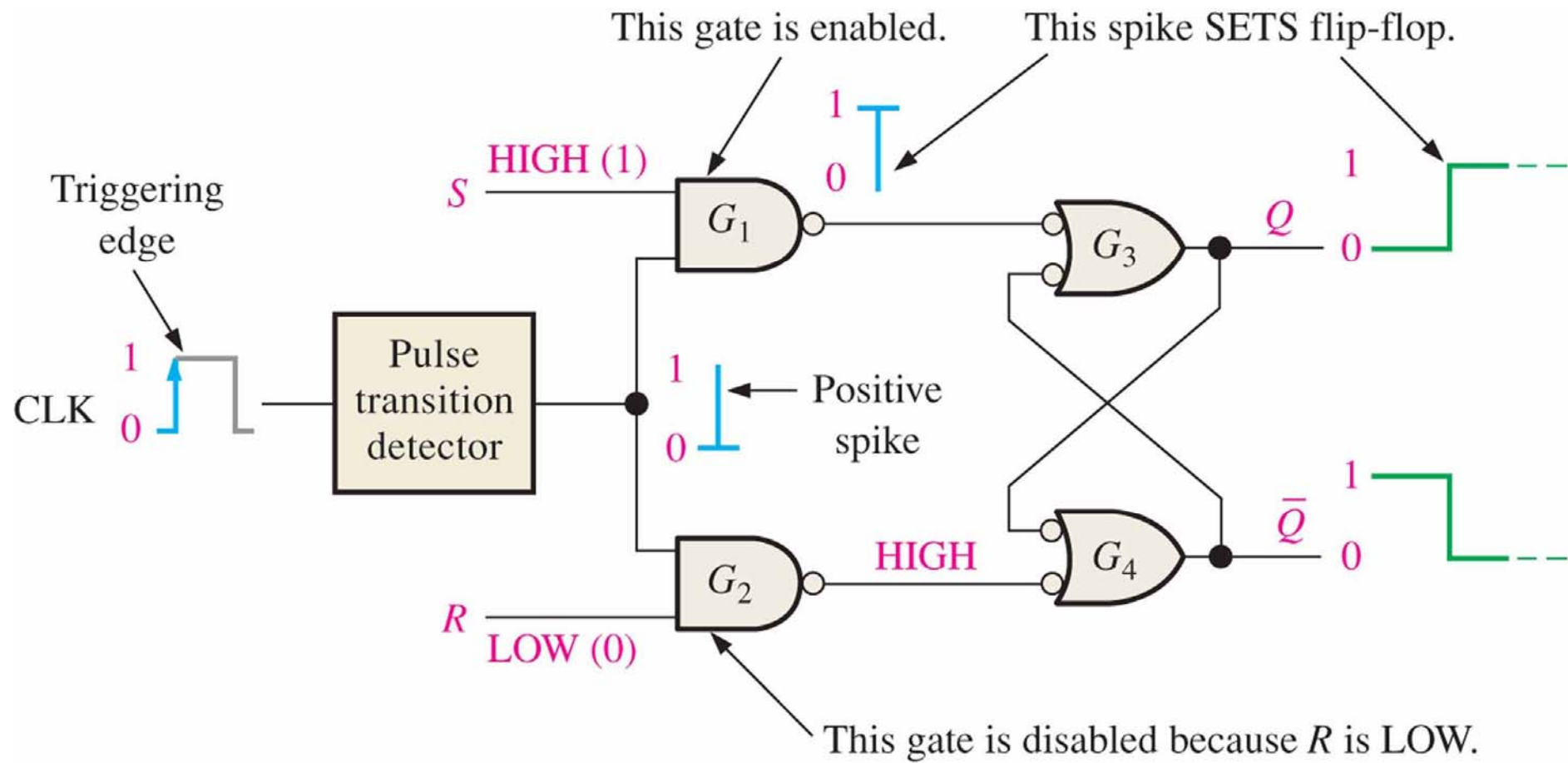
Pulse transition detector

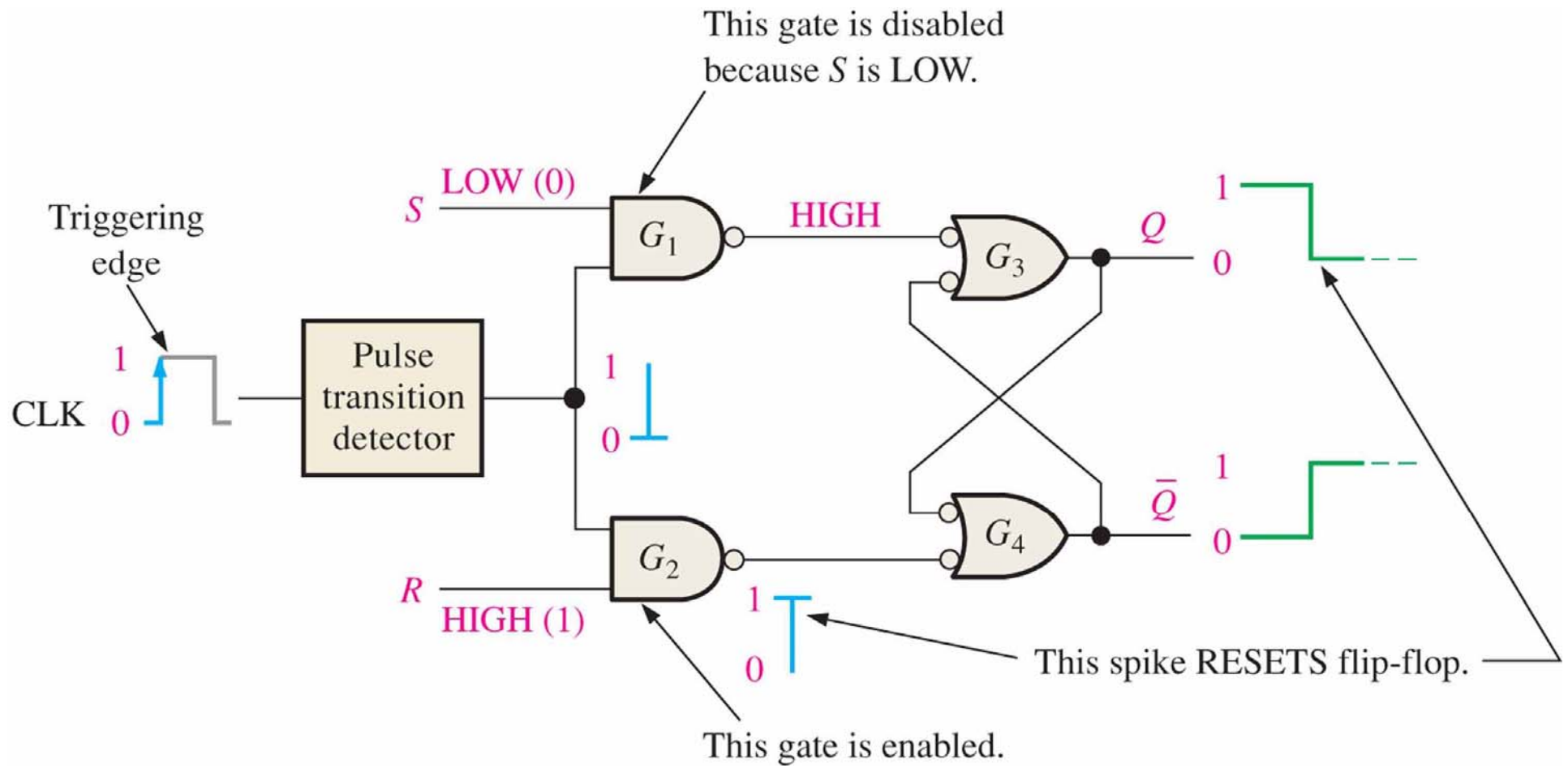
脉冲变换检测器



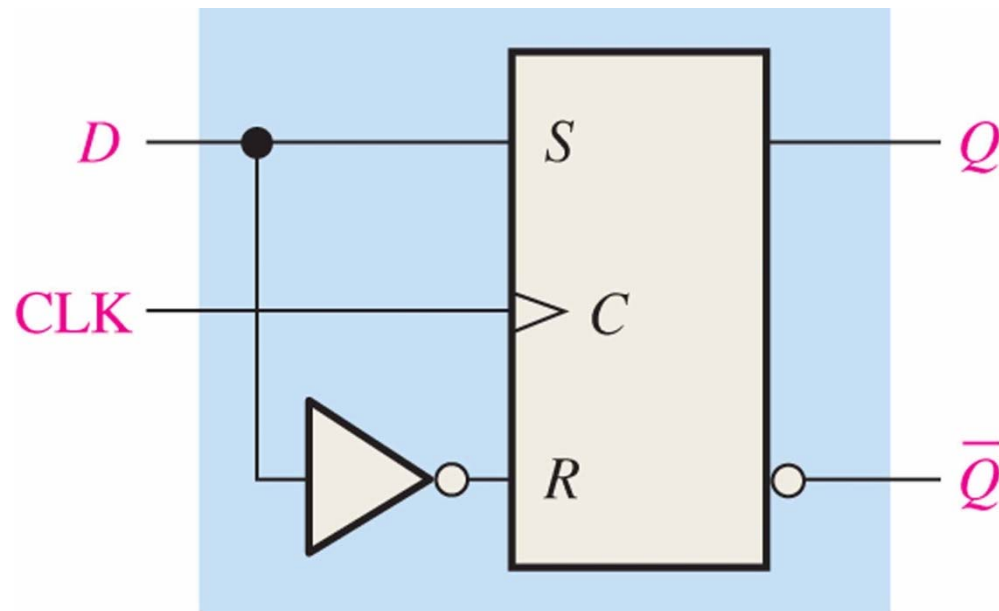
(b) A type of pulse transition detector

**Narrow
spike**





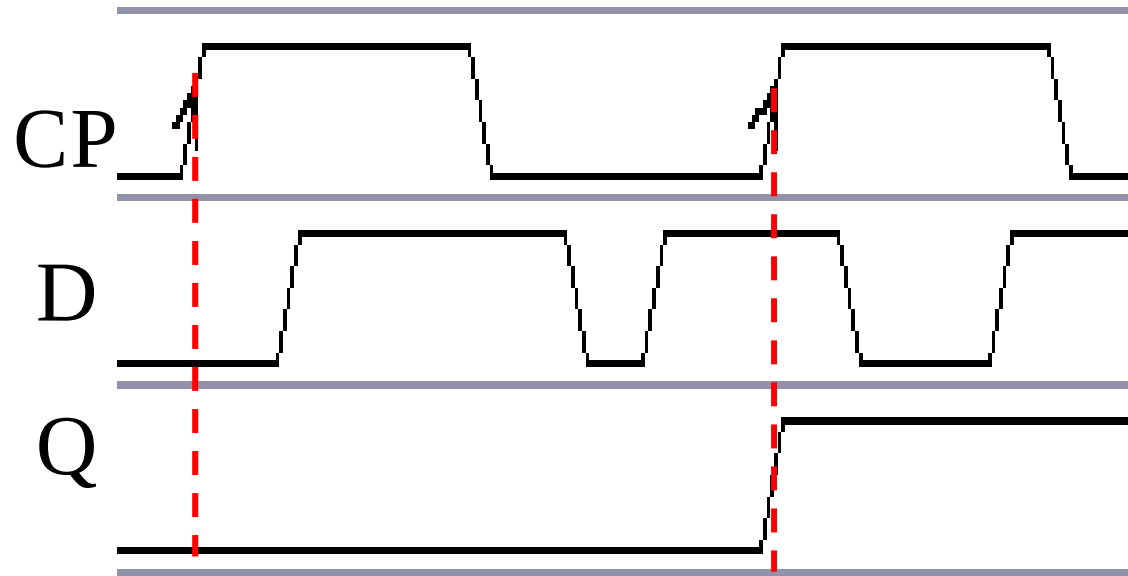
7.2.2 The Edge-Triggered D FF



Positive edge-triggered D flip-flop

The Q output of a D assumes the state of the D input on the triggering edge of the clock

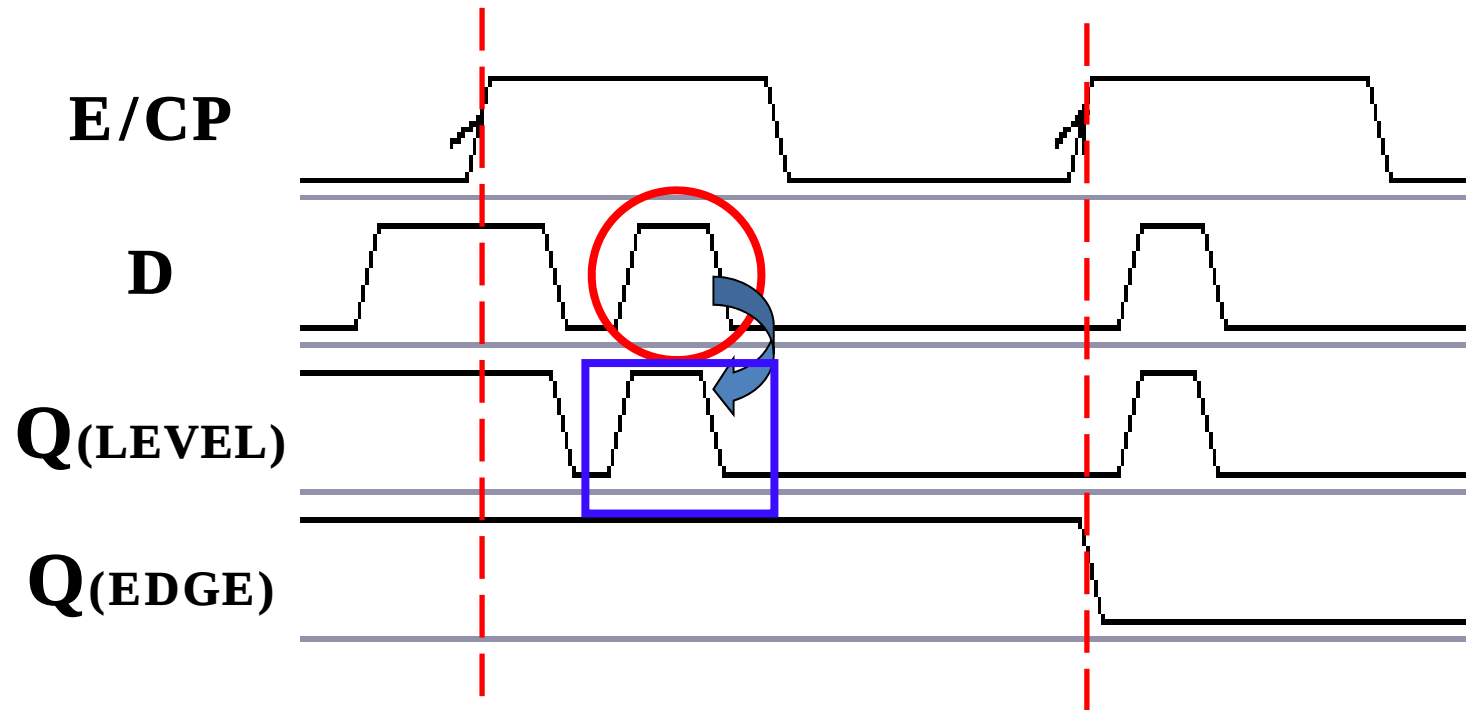
Timing diagram



Truth Table

CP	D	Q	$\overline{Q}$
$\uparrow$	D	D	$\overline{D}$

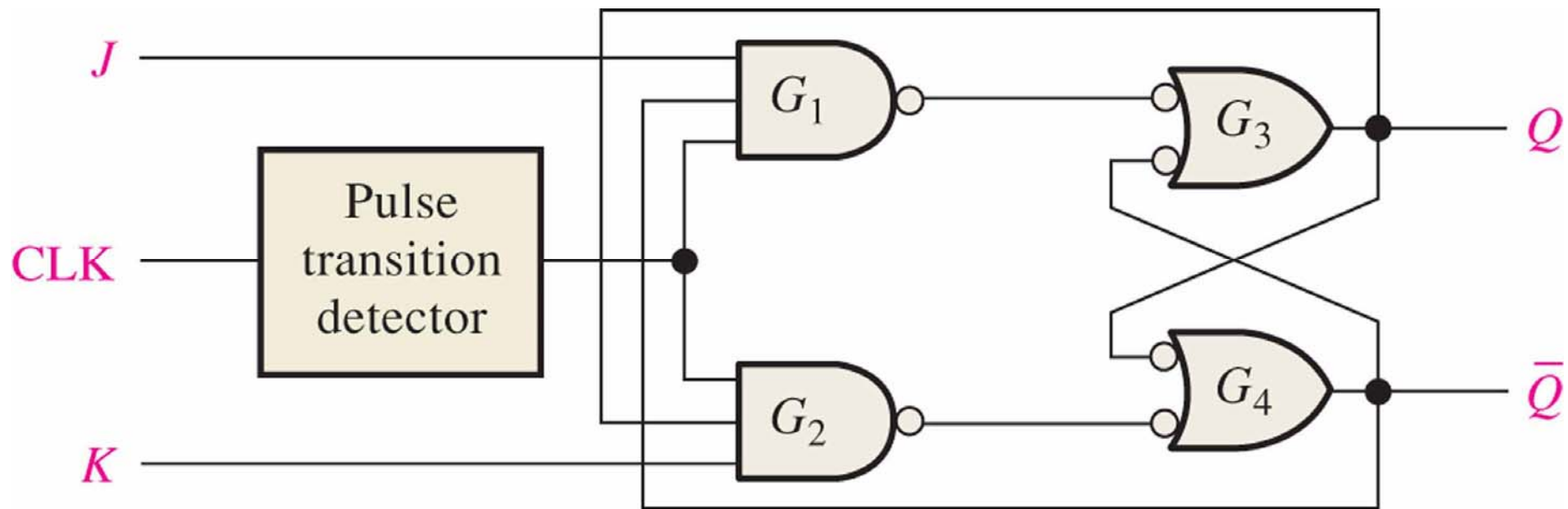
Comparison of edge-triggered and level-triggered



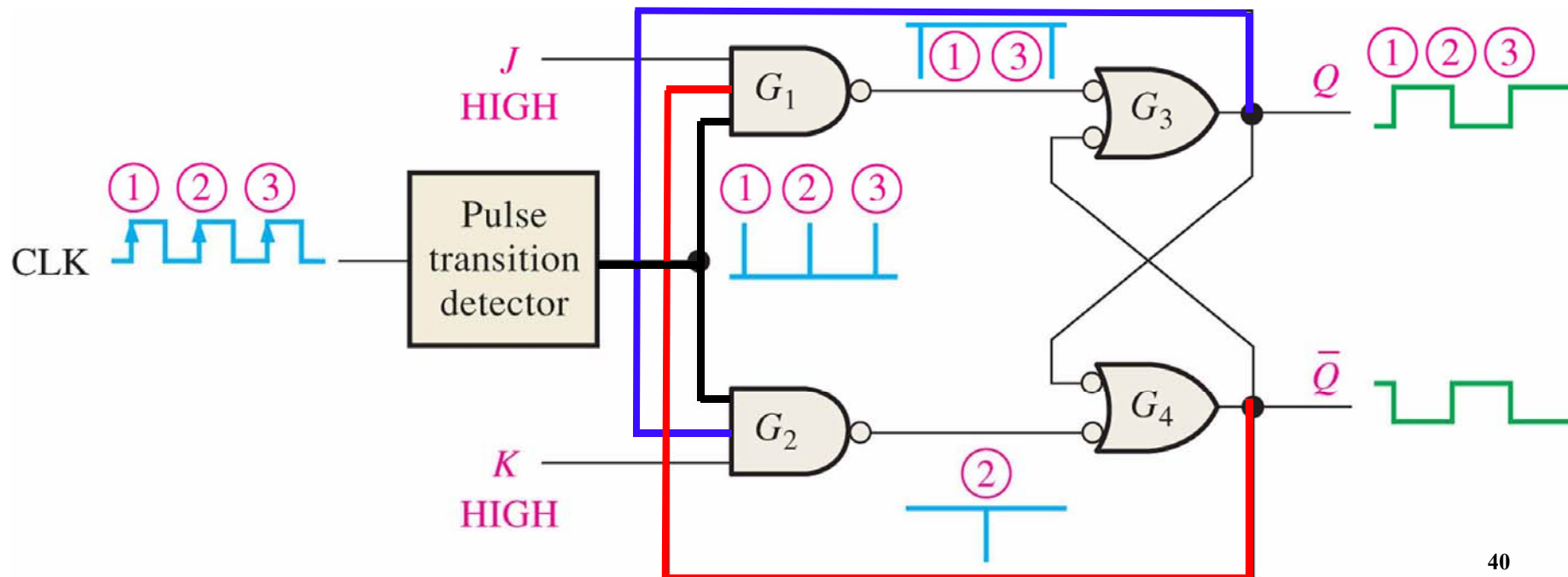
7.2.3 The Edge-Triggered J-K FF

****No invalid states.**

The two Set/Reset signals are labeled as J and K in honor of Jack Kilby.

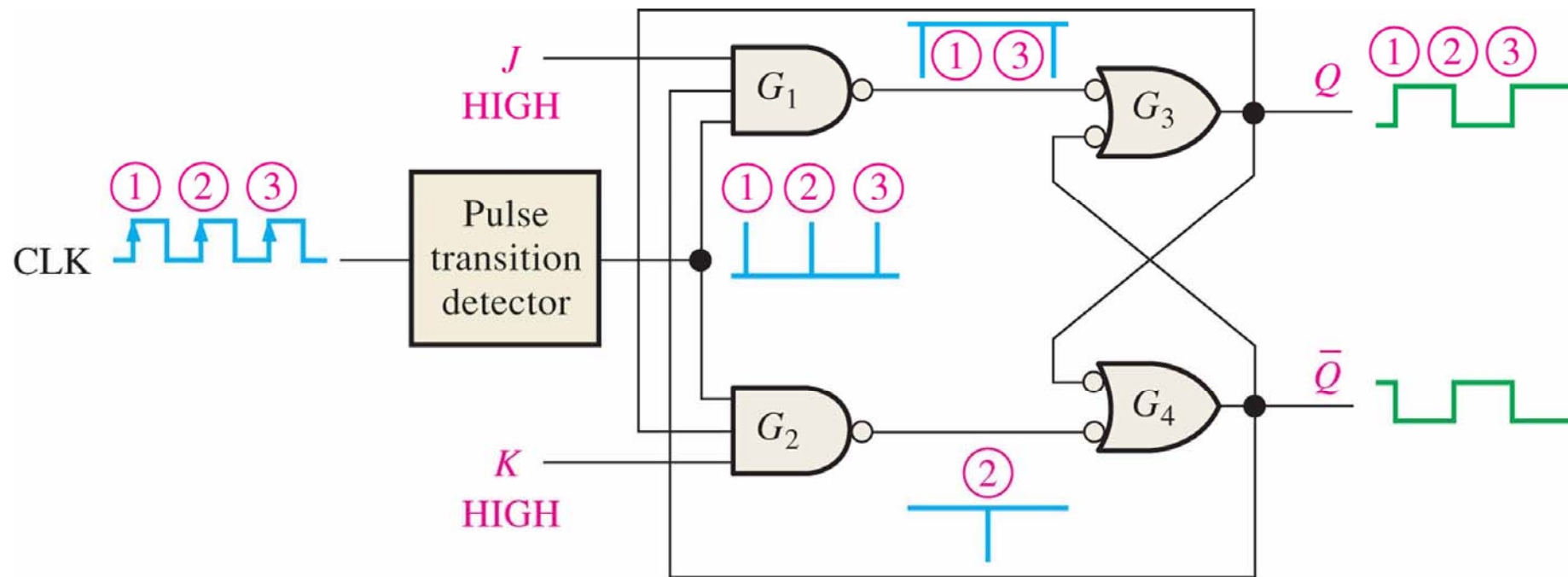


- ① IF $J=1, K=0, Q_0=0$; THEN G_1 enabled, $Q=1$ (SET)
- ② IF $J=0, K=1, Q_0=1$; THEN G_2 enabled, $Q=0$ (RESET)
- ③ IF $J=0, K=0$; THEN no change
- ④ IF $J=1, K=1$; THEN change to opposite state (*Toggle*)

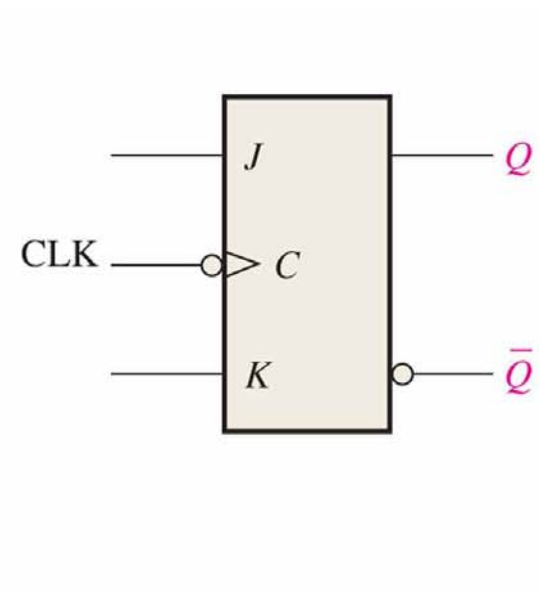
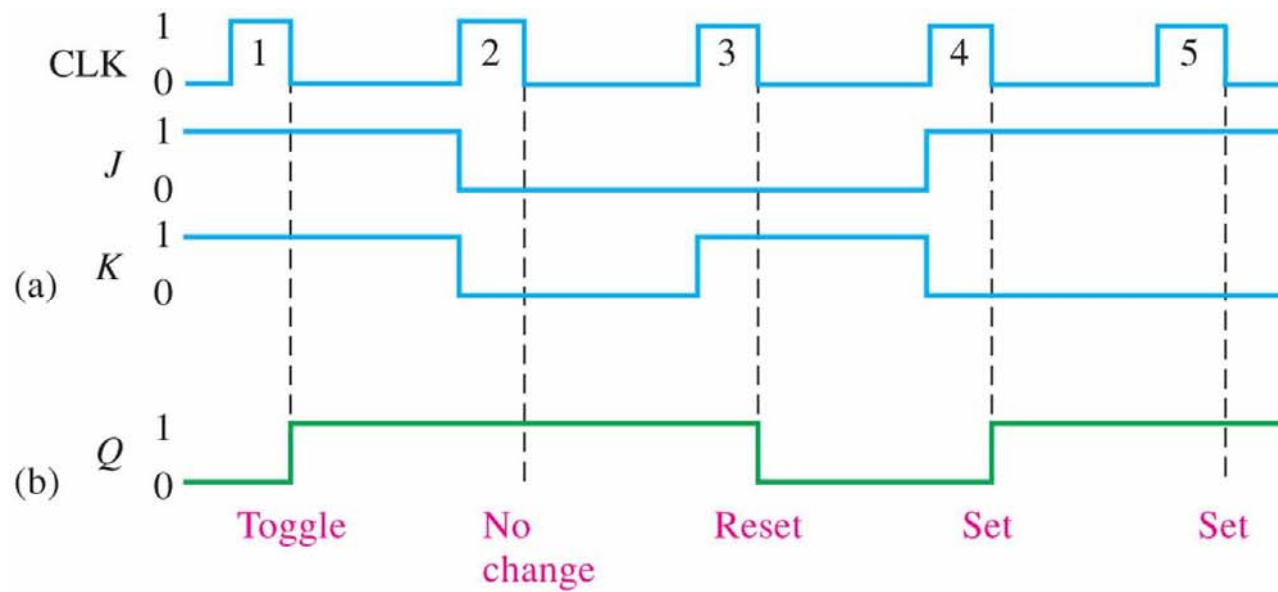


The Edge-Triggered Toggle FF

Toggle operation when $J=K=1$



Timing diagram



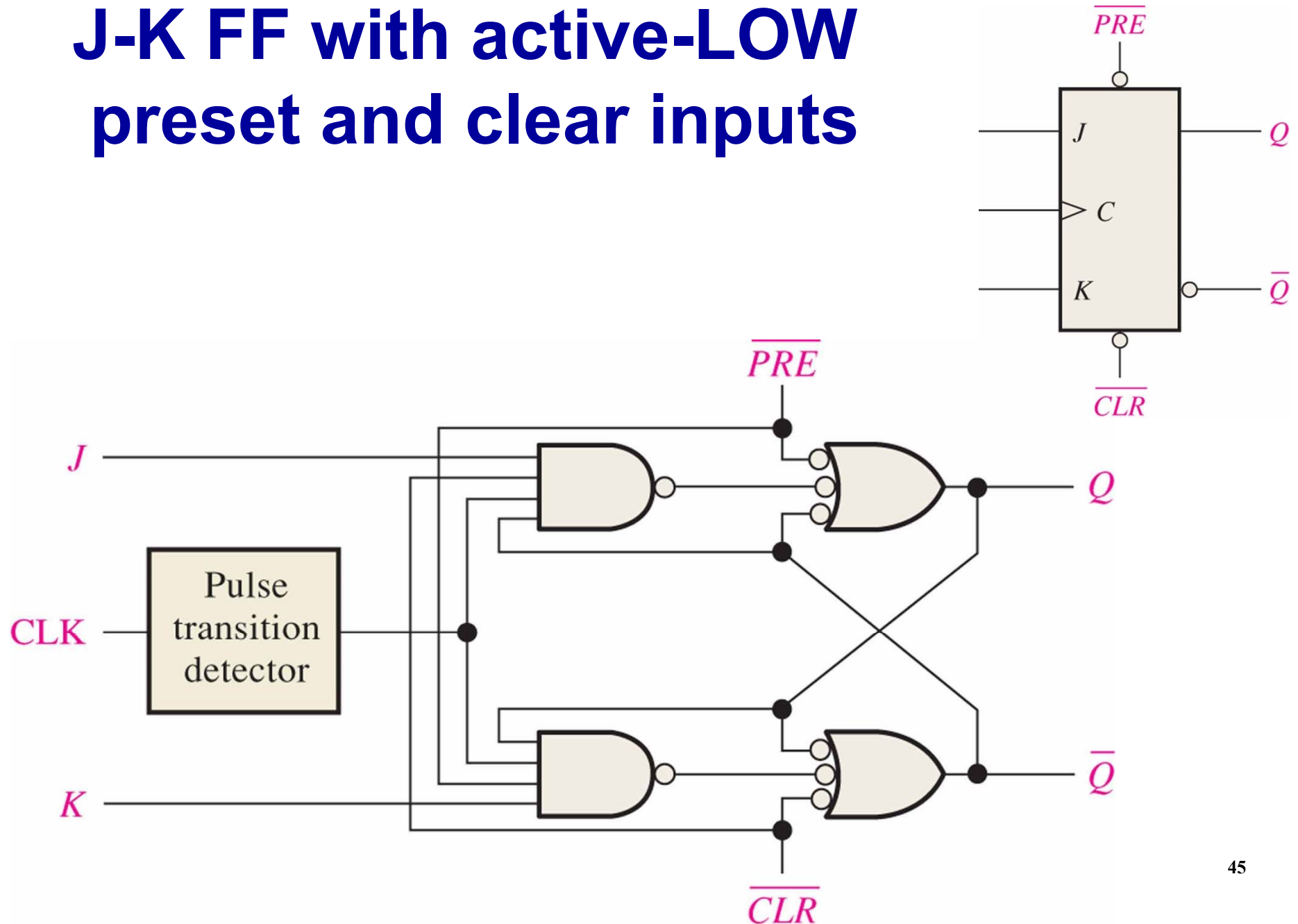
Truth Table

Inputs			Outputs		Comments
J	K	CLK	Q	$\bar{Q}$	
0	0	$\uparrow$	Q_0	$\bar{Q}_0$	No change
0	1	$\uparrow$	0	1	RESET
1	0	$\uparrow$	1	0	SET
1	1	$\uparrow$	$\bar{Q}_0$	Q_0	Toggle

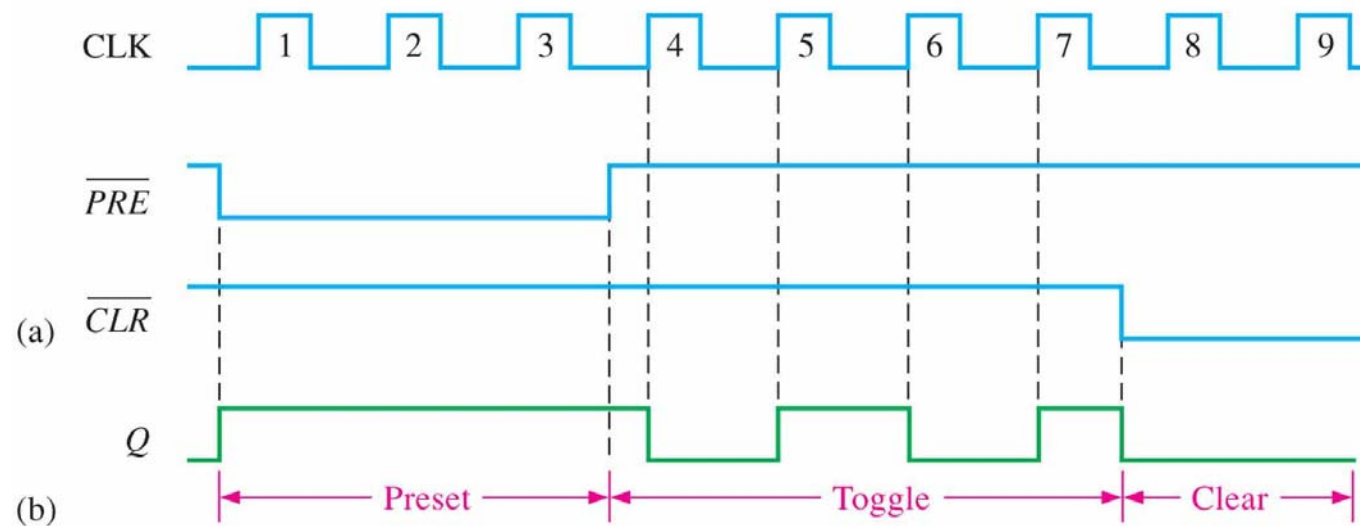
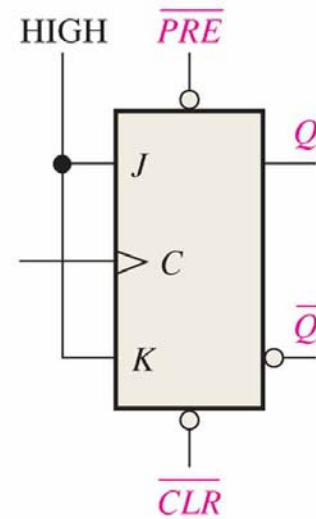
7.2.4. Asynchronous Preset and Clear Inputs (异步置位和复位)

- Asynchronous inputs are independent of the clock. Usually labeled *preset* (***PRE***) and *clear* (***CLR***), or *direct set* (***SD***) and *direct reset* (***RD***).
- An active level on the preset input will set the flip-flop, and an active level on the clear input will reset the flip-flop.

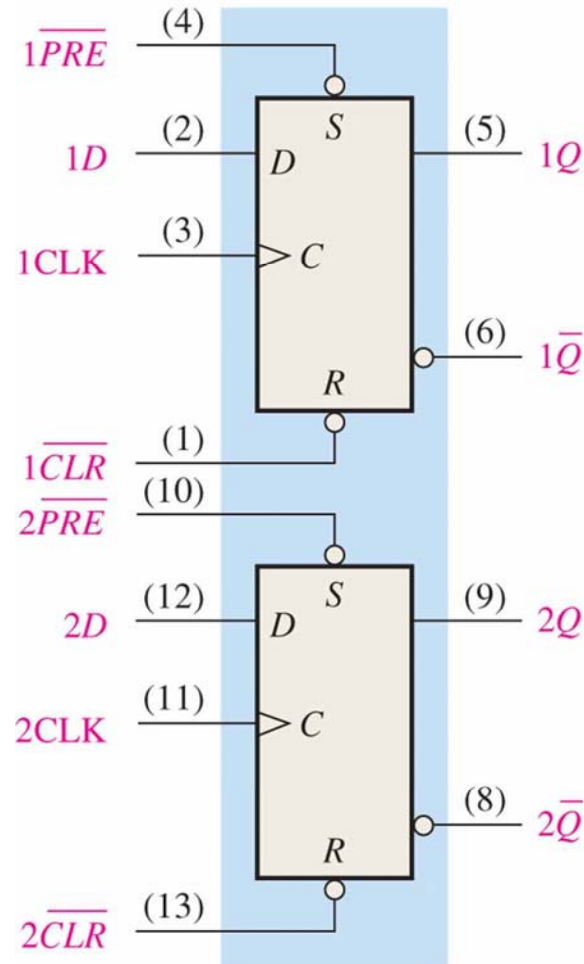
J-K FF with active-LOW preset and clear inputs



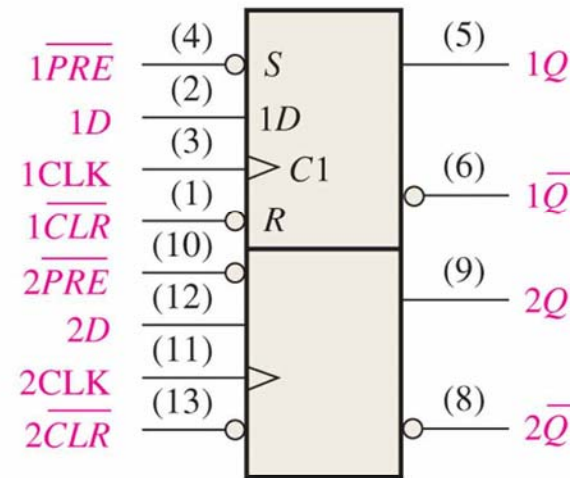
J-K FF with active-LOW preset and clear inputs



74AHC74 (dual D FF)



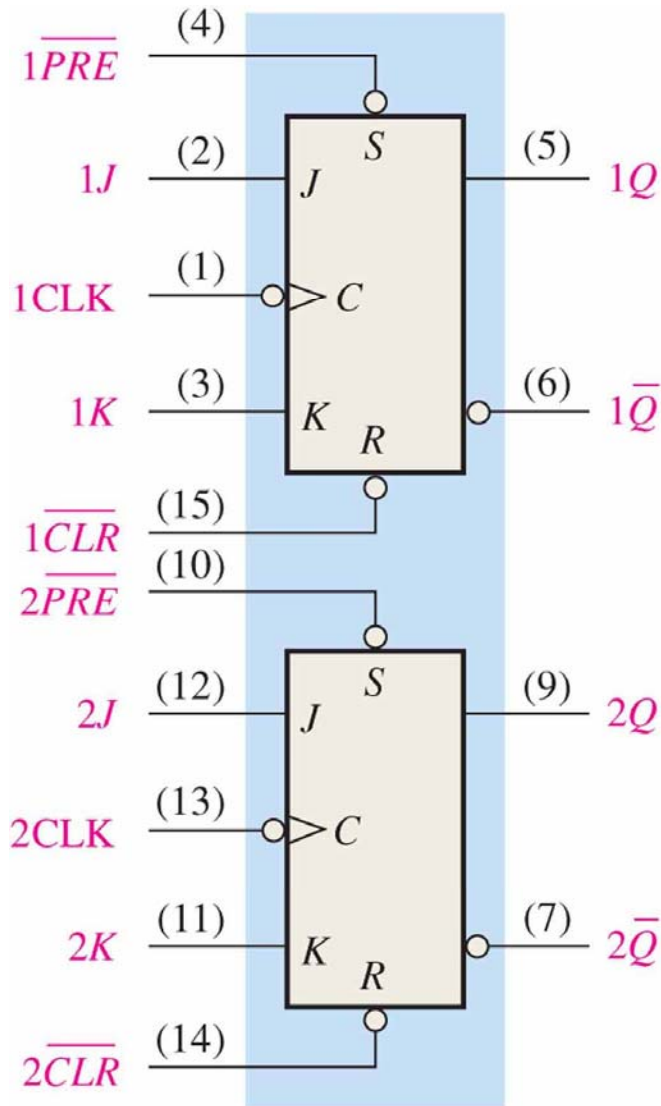
(a) Individual logic symbols



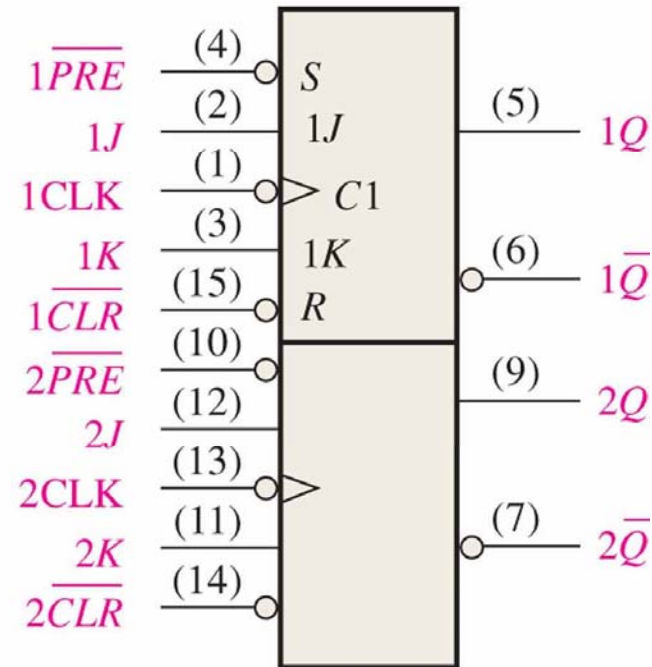
(b) Single block logic symbol

Note: The S and R inside the block indicate that $\overline{PRE}$ SETS and $\overline{CLR}$ RESETS.

74HC112 (dual JK FF)



(a) Individual logic symbols



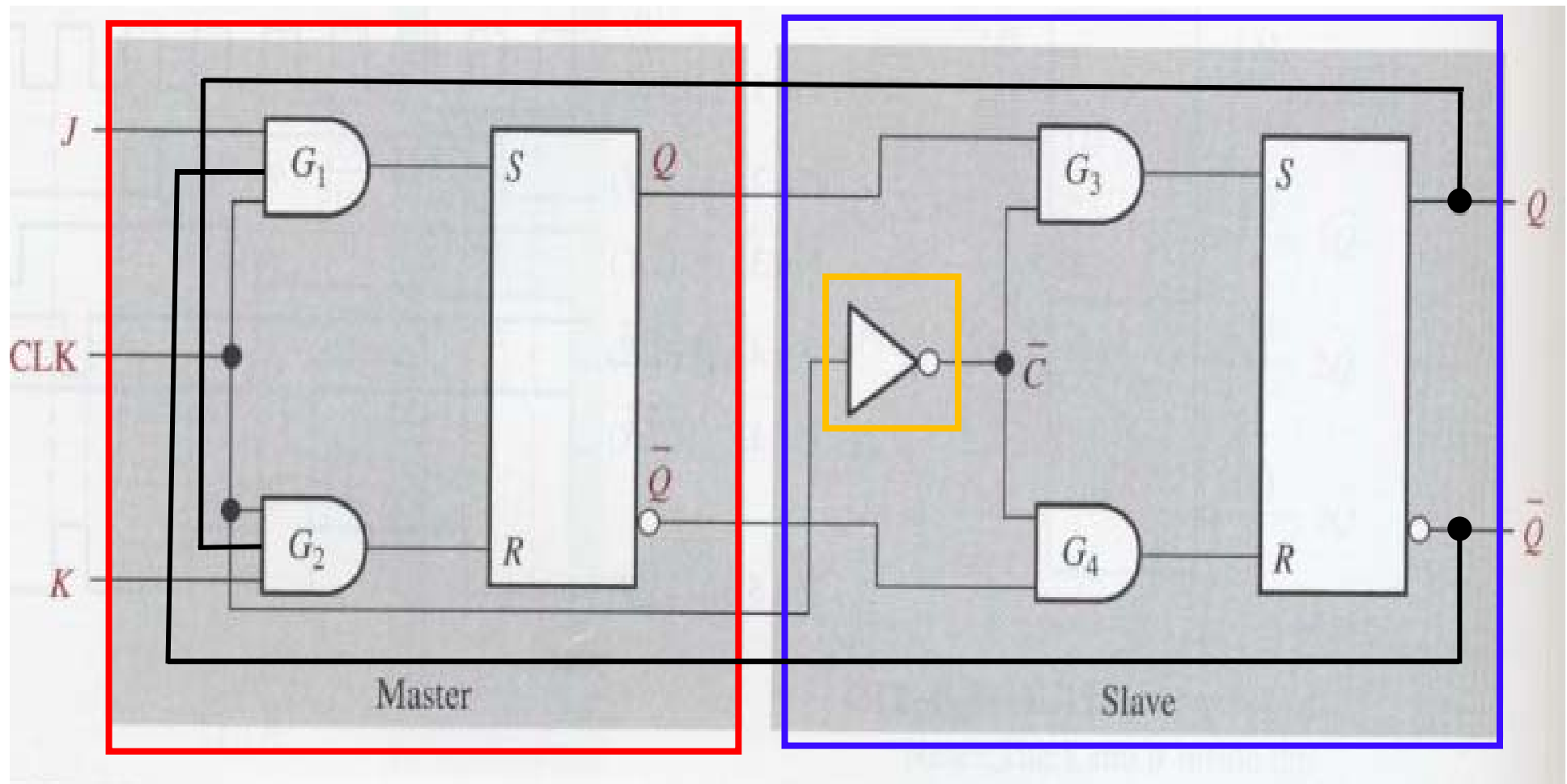
(b) Single block logic symbol

7.2.5 Master-slave Flip-Flops

主从触发器

- Composed of two section, the master section and slave section. The clock is inverted to the slave section.

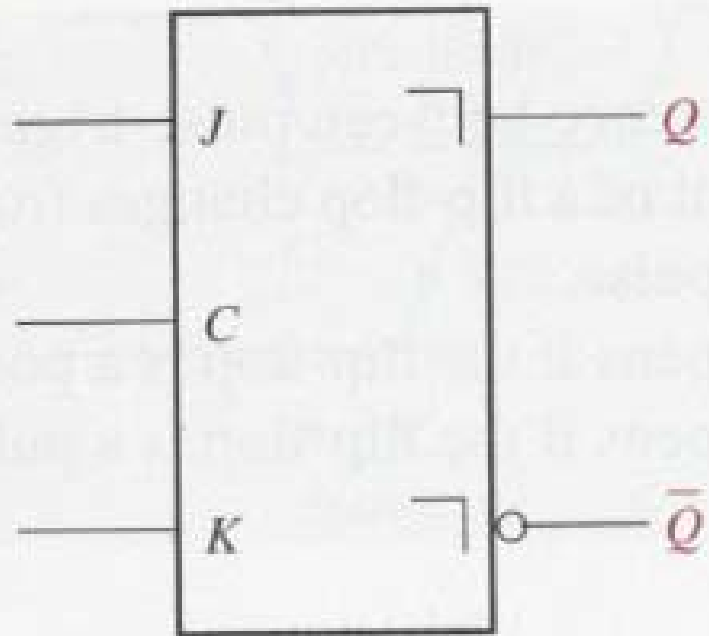
The Pulse-Triggered Master-Slave J-K Flip-Flop



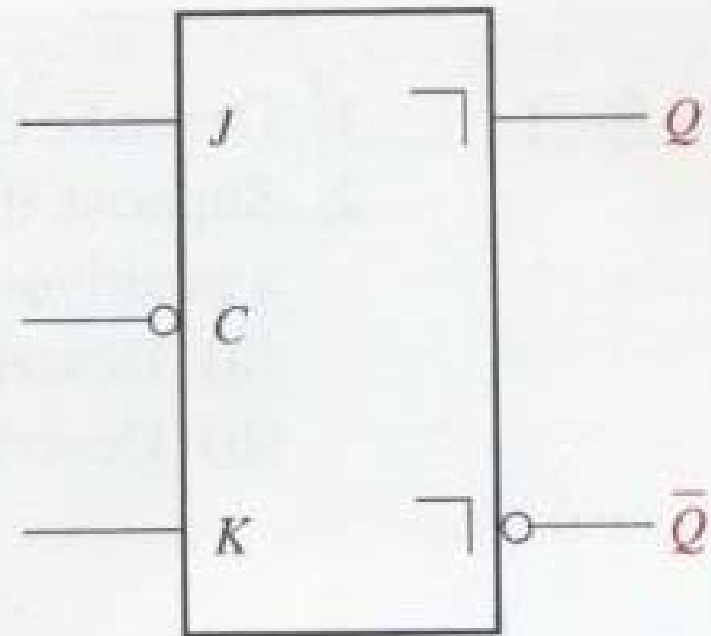
Truth Table

Inputs			Outputs		Comments
J	K	CLK	Q	$\bar{Q}$	
0	0		Q_0	$\bar{Q}_0$	No change
0	1		0	1	RESET
1	0		1	0	SET
1	1		$\bar{Q}_0$	Q_0	Toggle

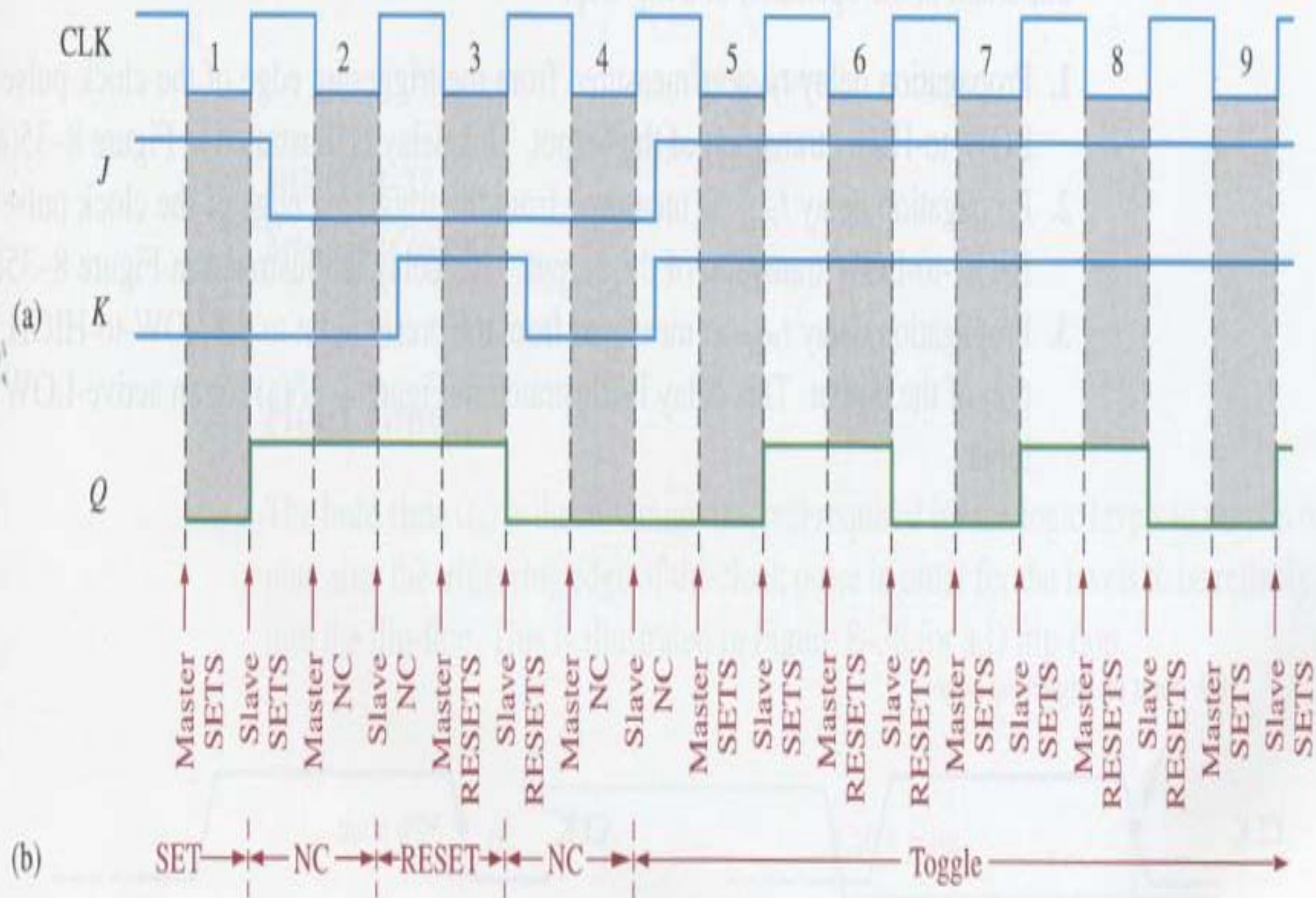
Logic Symbol



(a) Active-HIGH clock: Data are clocked in on positive-going edge of clock pulse and transferred to output on the following negative-going edge.



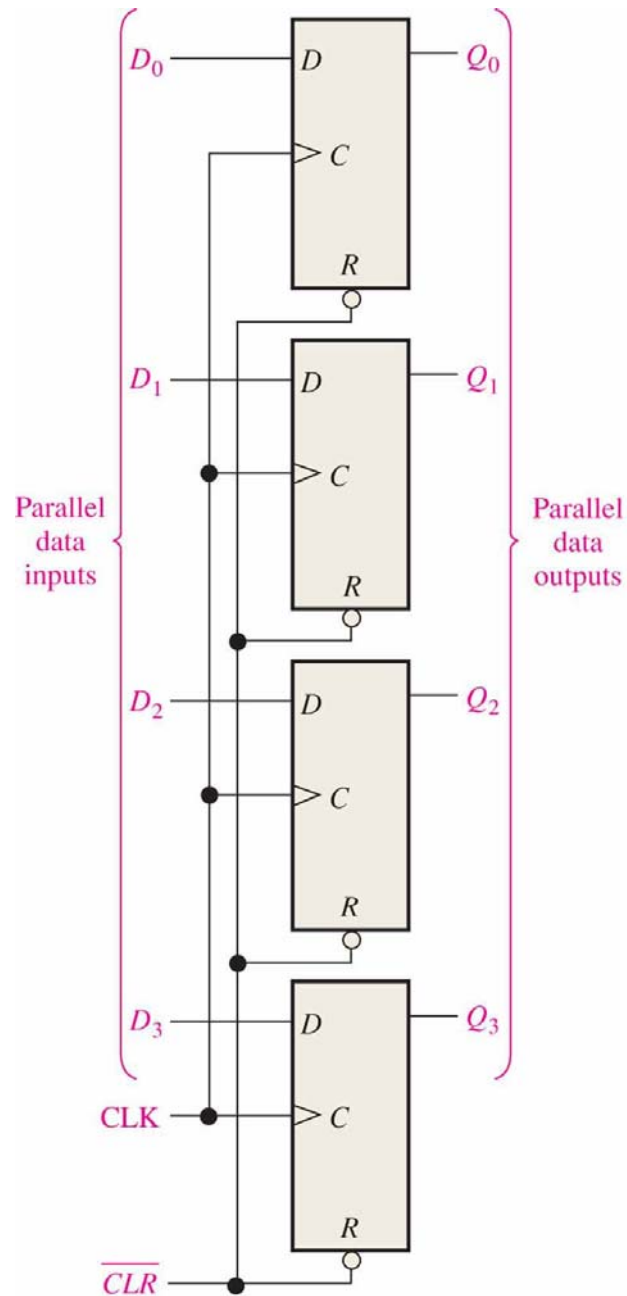
(b) Active-LOW clock: Data are clocked in on negative-going edge of clock pulse and transferred to output on the following positive-going edge.



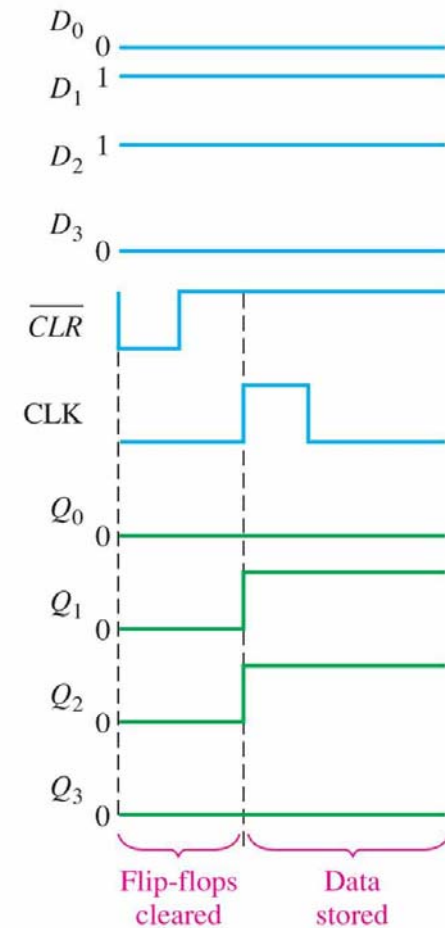
7.3 Flip-Flop Applications

- Parallel Data Storage (数据存储器)
- Frequency Division (分频器)
- Counter (计数器)

Application example: Basic register for parallel data storage

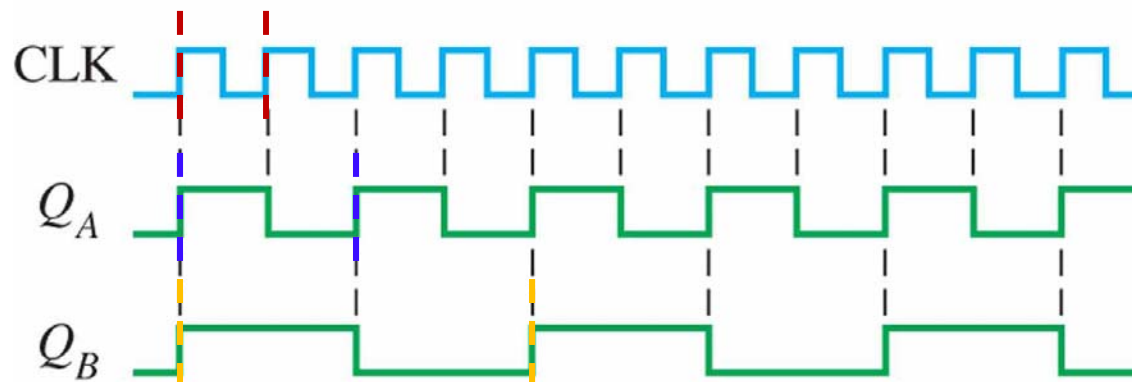
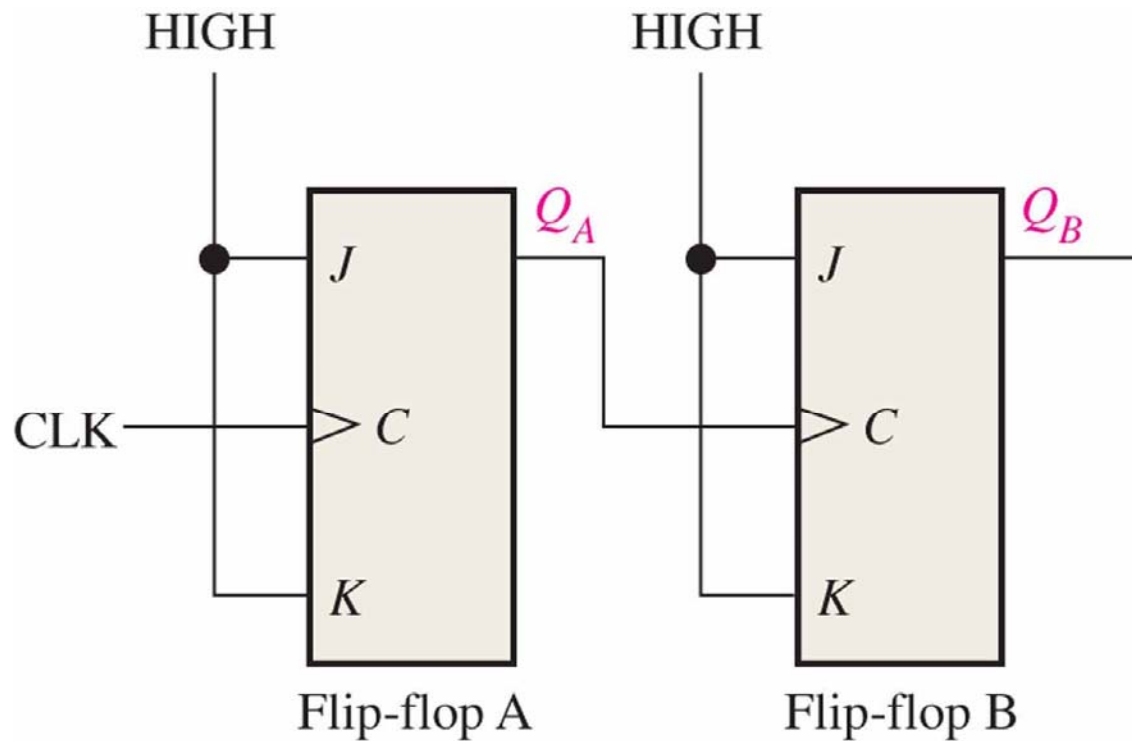


(a)

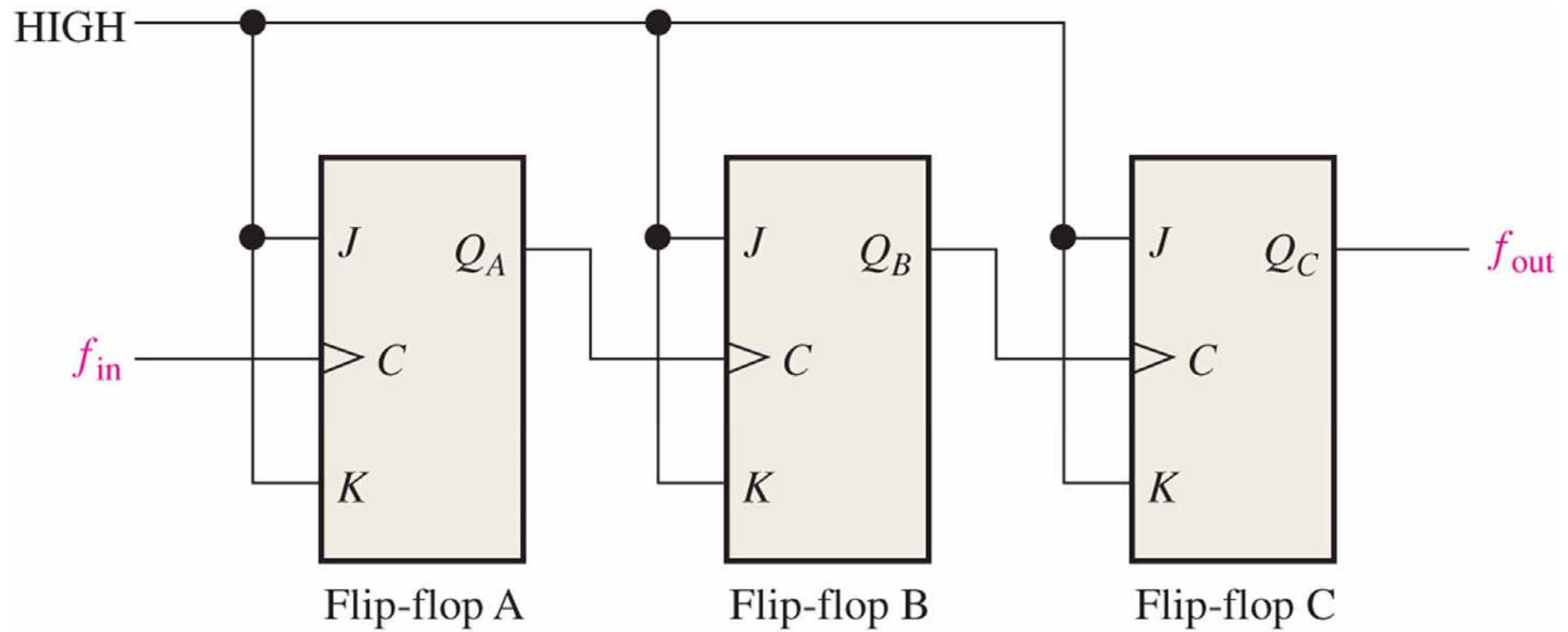


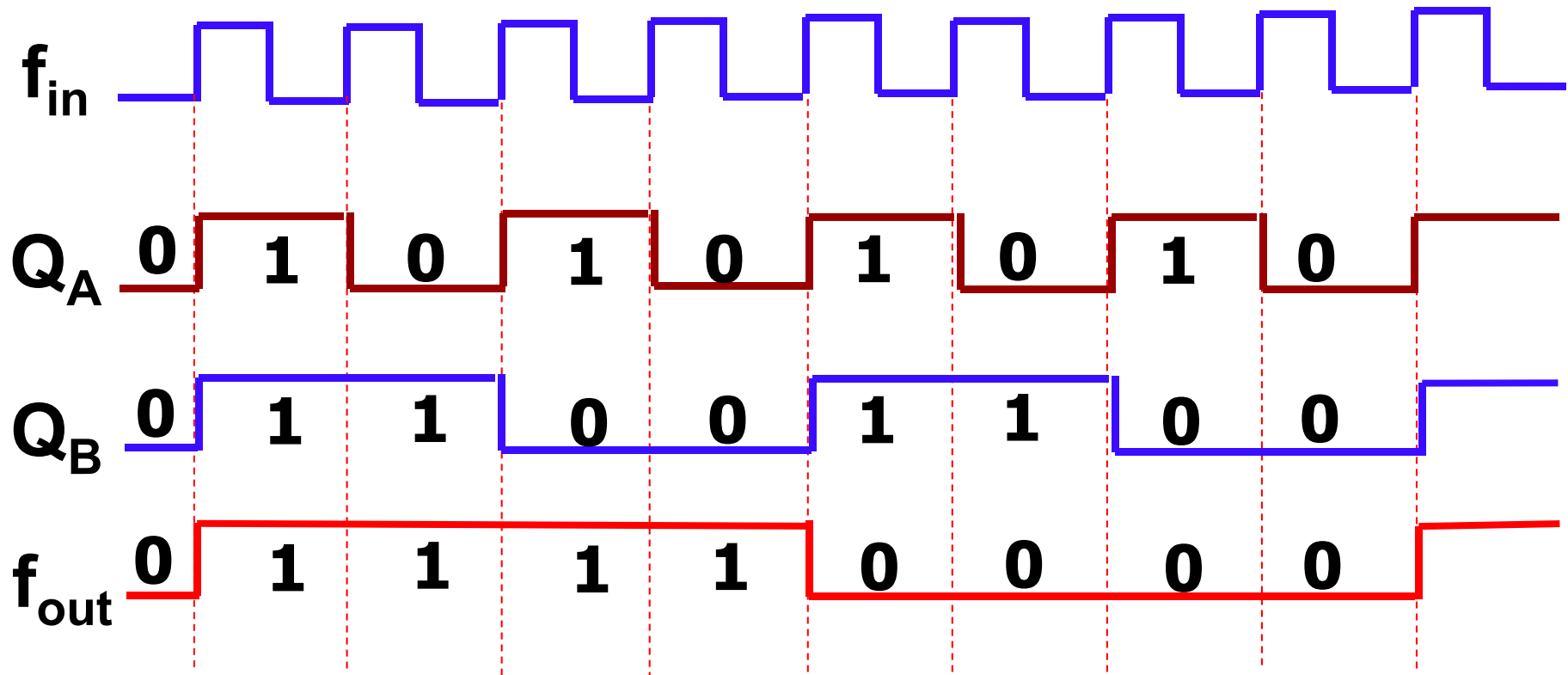
(b)

J-k FF as a divide-by-2 or 4 device

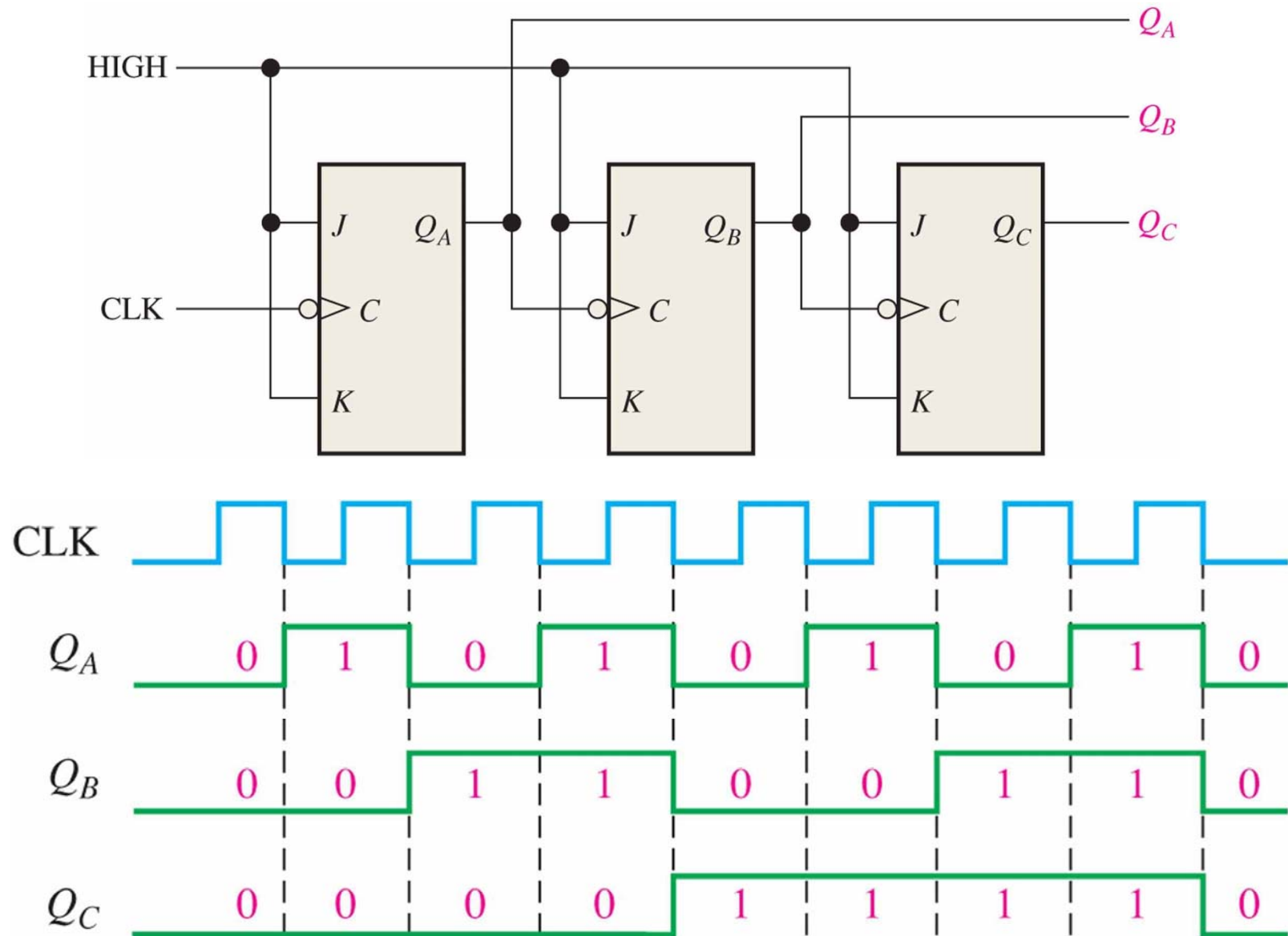


Exercise:





J-k FF as a counter

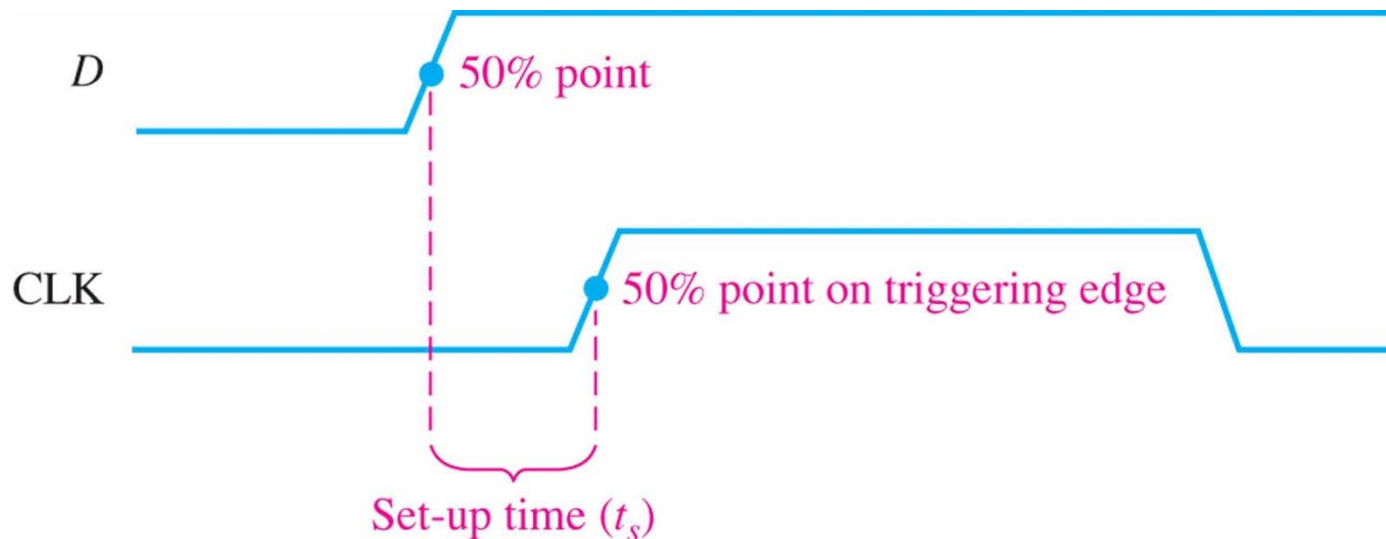


7.4 Flip-Flop Operating Characteristics

- Set-up time (建立时间)
- Hold time (保持时间)
- Propagation delay time (传播时延)
- Maximum of frequency (最大频率)
- Power dissipation (功耗)

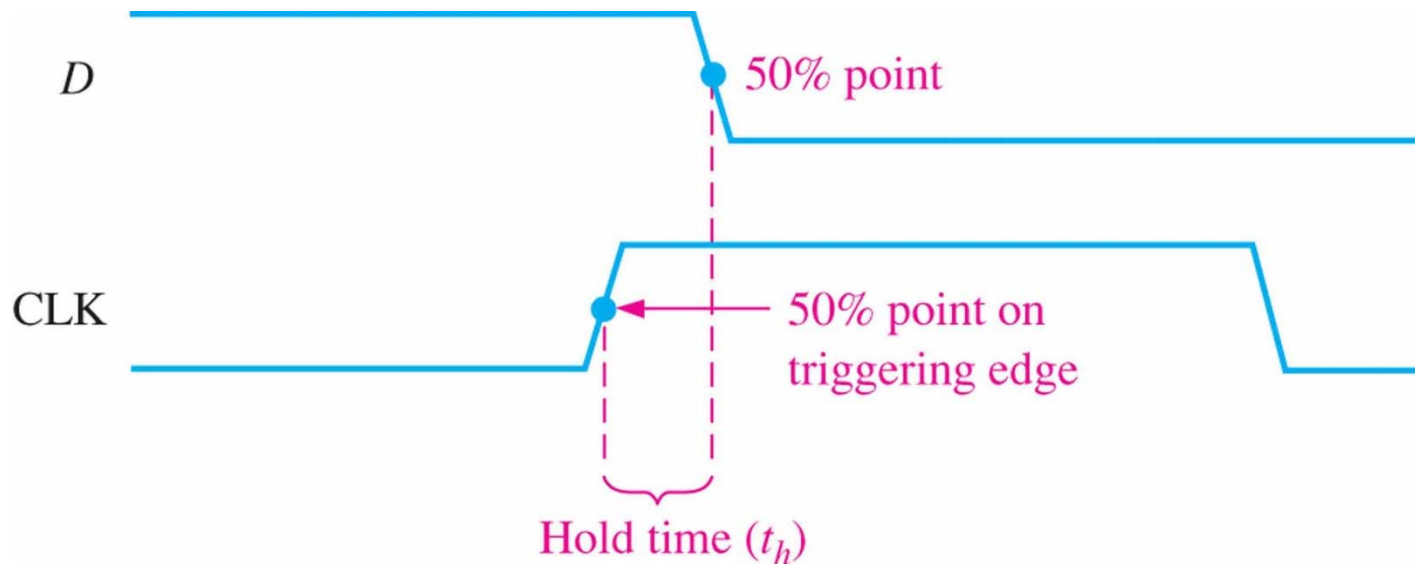
Set-up Time

The **set-up time** (t_s) is the **minimum interval of time** required for the logic levels to be maintained constantly on the inputs(J and K, or S and R, or D) prior to the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip-flop.



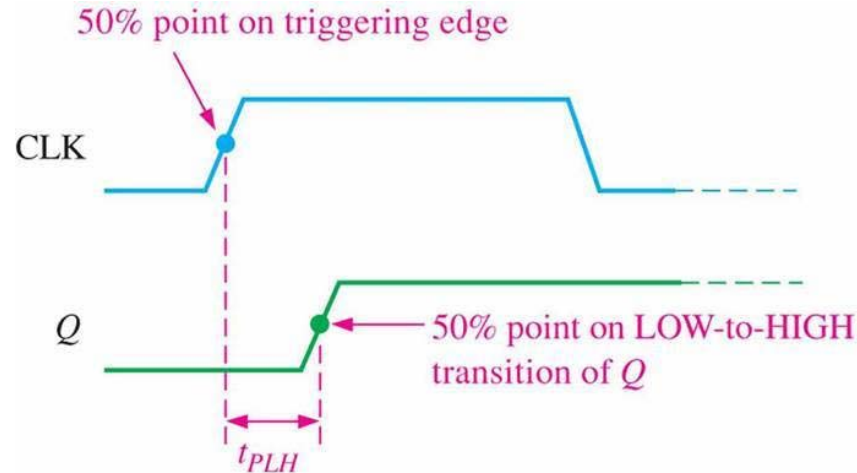
Hold Time

The **Hold time** (t_h) is the **minimum interval of time** required for the logic levels to remain constantly on the inputs after the triggering edge of the clock pulse in order for the levels to be reliably clocked into the flip-flop.

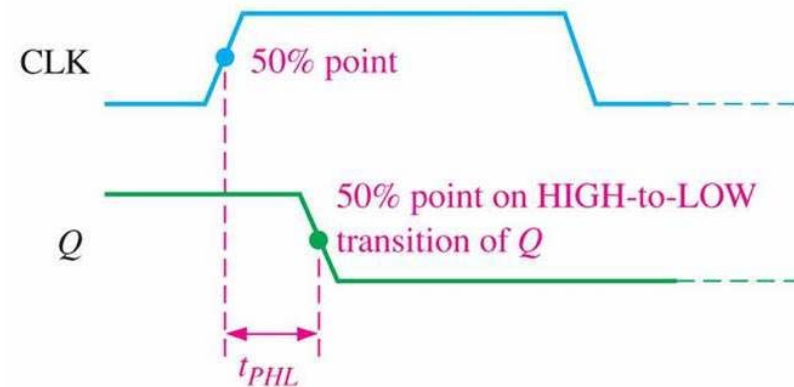


Propagation Delay Time for a flip-flop(I)

1. t_{PLH} : Measured from the **triggering edge of the clock** pulse to **the LOW-to-HIGH** transition of the output
2. t_{PHL} : Measured from the **triggering edge of the clock** pulse to the **HIGH-to-LOW** transition of the output.



(a)

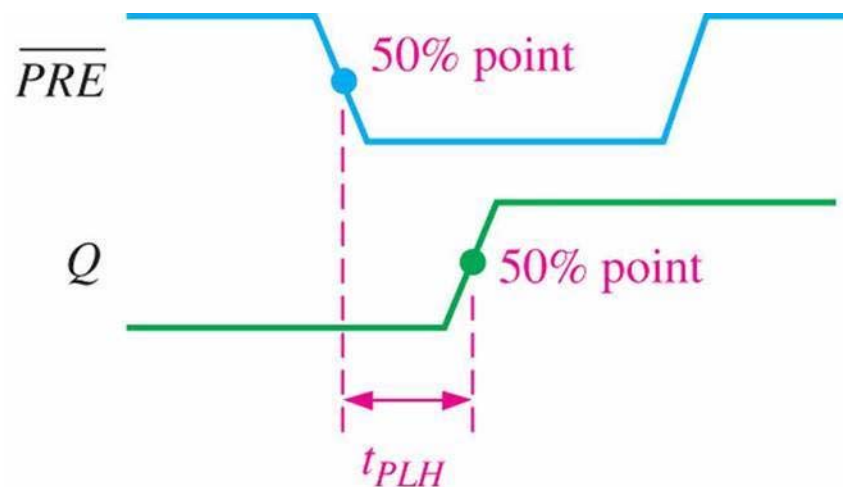


(b)

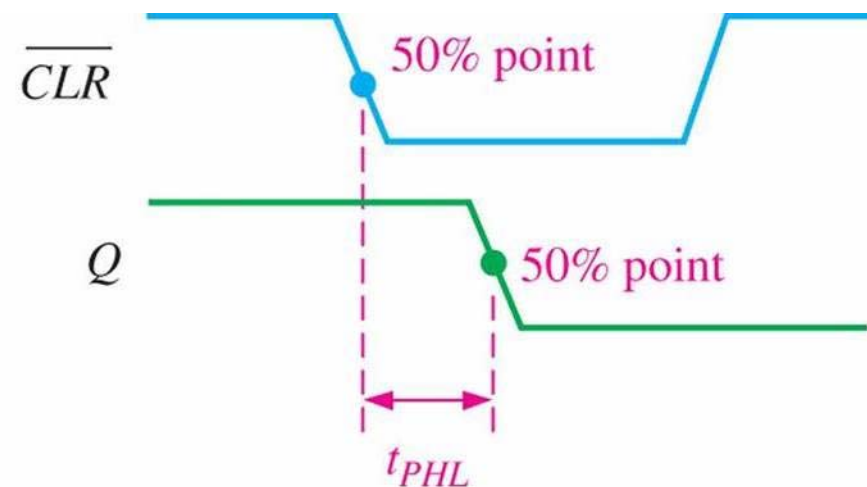
The propagation delay time is the interval of time required after an input signal has been applied for the resulting output change to occur.

Propagation Delay Time for a flip-flop(II)

3. t_{PLH} : Measured from the **leading edge of the preset input** to the **LOW-to-HIGH** transition of the output
4. t_{PHL} : Measured from the **leading edge of the clear input** to the **HIGH-to-LOW** transition of the output.



(a)



(b)

Maximum Clock Frequency

The **maximum clock frequency** ($f_{\max}$) is the **highest rate** at which a flip-flop can be reliably triggered. At clock frequencies above the maximum, the flip-flop would be unable to respond quickly enough and its operation would be impaired.

Power Dissipation

The **power dissipation** is the total power consumption of the device.

$$P = V_{cc} \times I_{cc} = 5V \times 5mA = 25mW$$

7.5 Schmitt Trigger (施密特触发器)

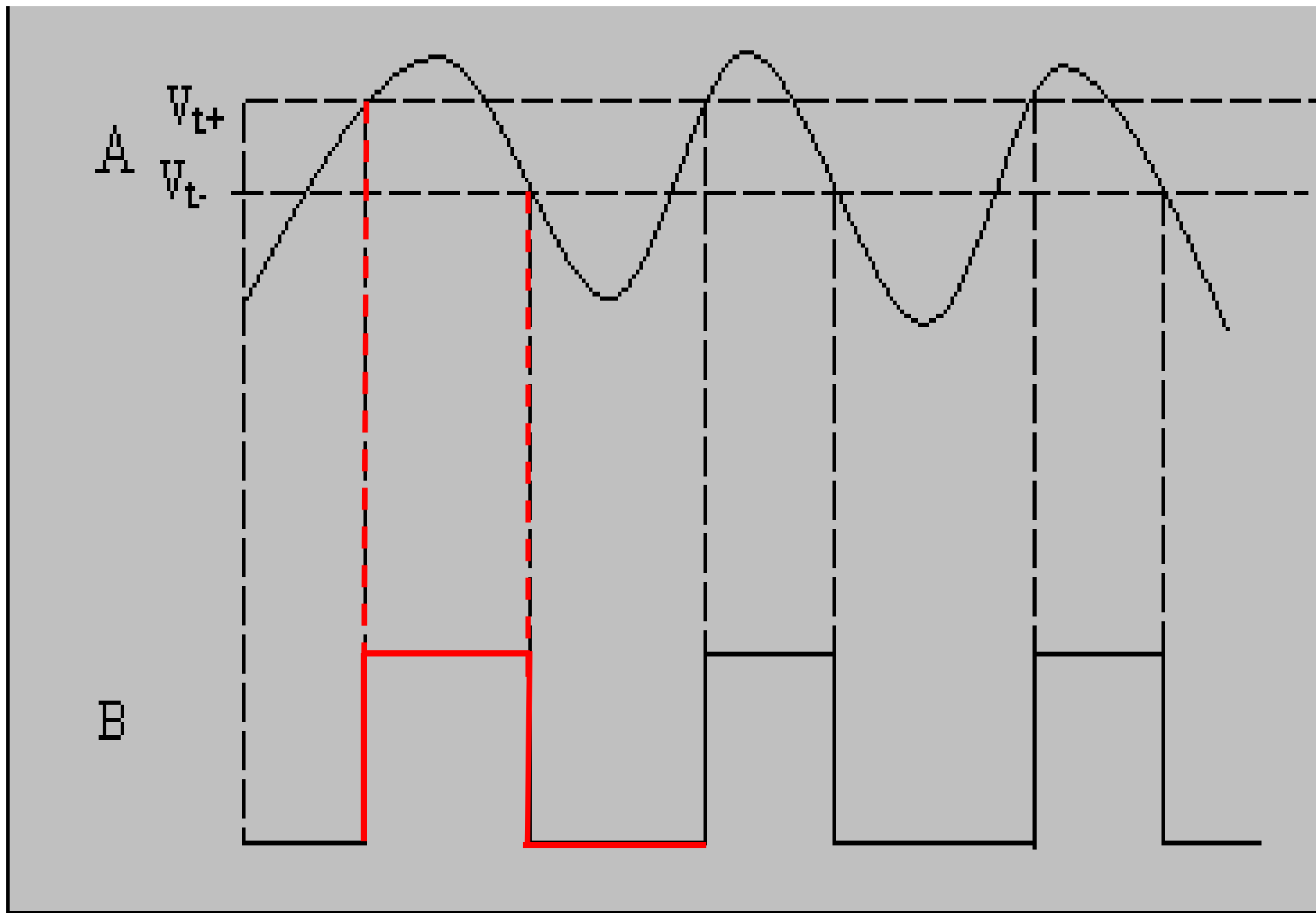
A *Schmitt trigger* is a special type of bi-stable device that has two threshold voltages (i.e. stable states).

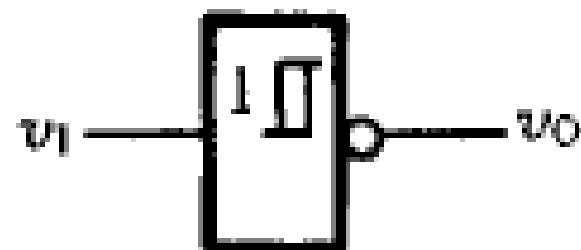
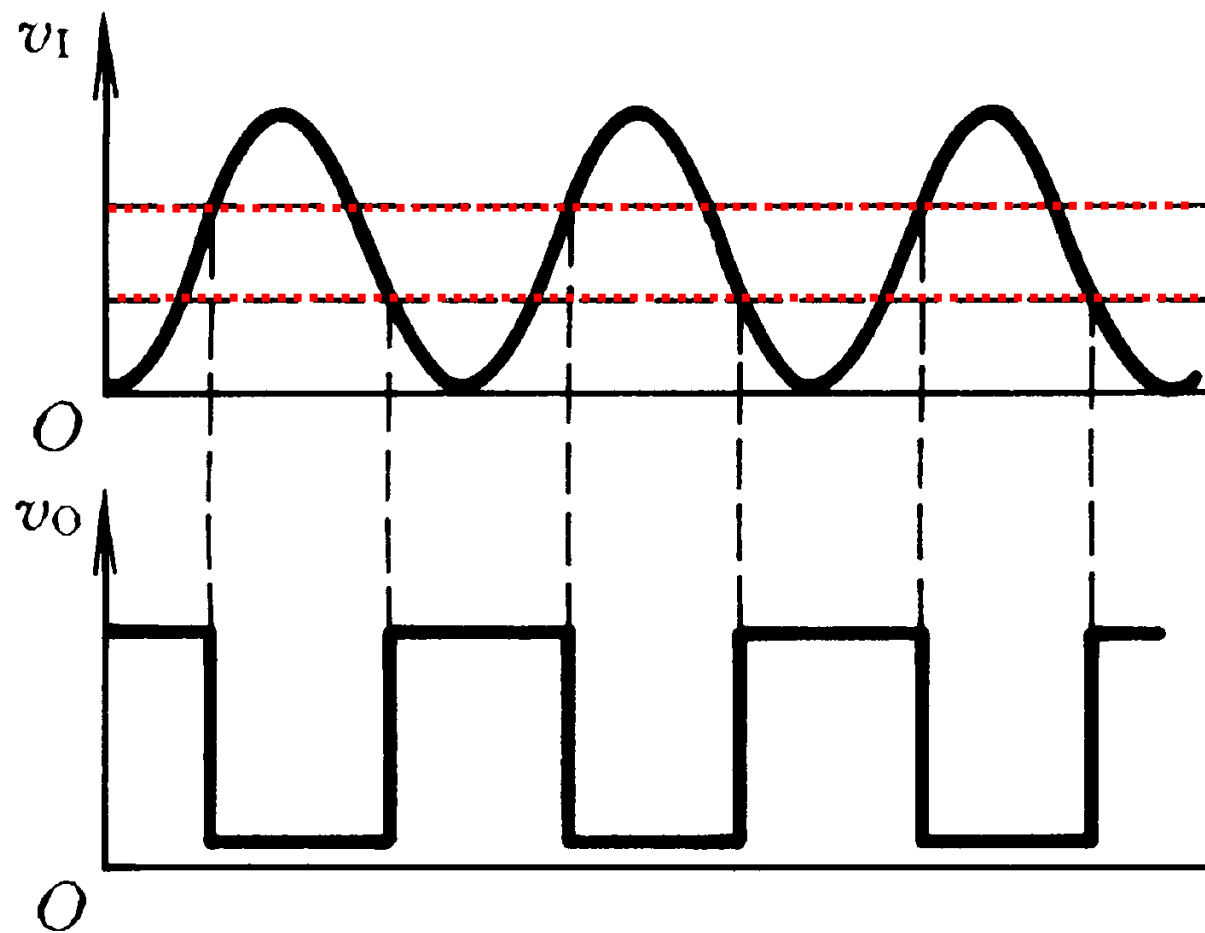
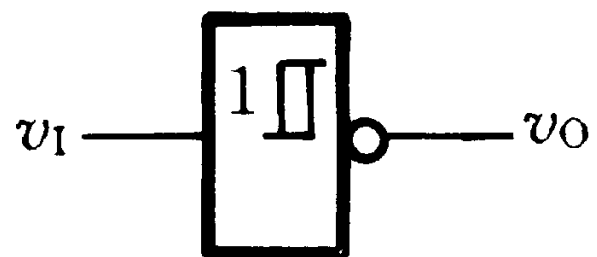
Characteristics:

- using two voltage thresholds: a high threshold to switch the circuit during low-to-high transitions and a lower threshold to switch the circuit during high-to-low transitions.

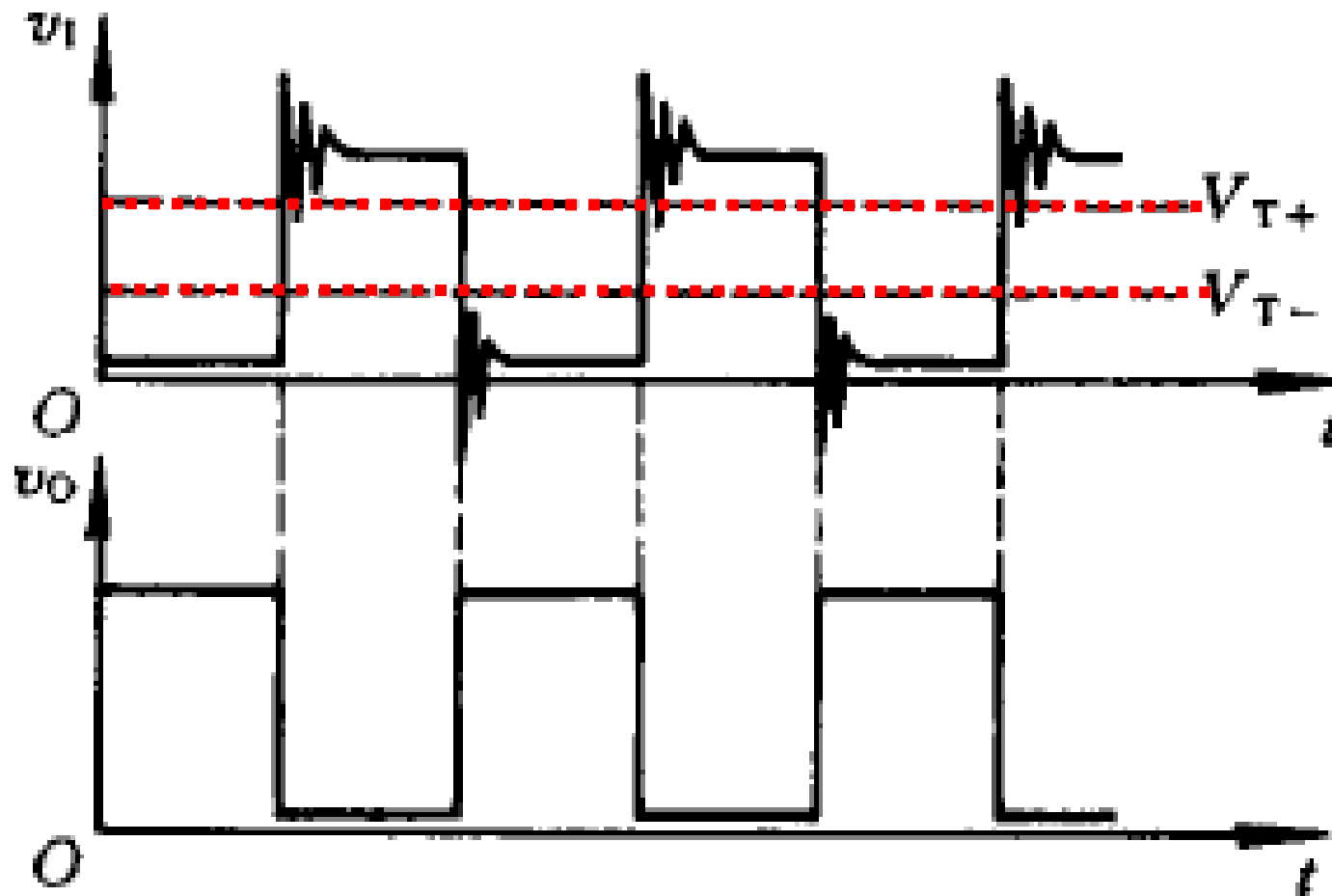
Advantages:

- Generating pulse with sharp edges
- increasing the noise immunity

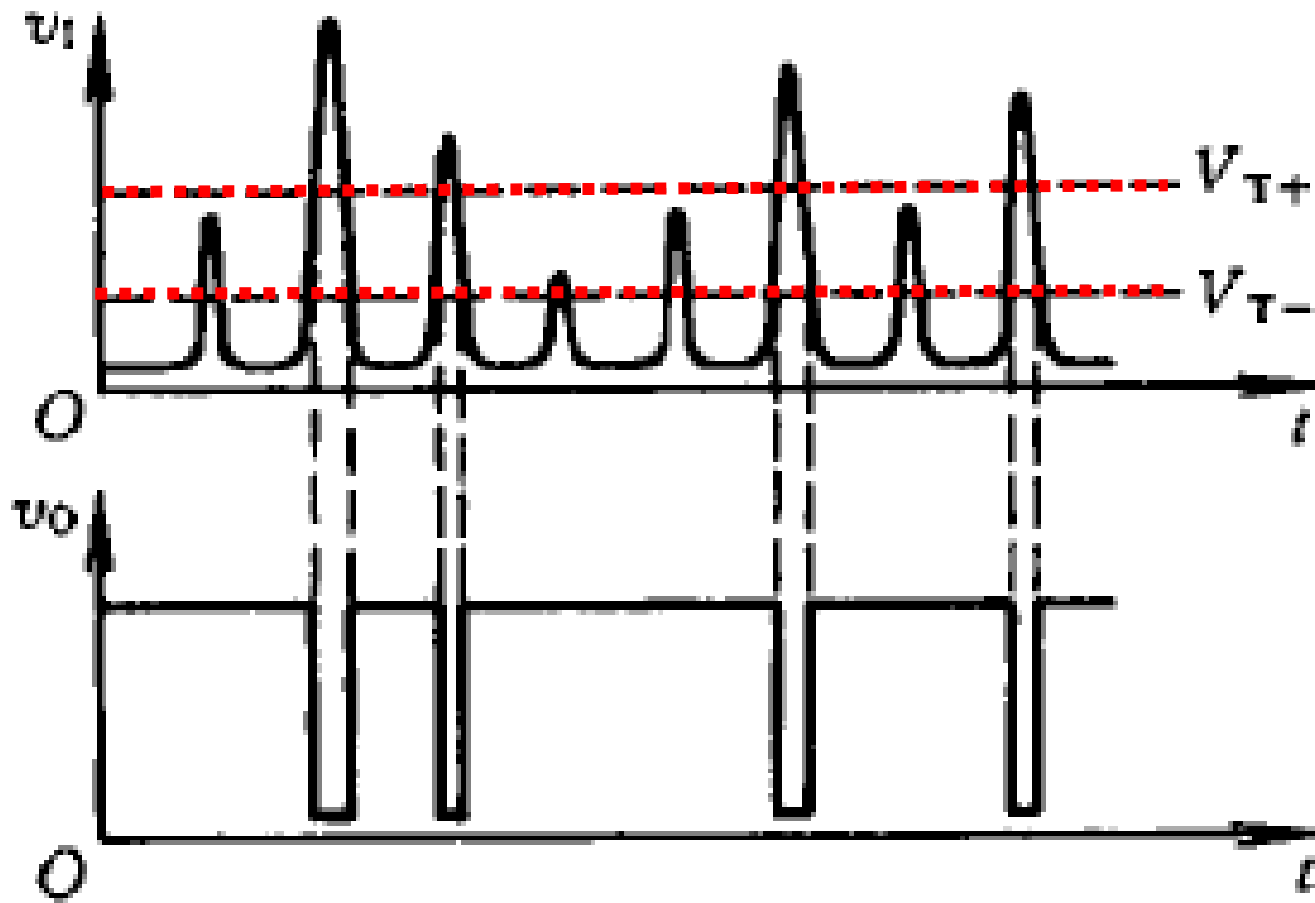




Noise immunity

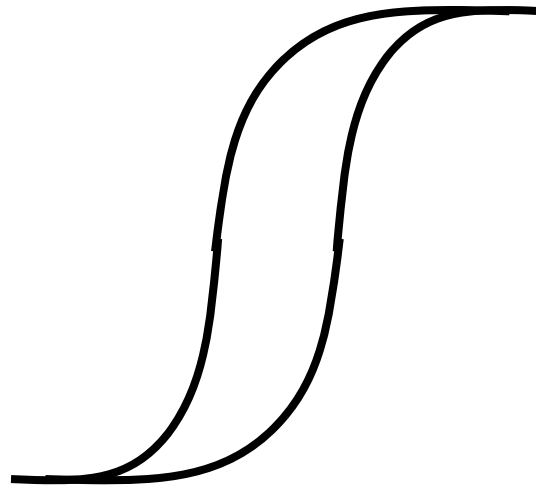


Amplitude Checking



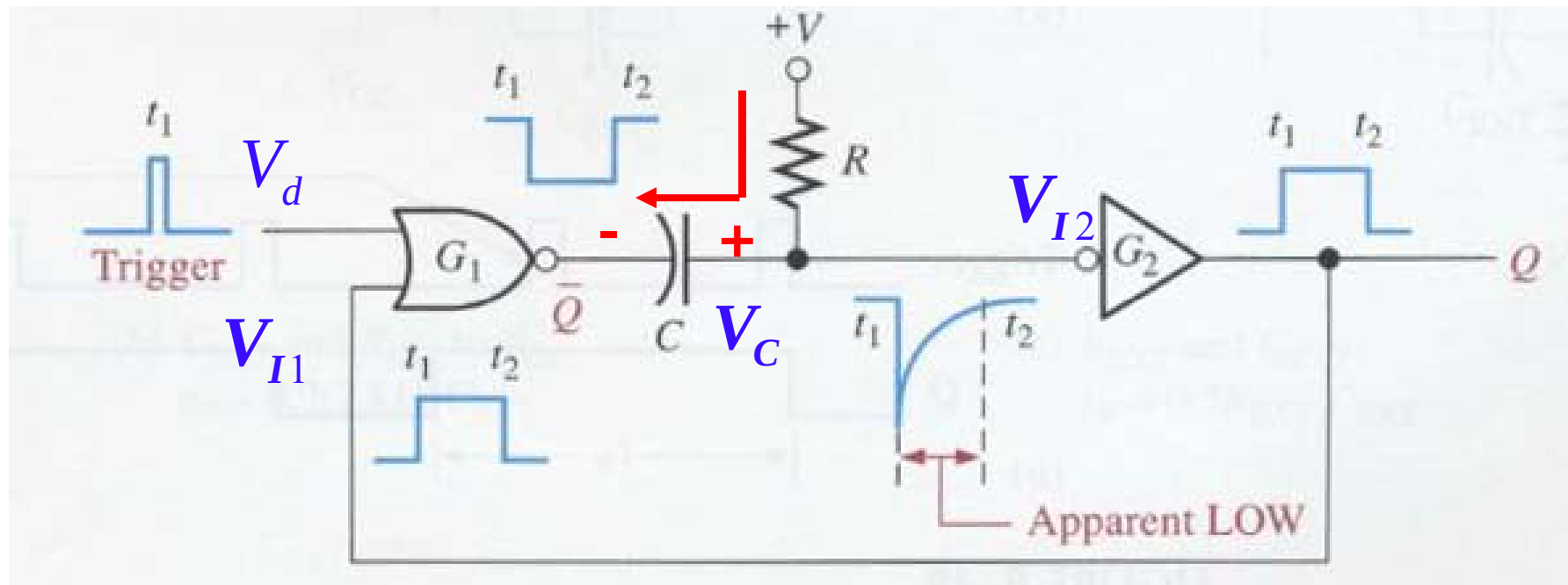
Symbol

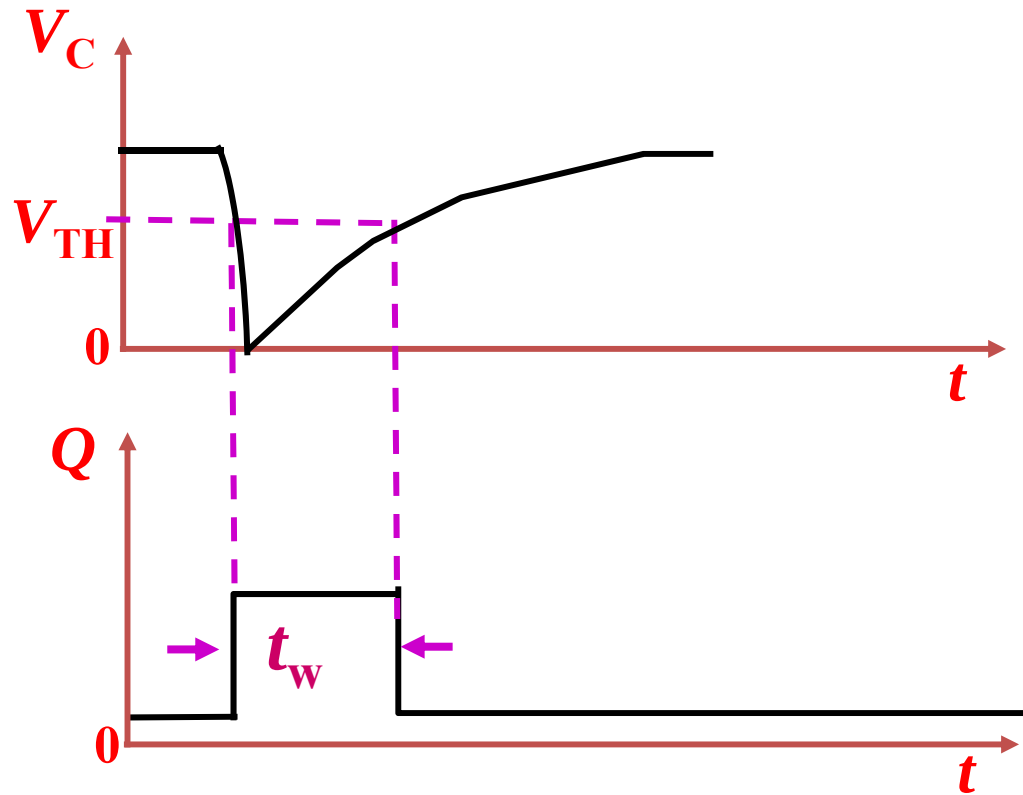
The ***hysteresis symbol*** indicates a Schmitt trigger input



7.6 One-Shots (单稳态触发器)

- **One shot** (monostable multivibrator) only has one stable state and one unstable state.
- When triggered, the device changes from its stable state to its unstable state and remains there for a fixed period of time, known as the **pulse width**, before returning to its stable state.
- The duration time of the unstable state determined by the circuit parameters, no related with the triggered pulse.
- The duration time of the unstable state determines the pulse width of the output pulse.





$$v_c(t) = v_c(\infty) + [v_c(0_+) - v_c(\infty)]e^{-\frac{t}{\tau}}$$

$$v_c(\infty) = V_{DD} \quad v_c(0_+) = 0 \quad \tau = RC$$

$$v_c(t_w) = V_{TH} \quad \longrightarrow \quad t_w = RC \ln \frac{V_{DD}}{V_{DD} - V_{TH}}$$

- Stable state: $Q=0$
- Unstable state: $Q=1$
- The time duration determined by the charge time of C and R , i.e. the RC time constant.

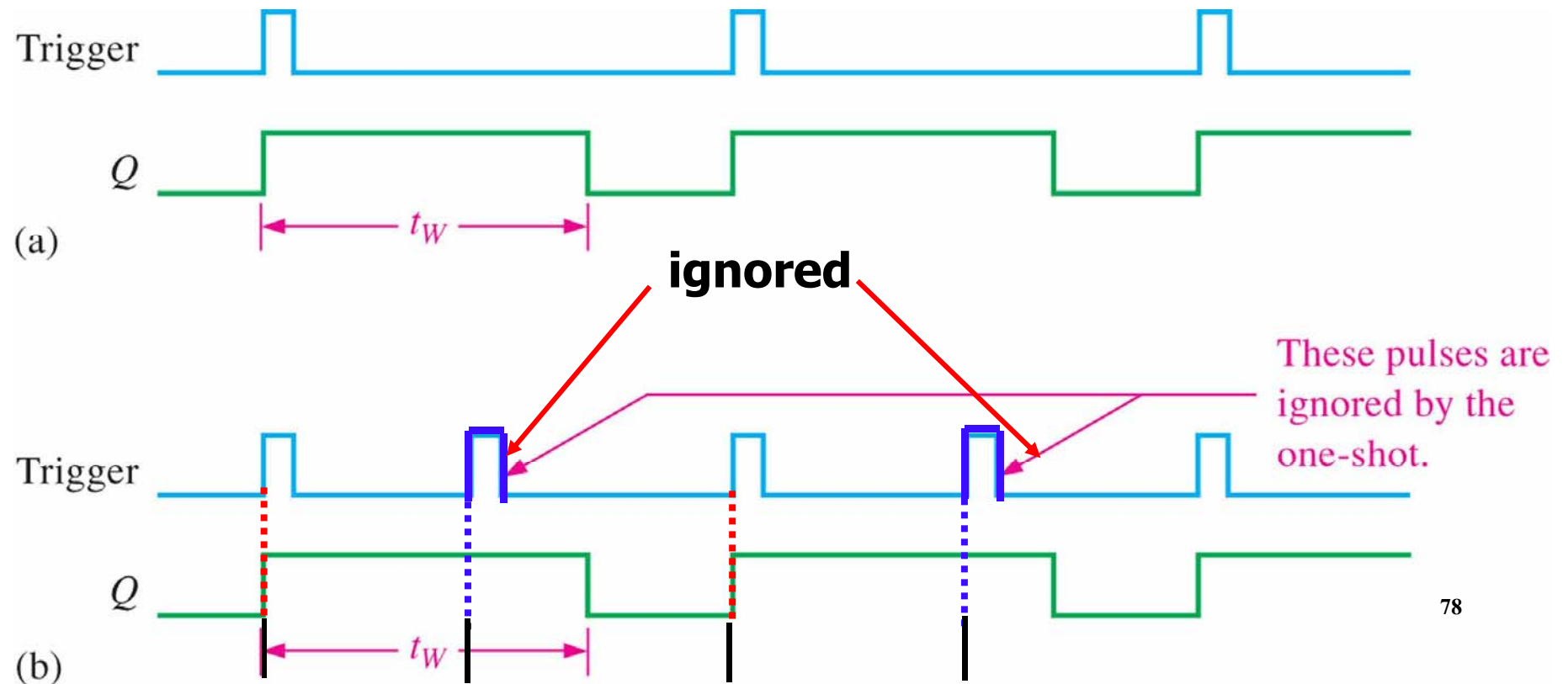
$$t_w = RC \ln \frac{V_{DD}}{V_{DD} - V_{TH}} = RC \ln 2 \approx 0.69 RC$$
$$\approx 0.7 RC$$

Two basic types of IC one-shot

- *Nonretriggerable* one-shot
- *Retriggerable* one-shot

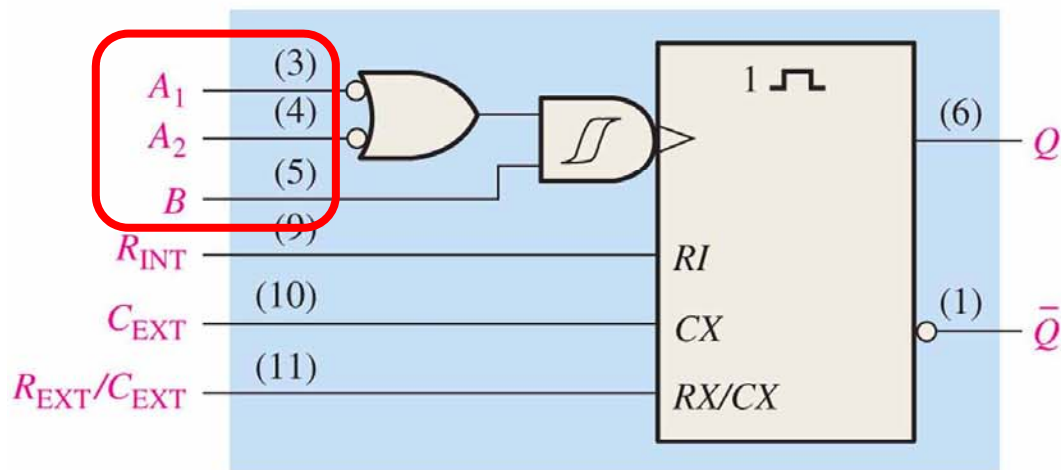
Non-retriggerable one-shot

Not respond to any additional trigger pulse from the time it is triggered until it returns to stable state.

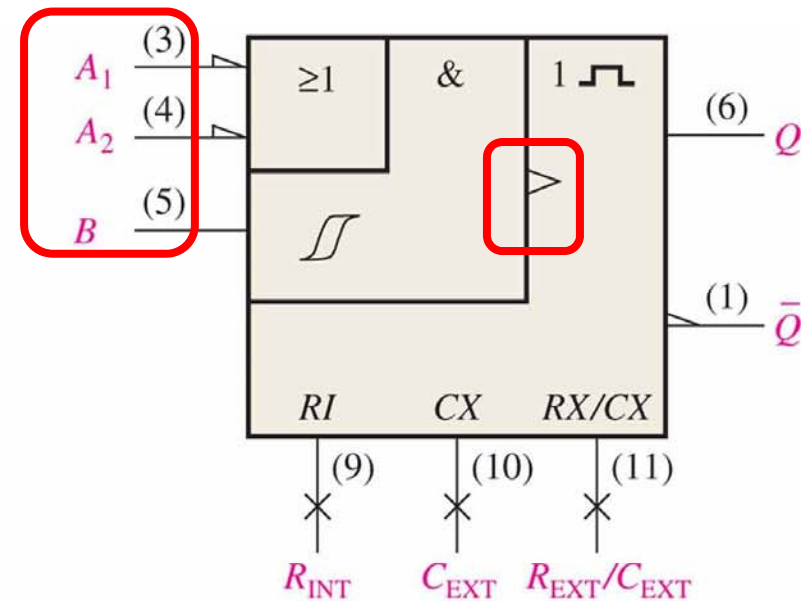


74LS121: non-retriggered one-shot

trigger

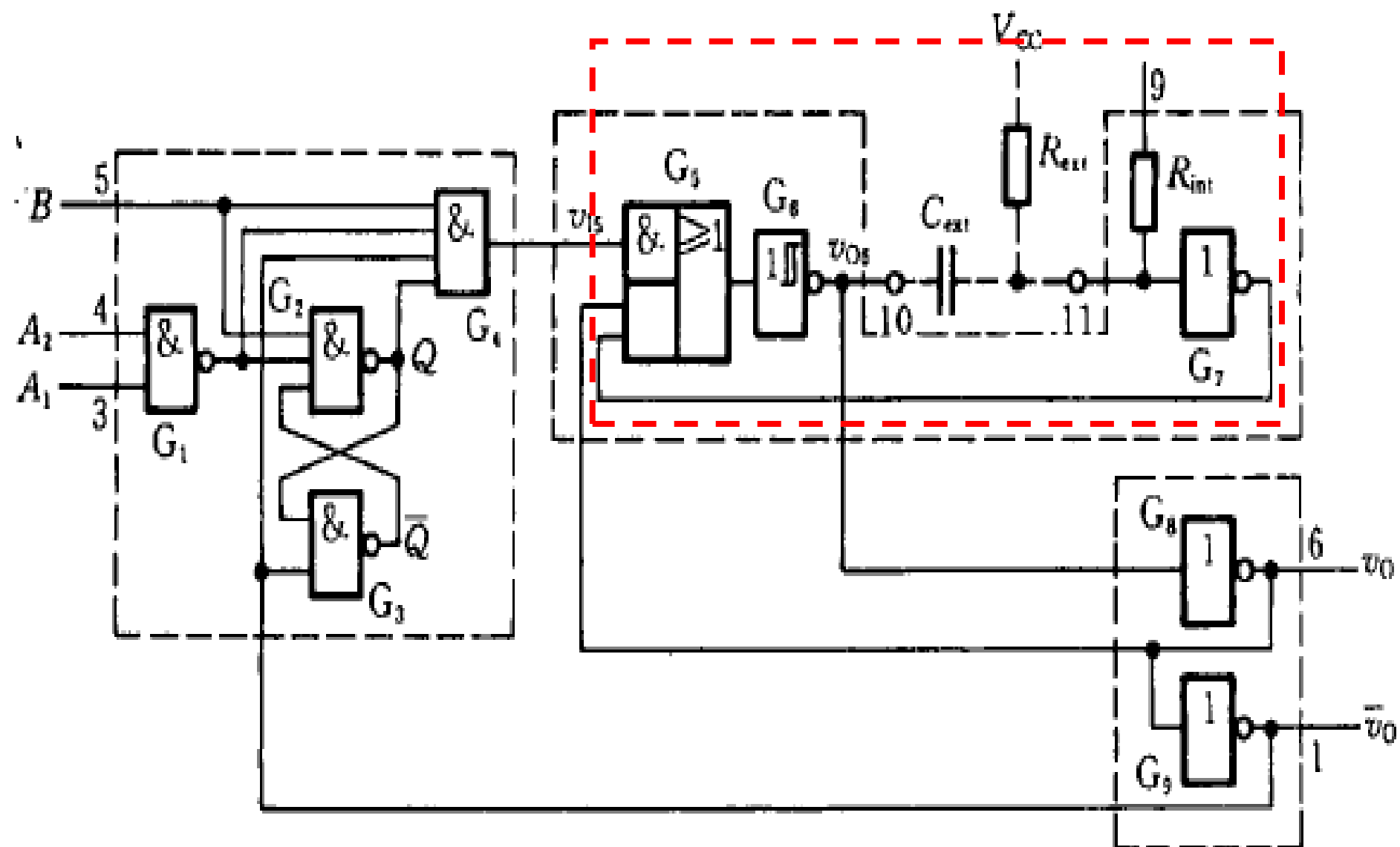


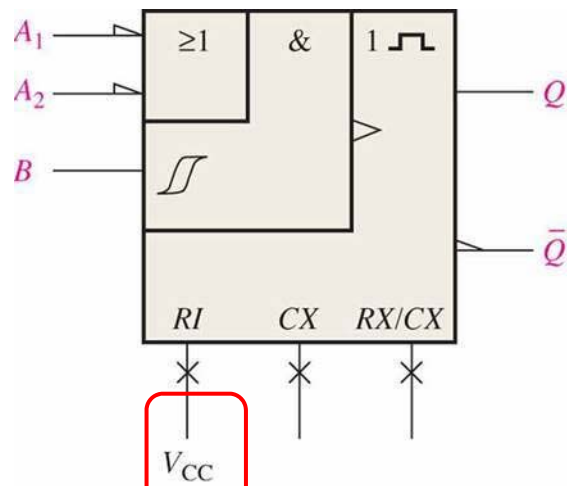
(a) Traditional logic symbol



(b) ANSI/IEEE std. 91–1984 logic symbol
($\times$ = nonlogic connection). “1 

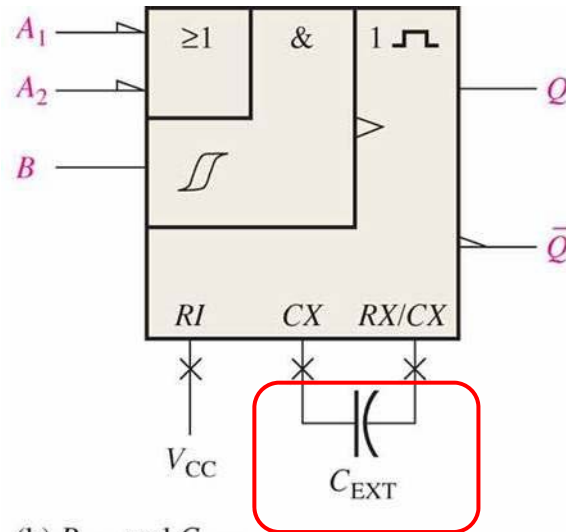
$$R_{INT}=2K\Omega$$





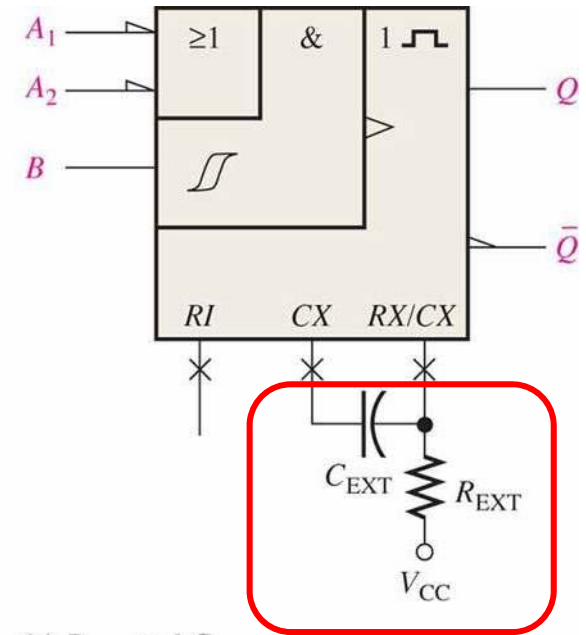
(a) No external components
 R_{INT} to V_{CC}
 $t_W \cong 30 \text{ ns}$

$$t_W \approx 30 \text{ ns}$$



(b) R_{INT} and C_{EXT}
 $t_W = 0.7(2 \text{ k}\Omega)C_{EXT}$

$$t_W \approx 0.7(2 \text{ k}\Omega) C_{EXT}$$



(c) R_{EXT} and C_{EXT}
 $t_W = 0.7R_{EXT}C_{EXT}$

$$t_W \approx 0.7R_{EXT} C_{EXT}$$

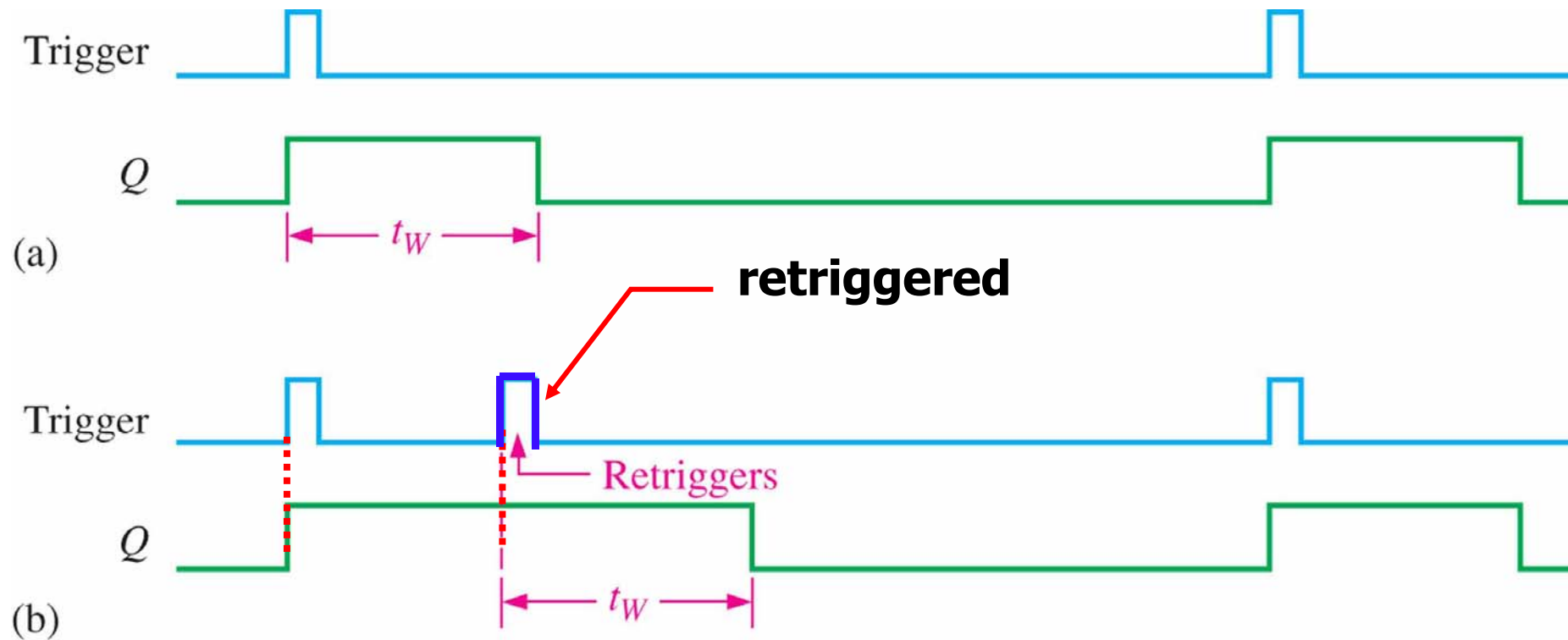
Homework: Exercise

To achieve a one-shot with a pulse width of approximately 10ms, using a 74121.

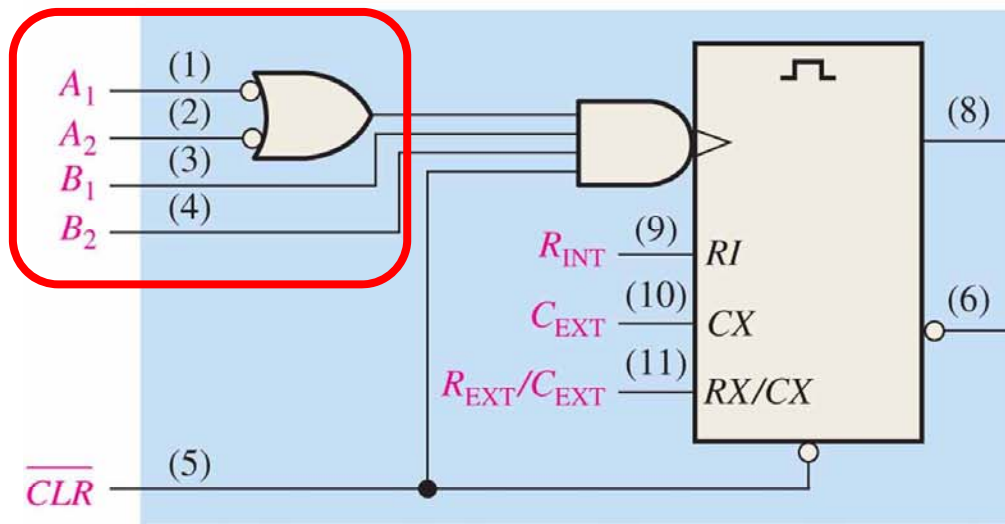
If select $R_{EXT}=20K\Omega$, calculate the capacitance:

$$C_{EXT}=?$$

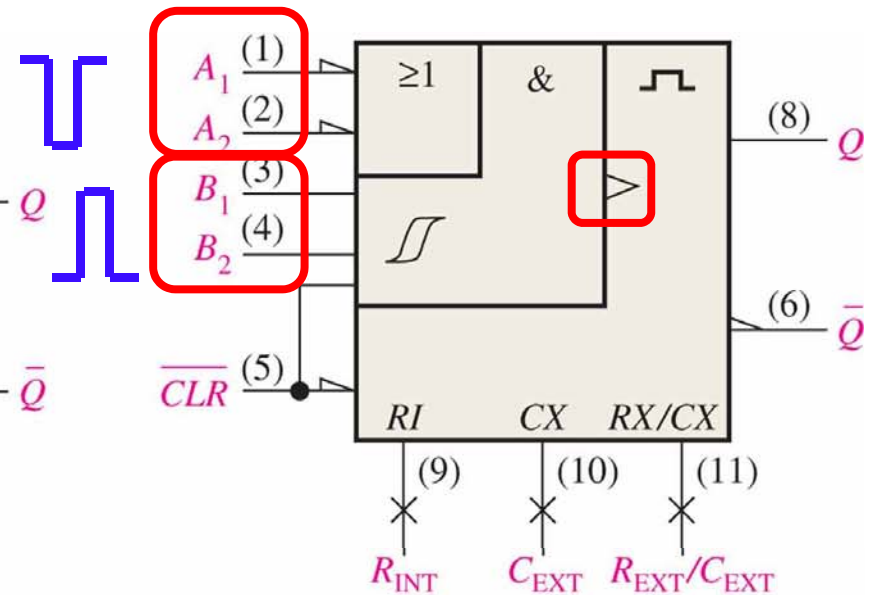
Retriggerable One-shots



74LS122



(a) Traditional logic symbol



(b) ANSI/IEEE std. 91–1984 logic symbol
($\times$ = nonlogic connection). ⏏ is the qualifying symbol for a retriggerable one-shot.

$$R_{INT}=10K\Omega$$

Calculation

With no external resistor and capacitor:

$$t_w \approx 45\text{ns}$$

With external resistor and capacitor:

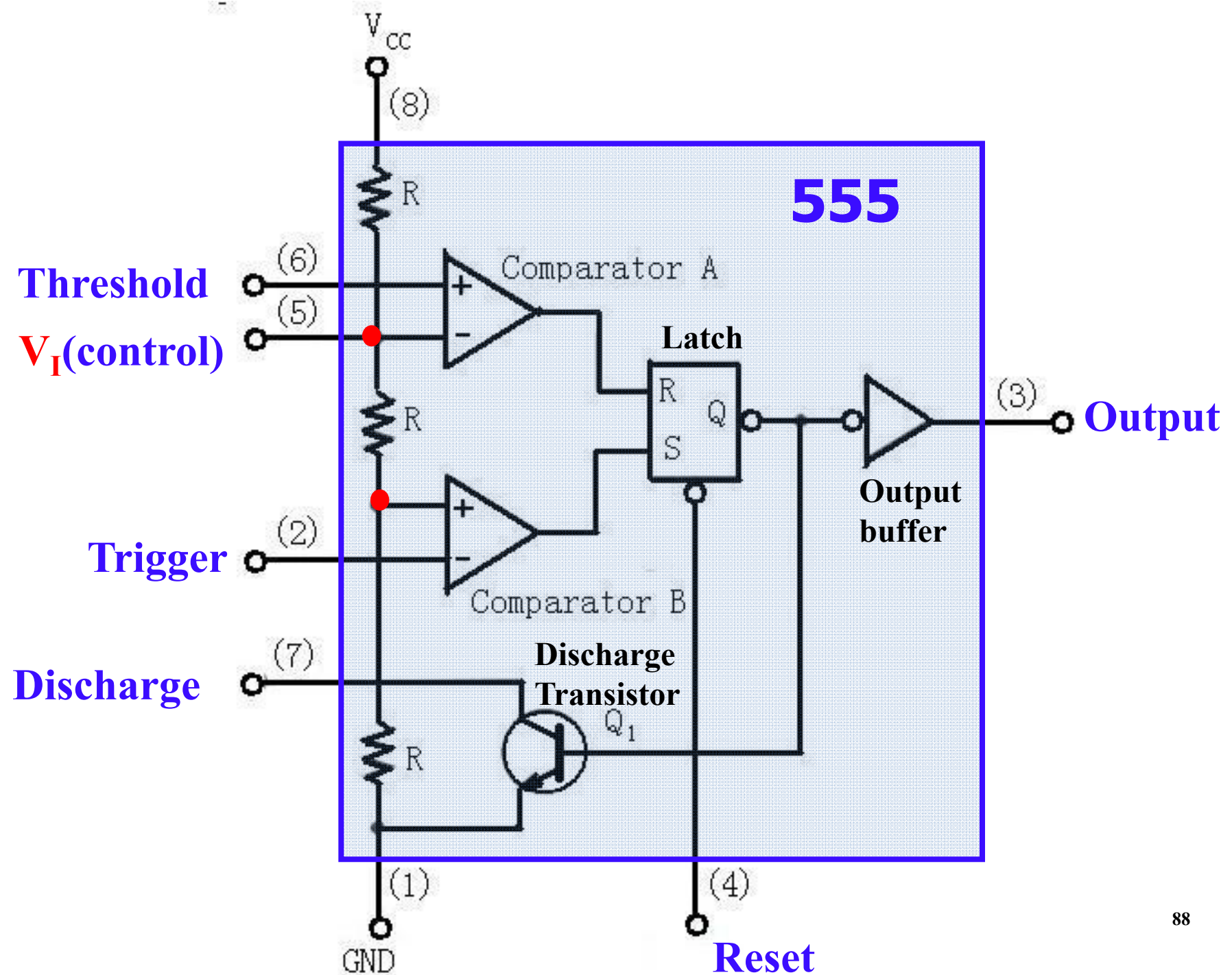
$$t_w = 0.32RC_{EXT} \left(1 + \frac{0.7}{R}\right)$$

Exercise:

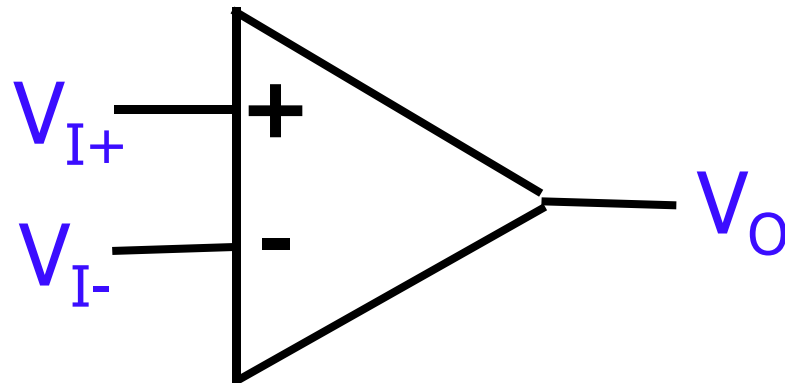
$$\begin{aligned}t_W &= 0.32 \mathbf{RC}_{EXT} \left(1 + \frac{0.7}{R}\right) \\&= 0.32 \times 47 \mathbf{K\Omega} \times 68 \mathbf{\mu F} \times \left(1 + \frac{0.7}{47 \mathbf{K\Omega}}\right) \\&= 1.0227(\mathbf{s}) \\&\approx 1(\mathbf{s})\end{aligned}$$

7.7 555 Timer

- Basic Operation
- One-shot Operation
- Schmitt Trigger Operation
- Astable Operation

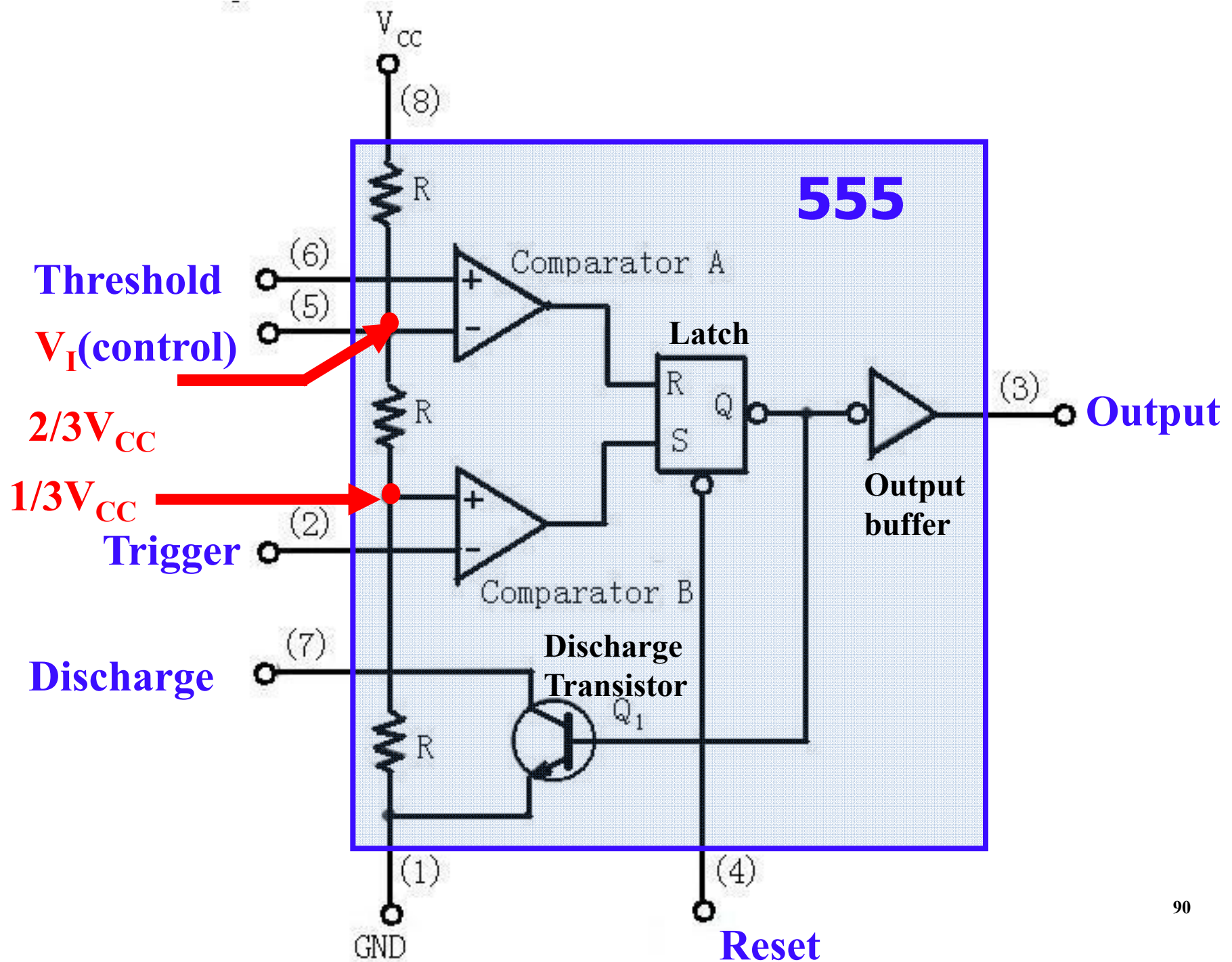


Comparator



If $V_{I+} > V_{I-}$, then $V_O = \text{HIGH}$

If $V_{I+} < V_{I-}$, then $V_O = \text{LOW}$



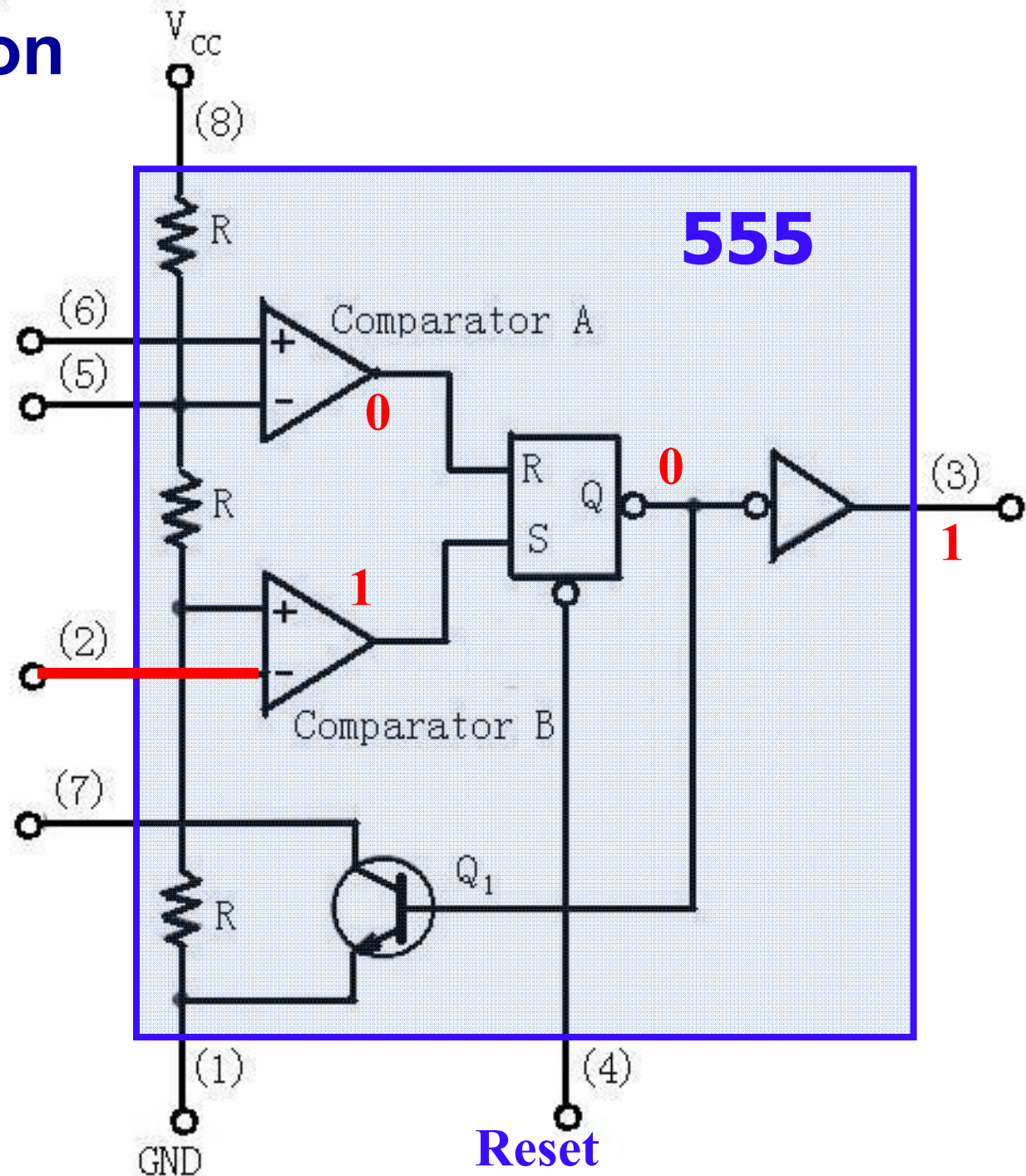
Basic Operation

(施密特锁存器)

0 → V_I Threshold
 V_I (control)

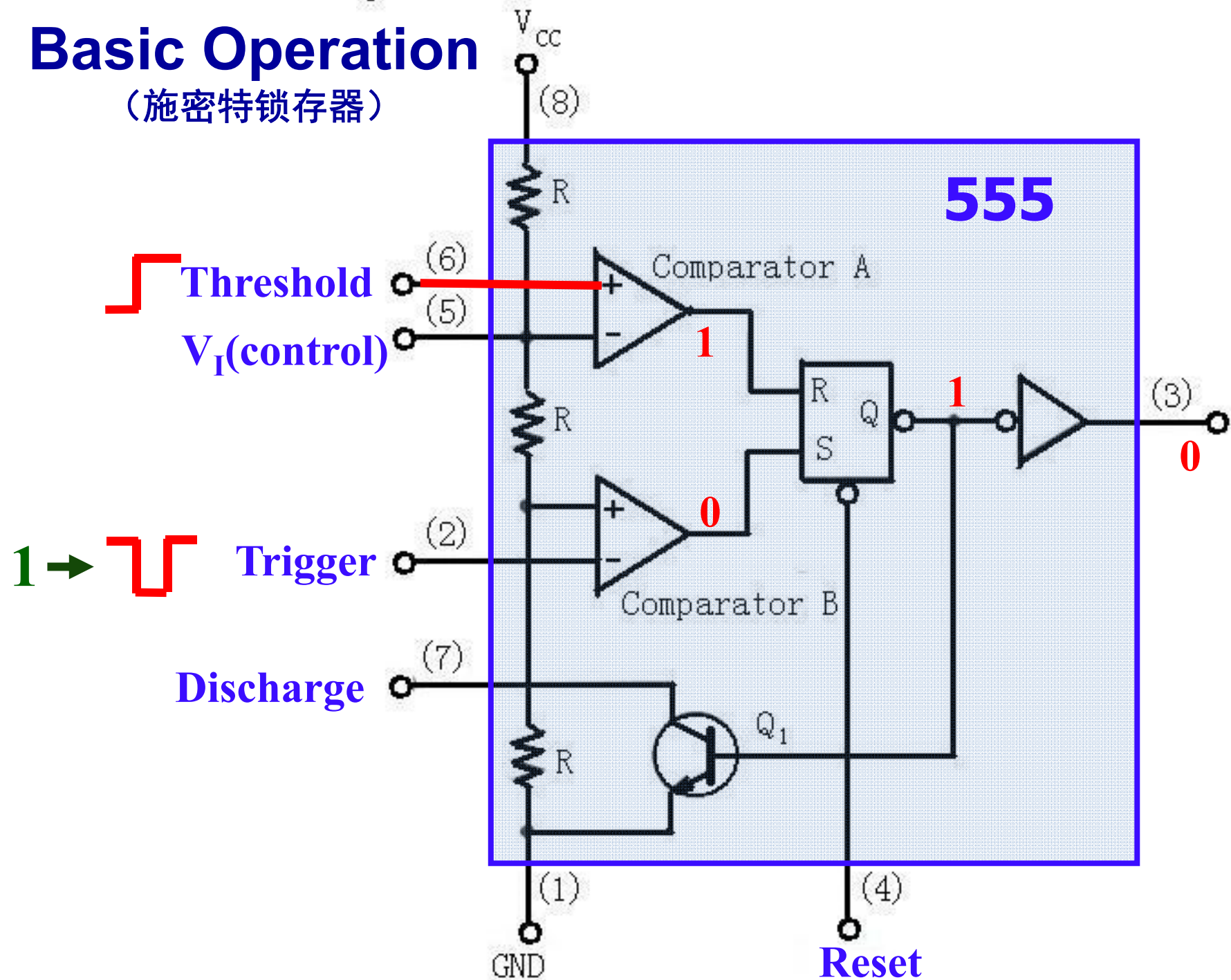
Trigger

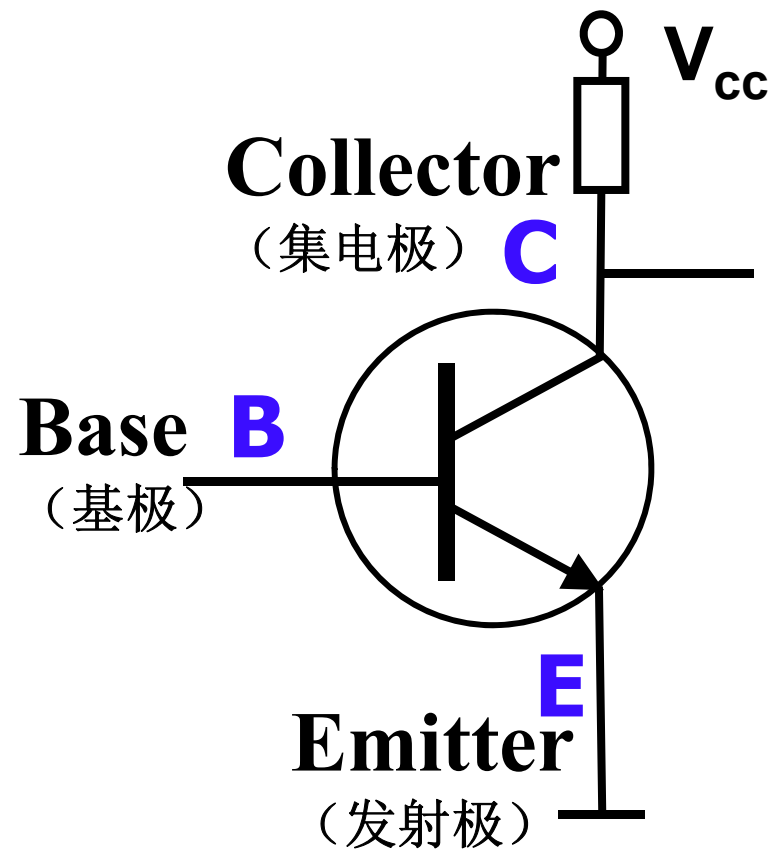
Discharge



Basic Operation

(施密特锁存器)

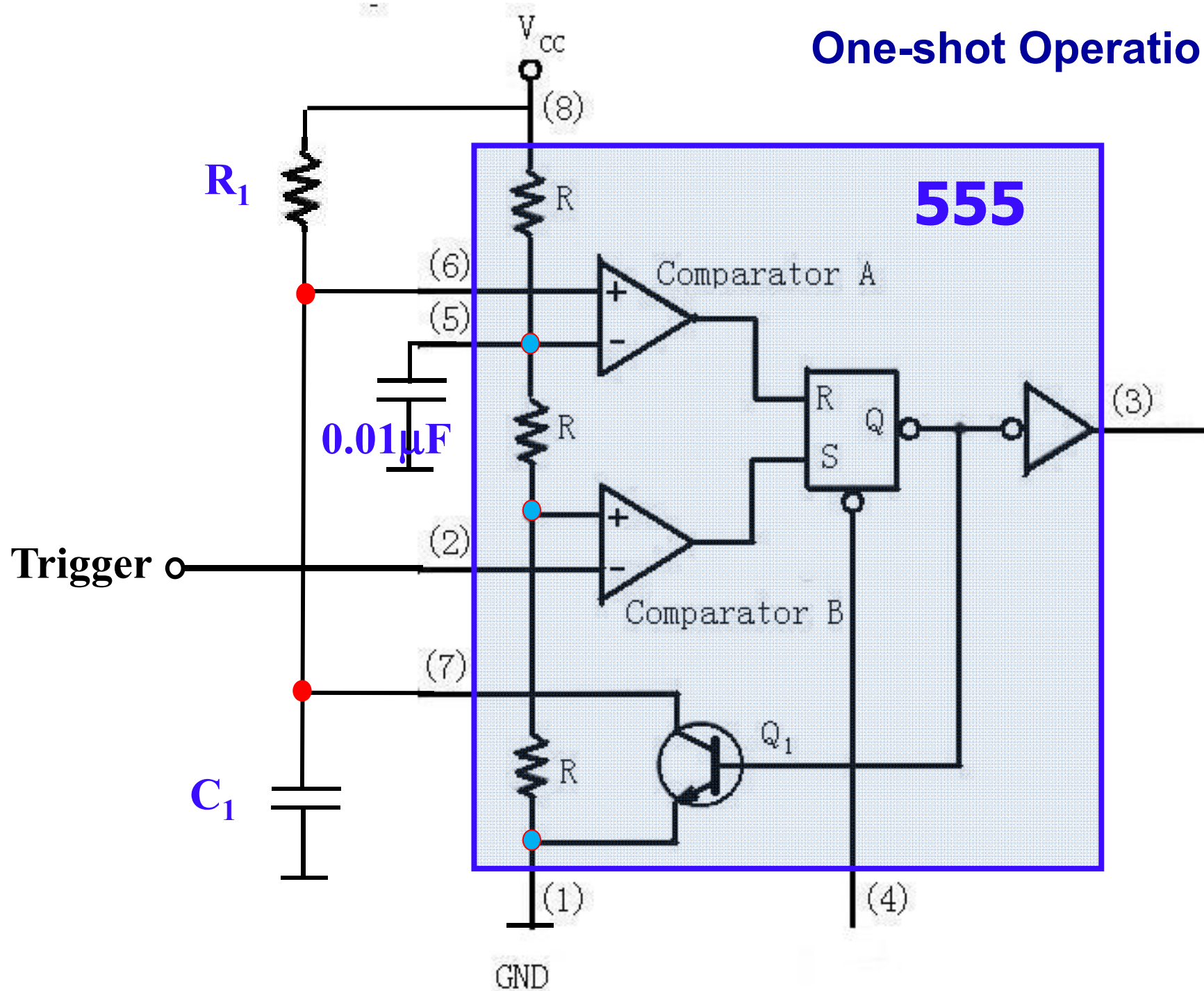




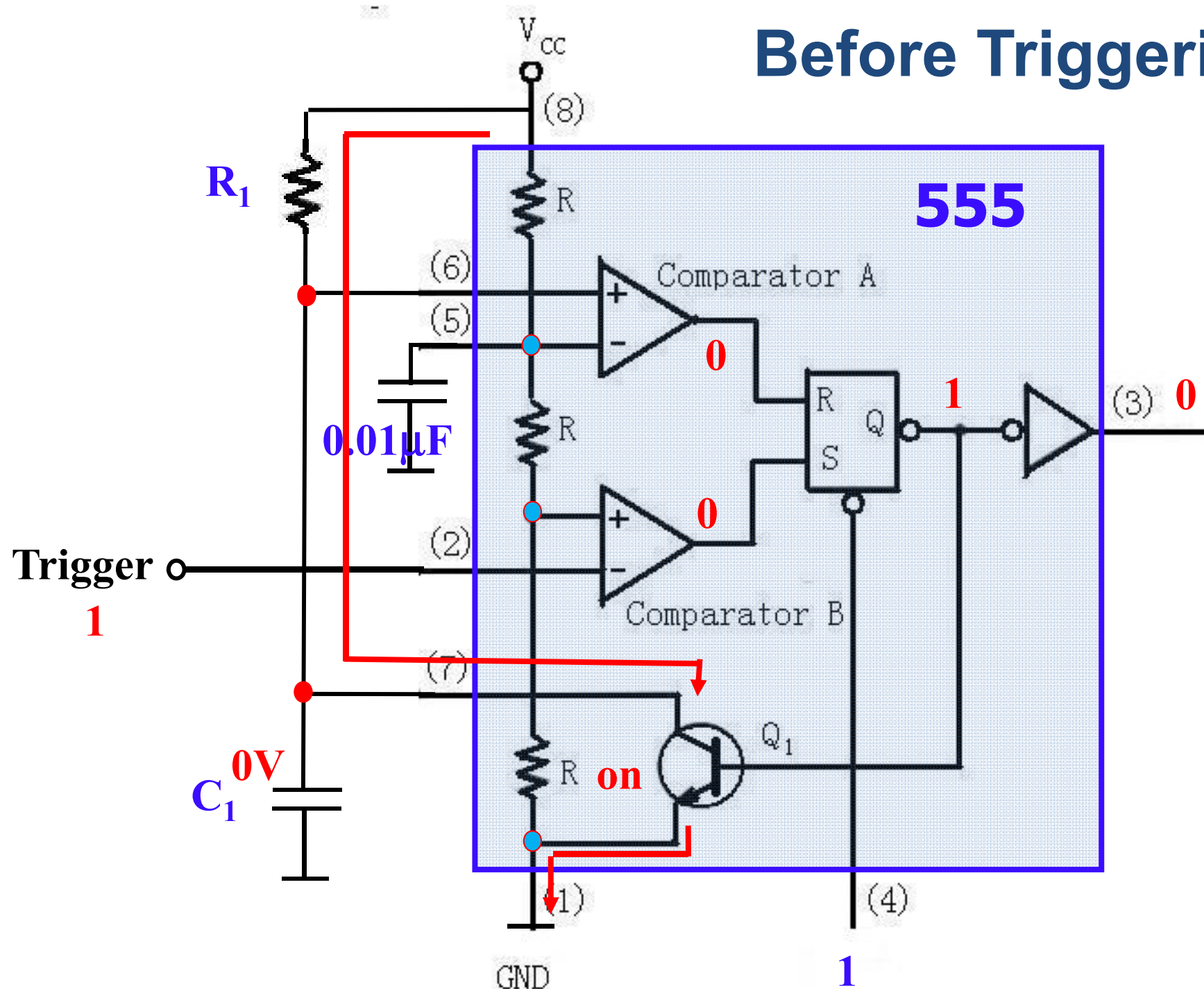
If $V_{BE} > V_{bias}$, $V_c = \text{LOW}$

If $V_{BE} < V_{bias}$, $V_c = \text{HIGH}$

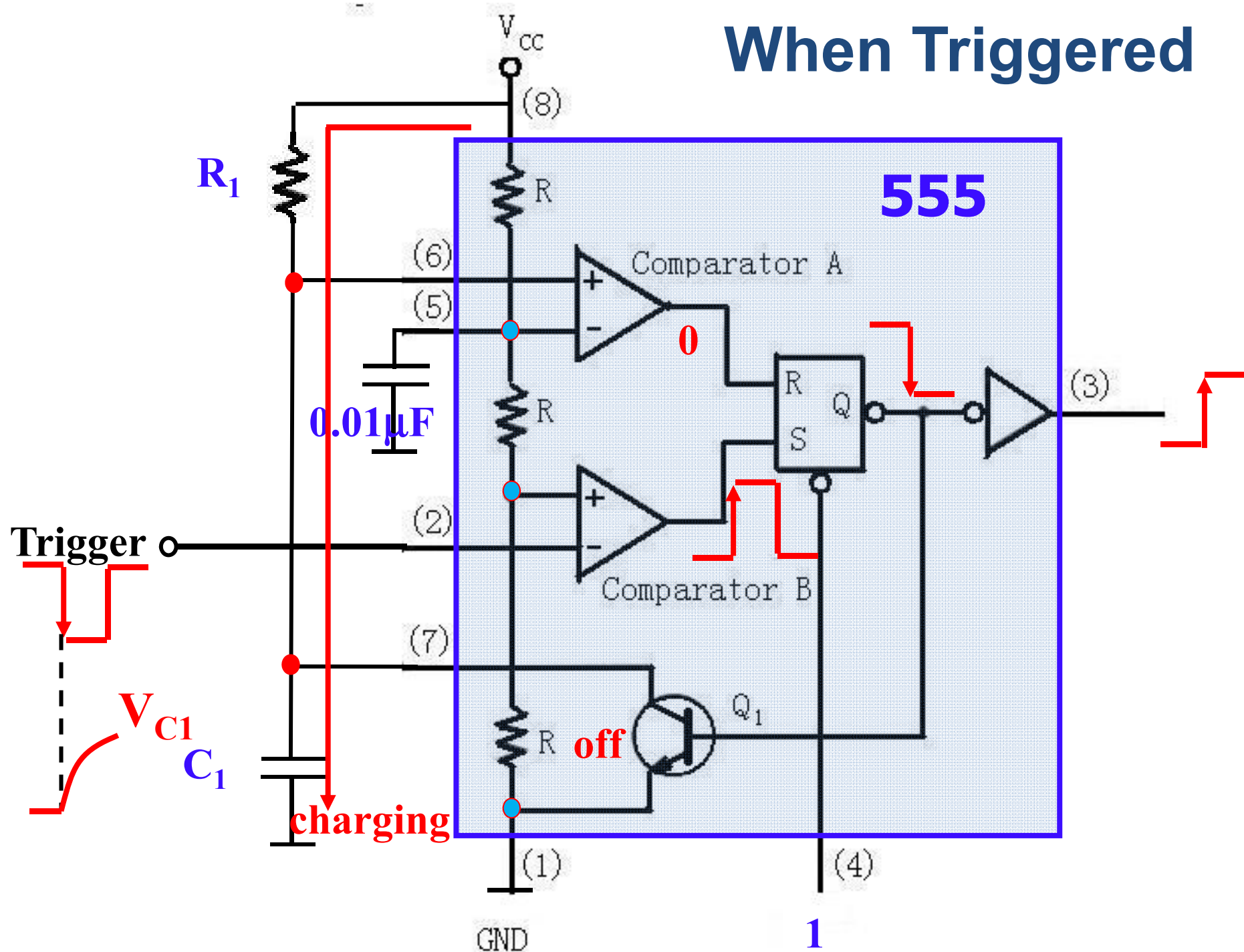
One-shot Operation



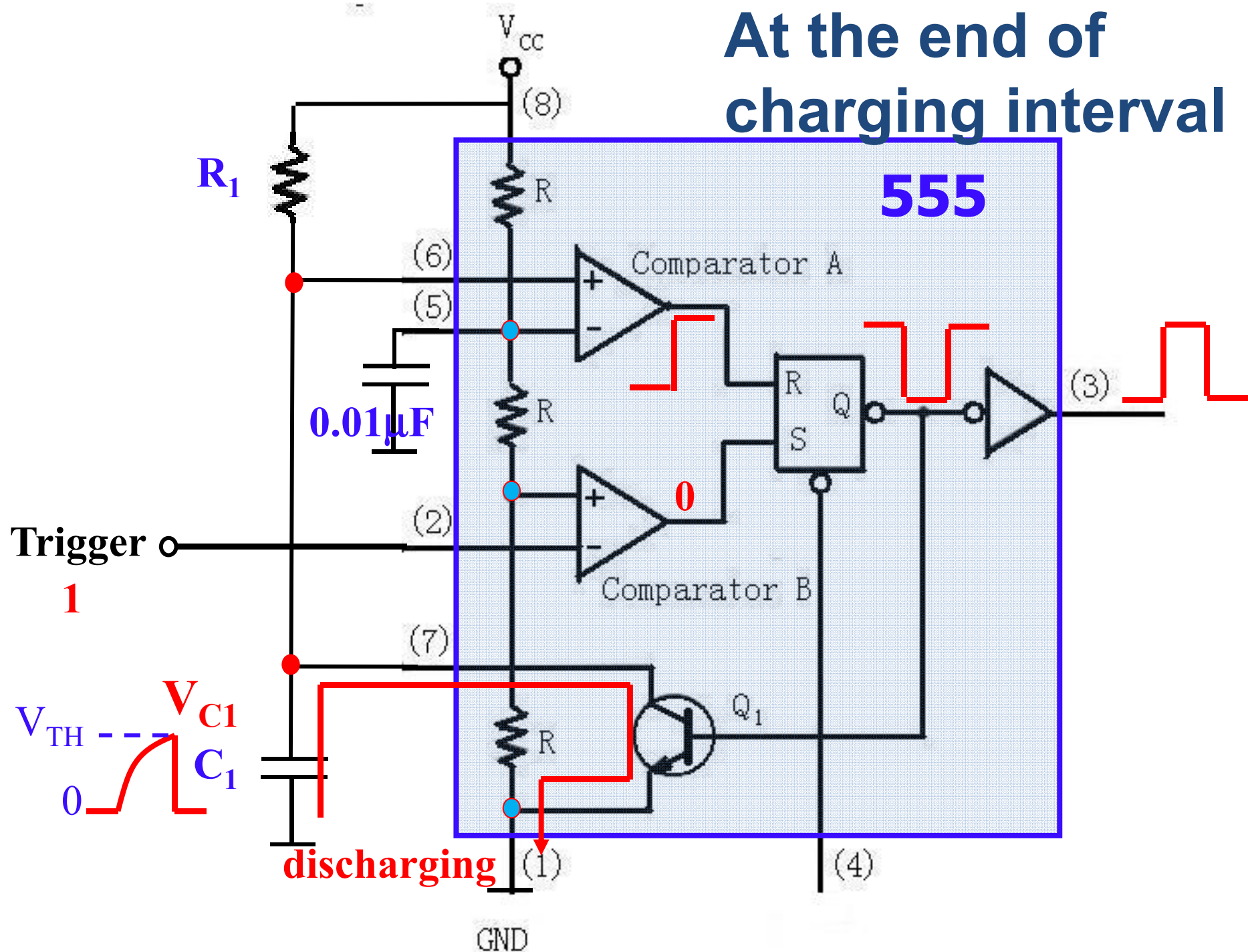
Before Triggering

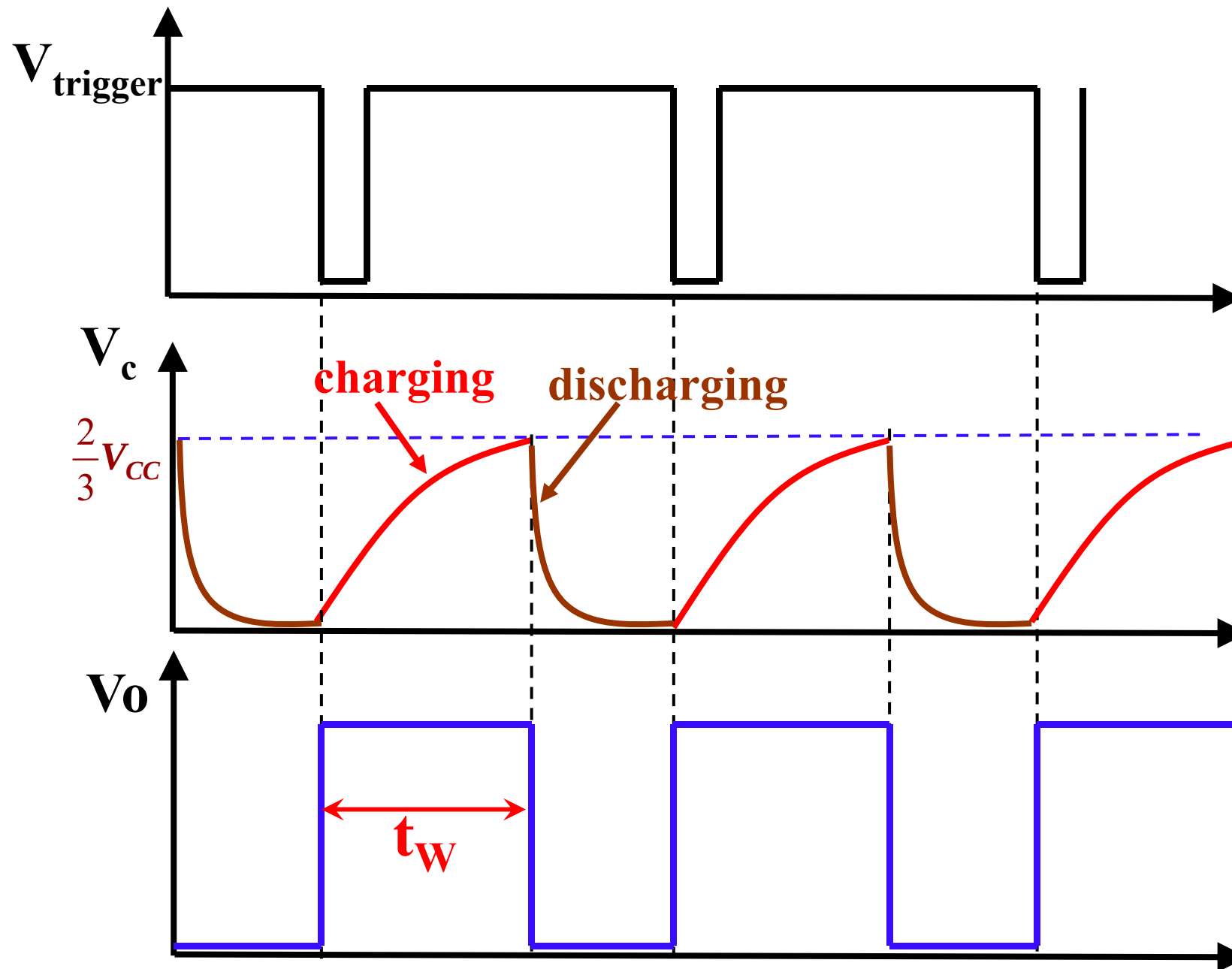


When Triggered



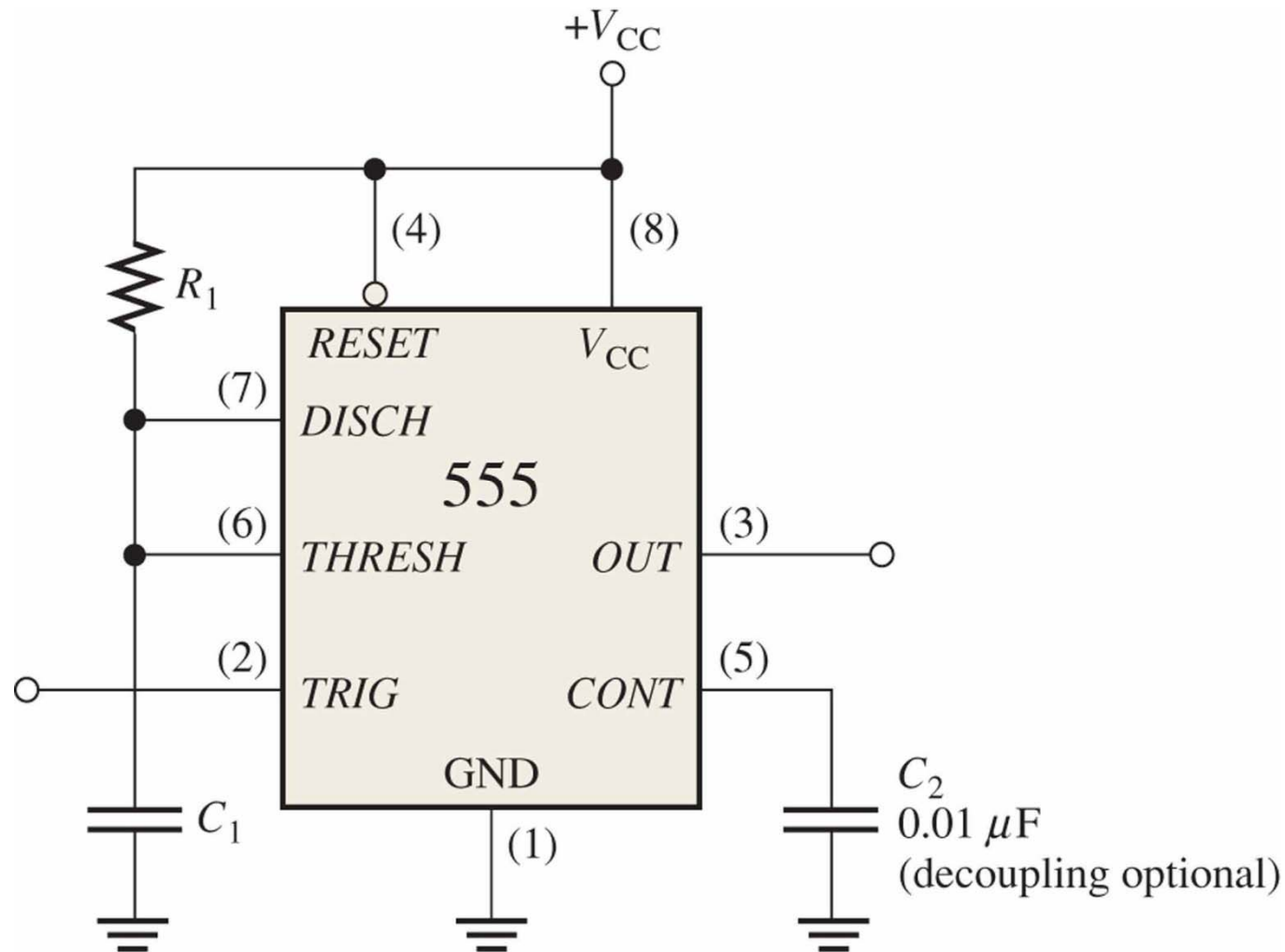
At the end of
charging interval





$$\begin{aligned}
 t_w &= R_1 C_1 \ln \frac{V_{CC} - 0}{V_{CC} - \frac{2}{3} V_{CC}} \\
 &= R_1 C_1 \ln 3 \approx 1.1 R_1 C_1
 \end{aligned}$$

One-shot Operation

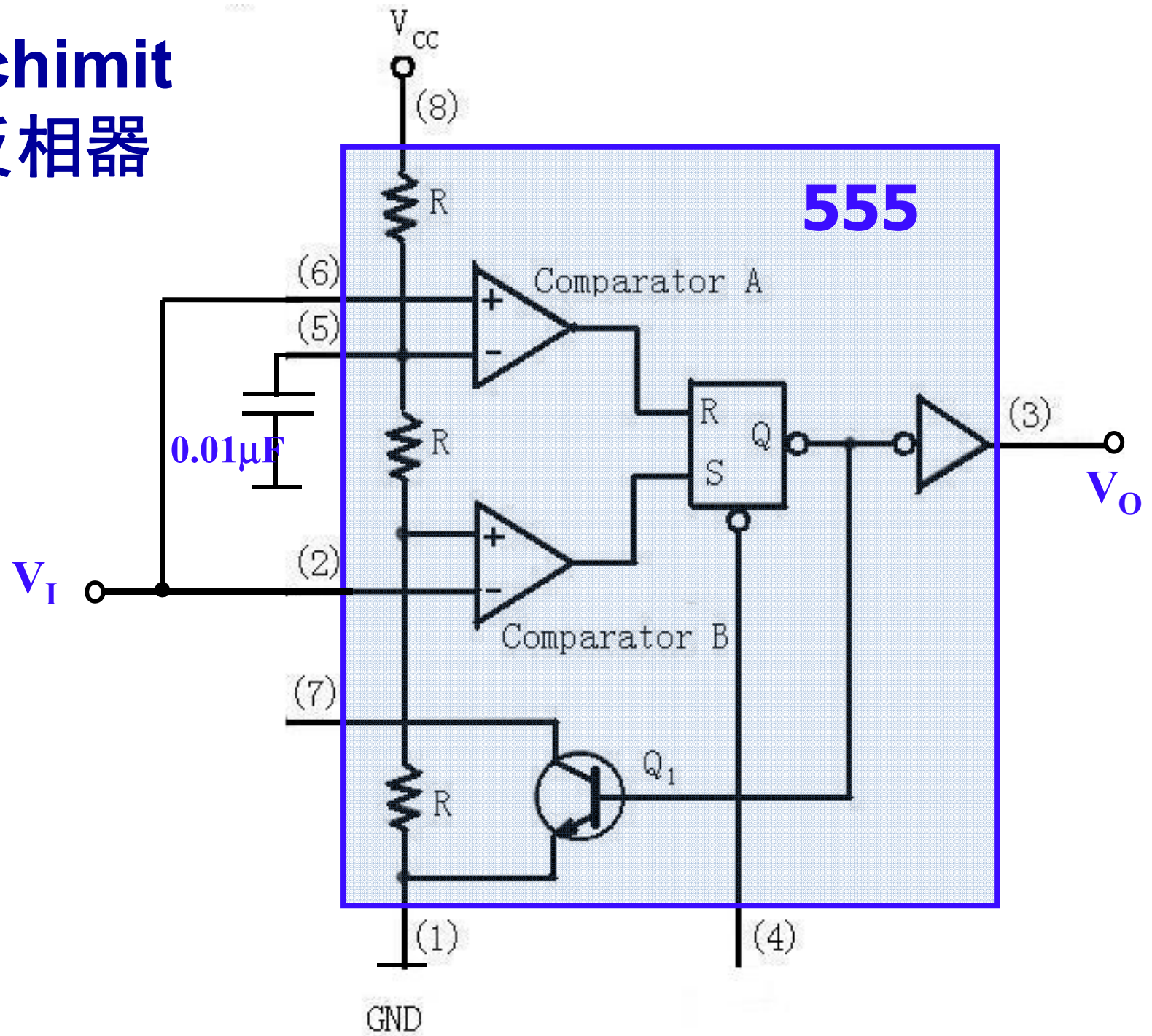


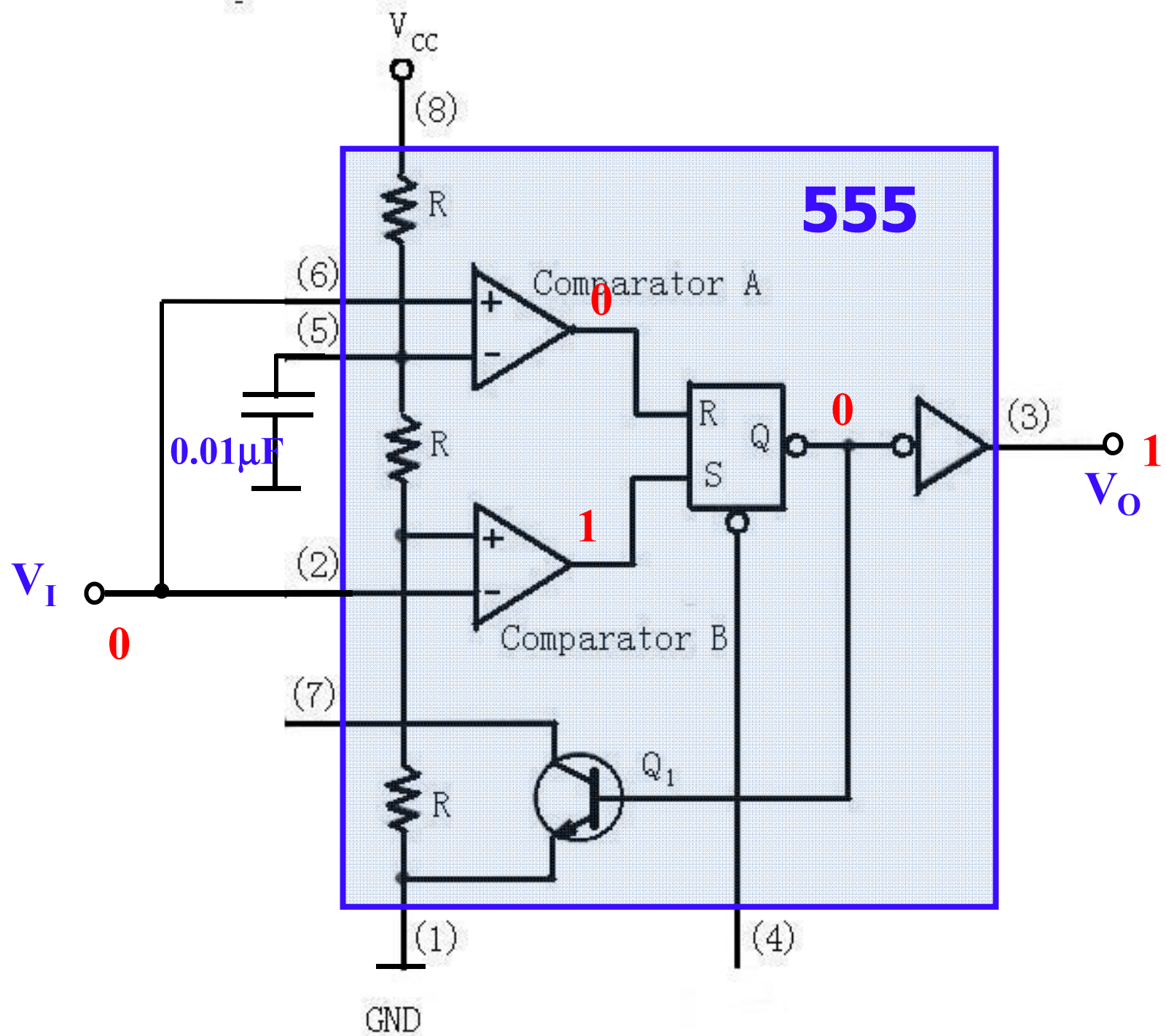
Example

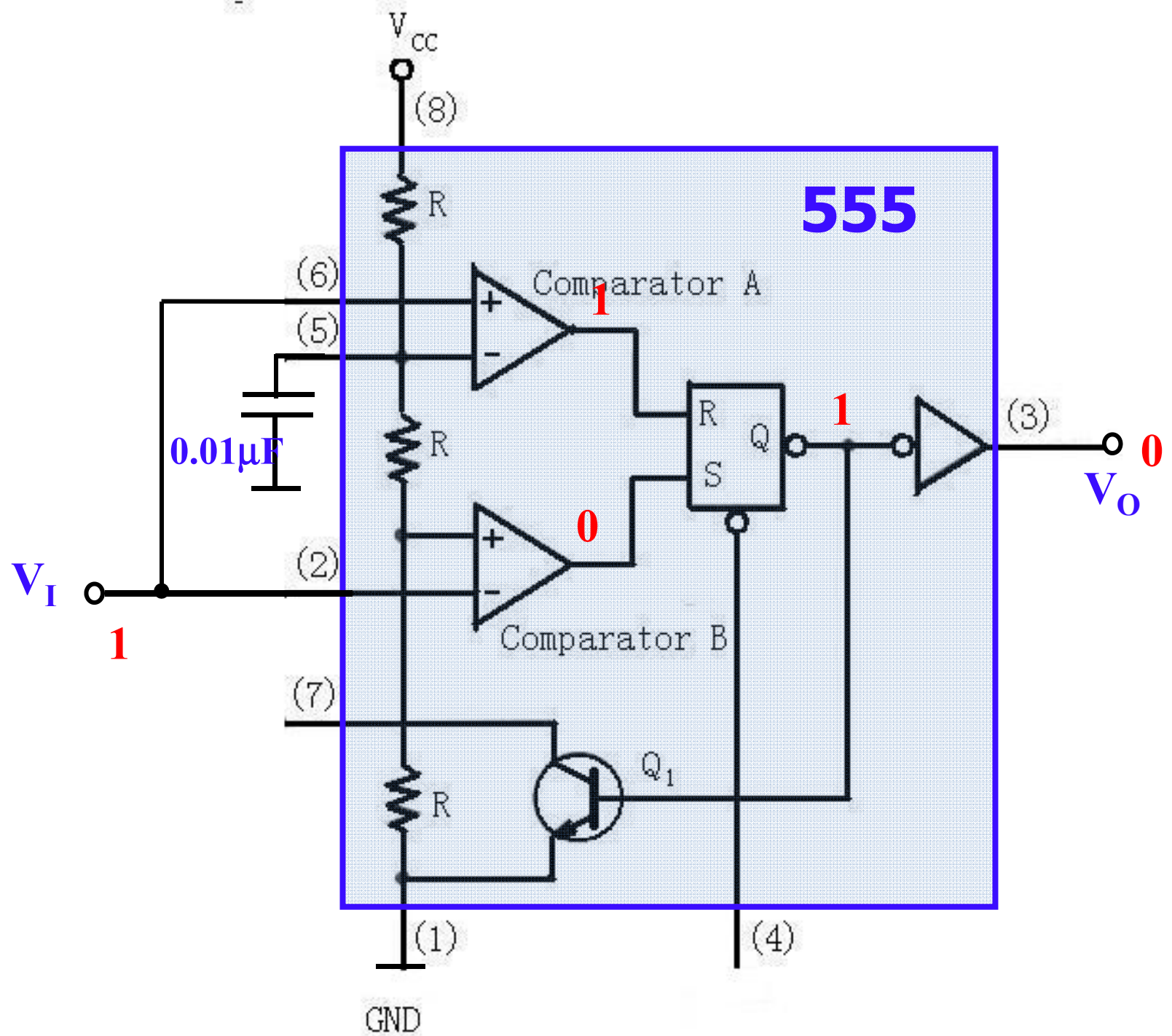
For $C_1=0.01\mu\text{F}$, determine the value of R_1 for a pulse width of 1ms.

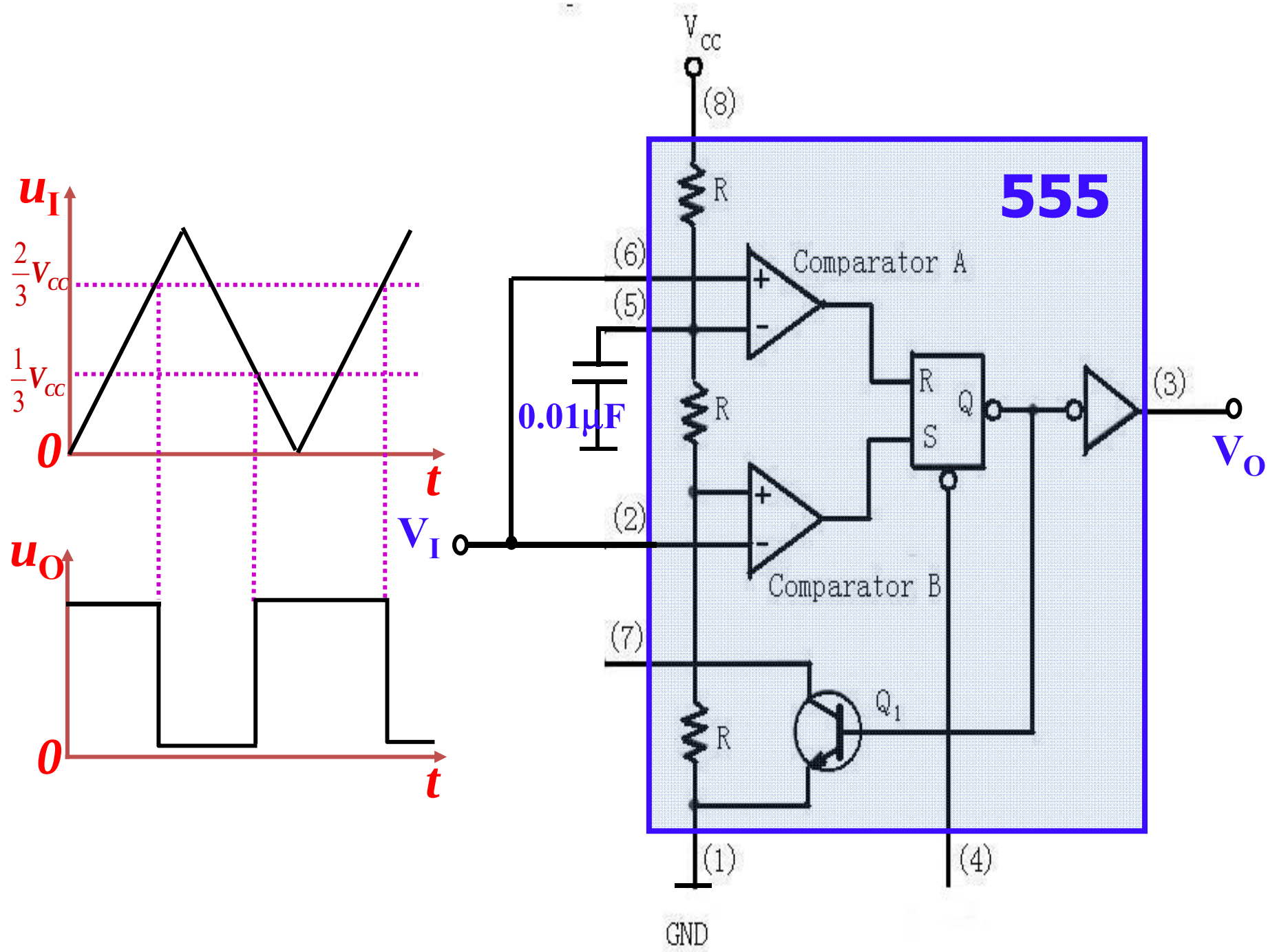
$$R_1 = \frac{t_w}{1.1C_1} = \frac{1 \times 10^{-3}}{1.1 \times 0.01 \times 10^{-6}} \approx 91 K\Omega$$

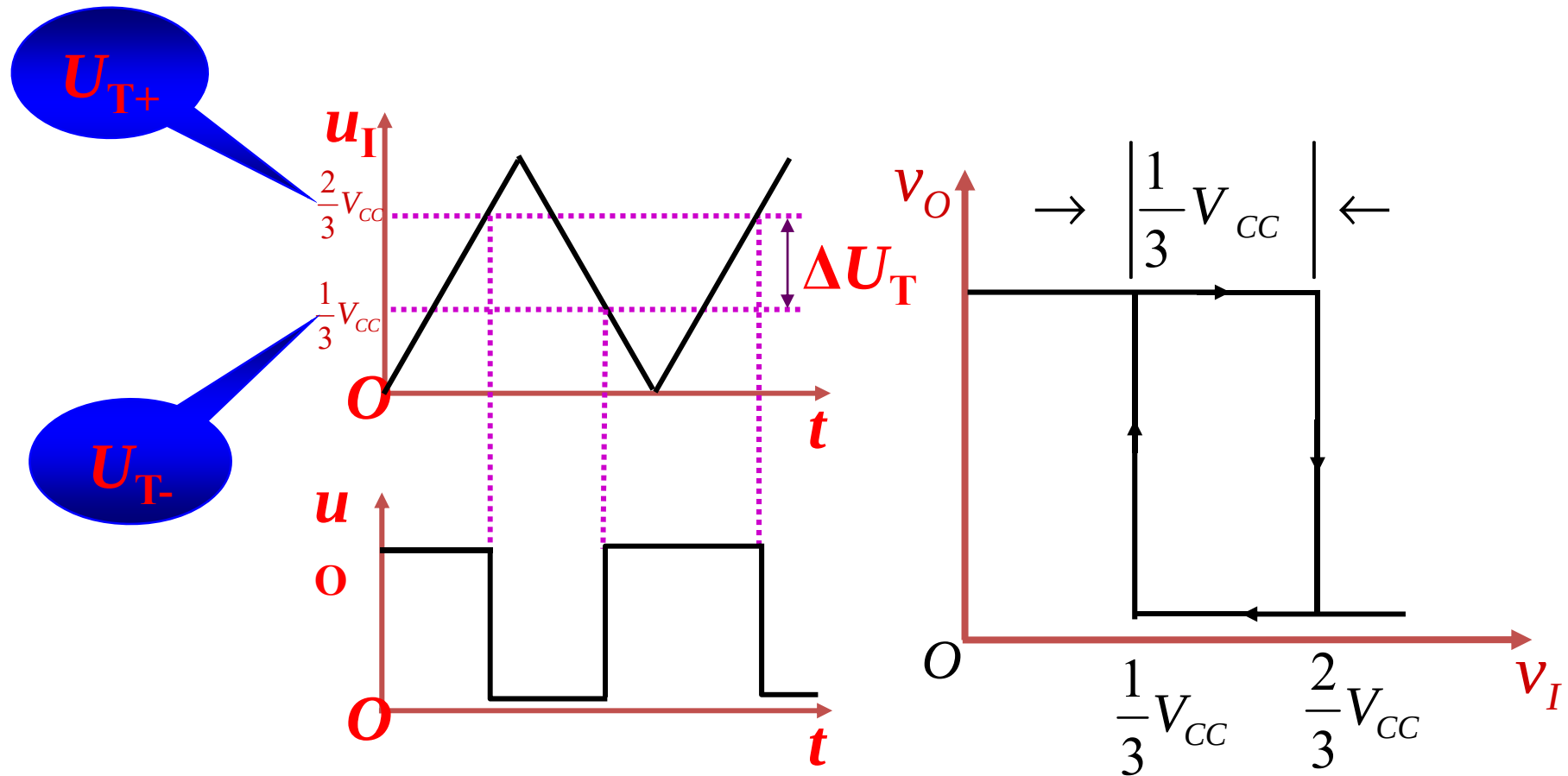
Schmitt 反相器











Note: if control voltage V_{CO} (PIN 5) is given, then

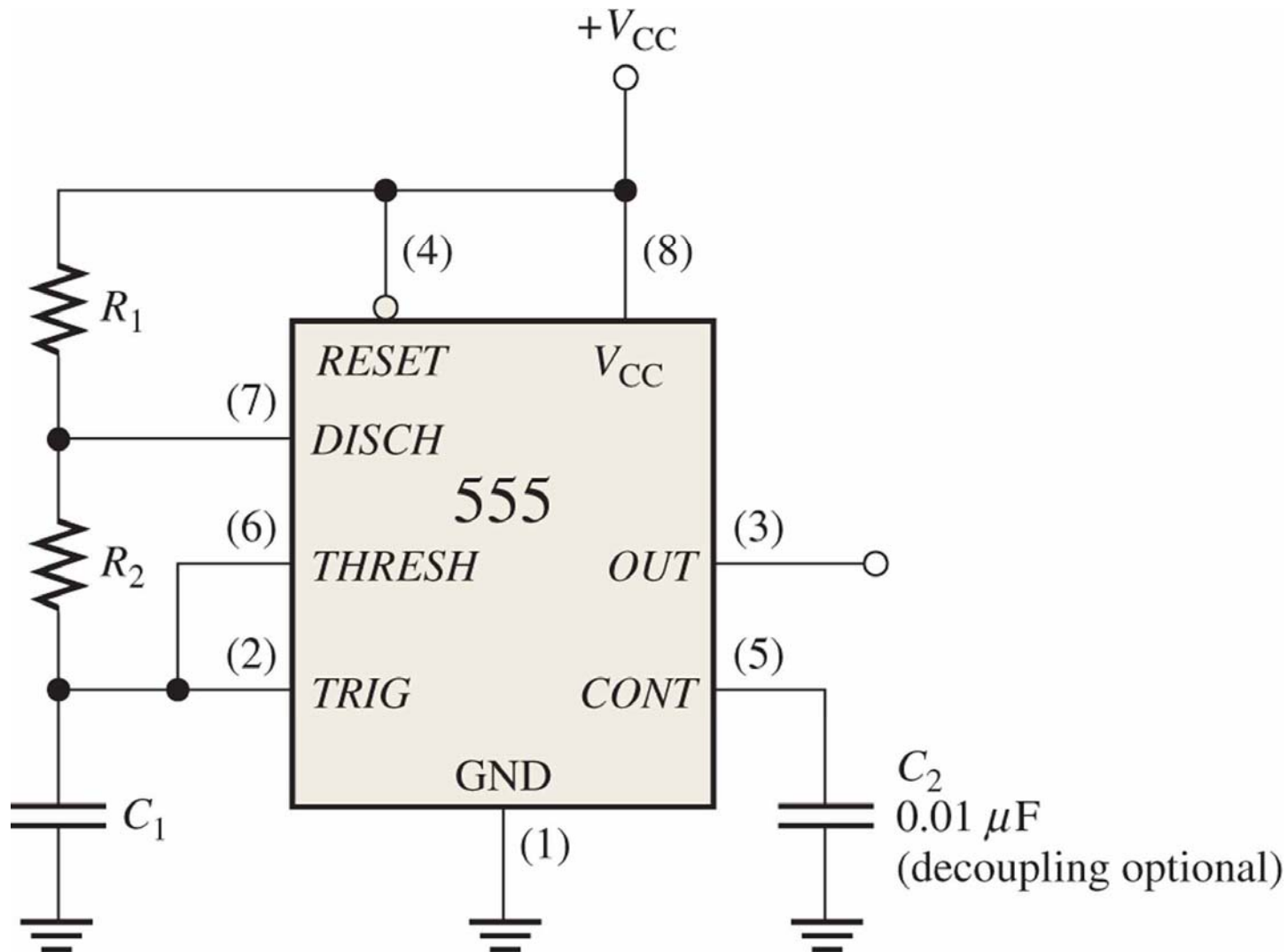
$$U_{T+} = V_{CO} \quad U_{T-} = 1/2 V_{CO} \quad \Delta U_{T-} = 1/2 V_{CO}$$

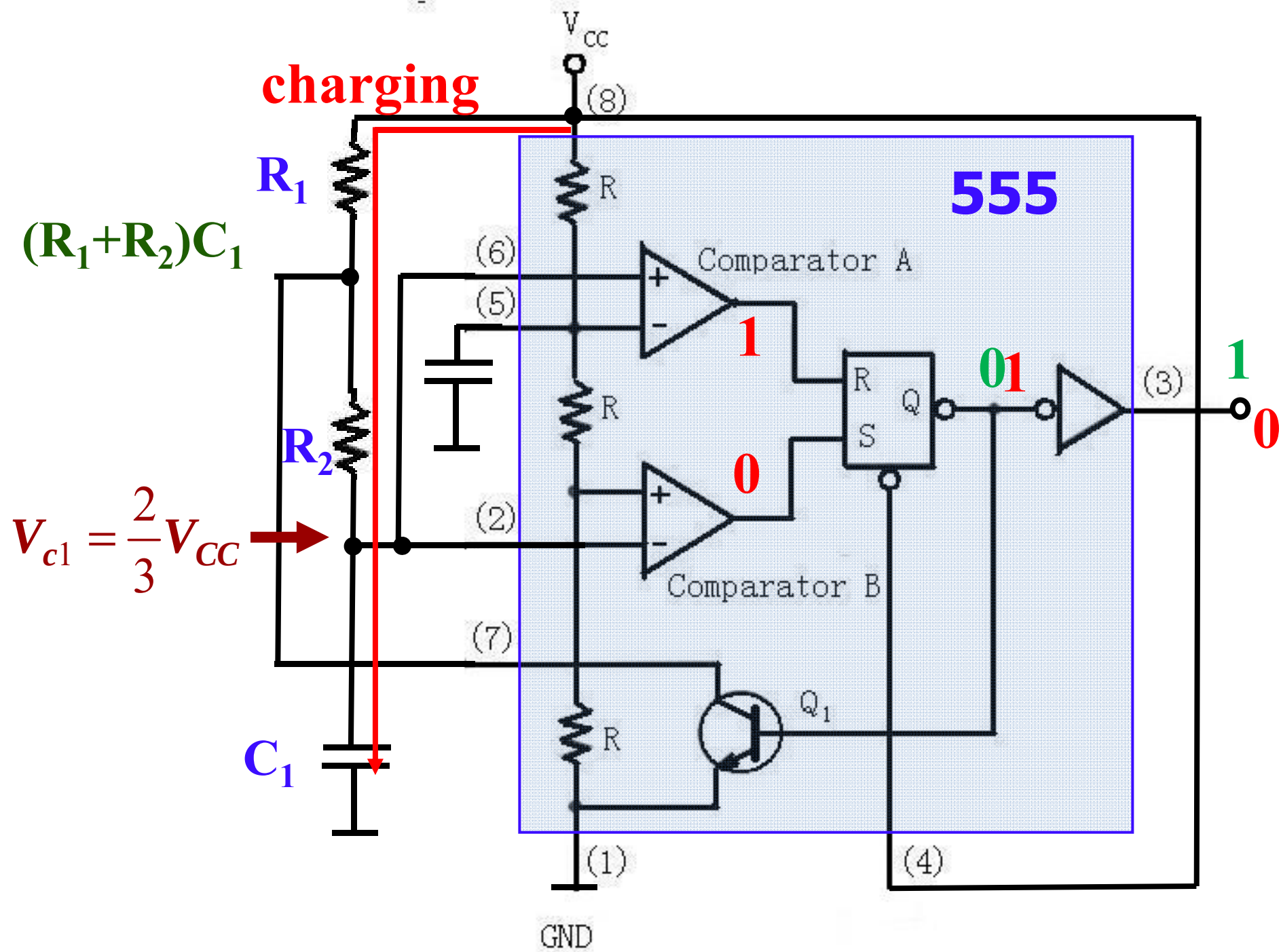
7.8 Astable Multivibrators

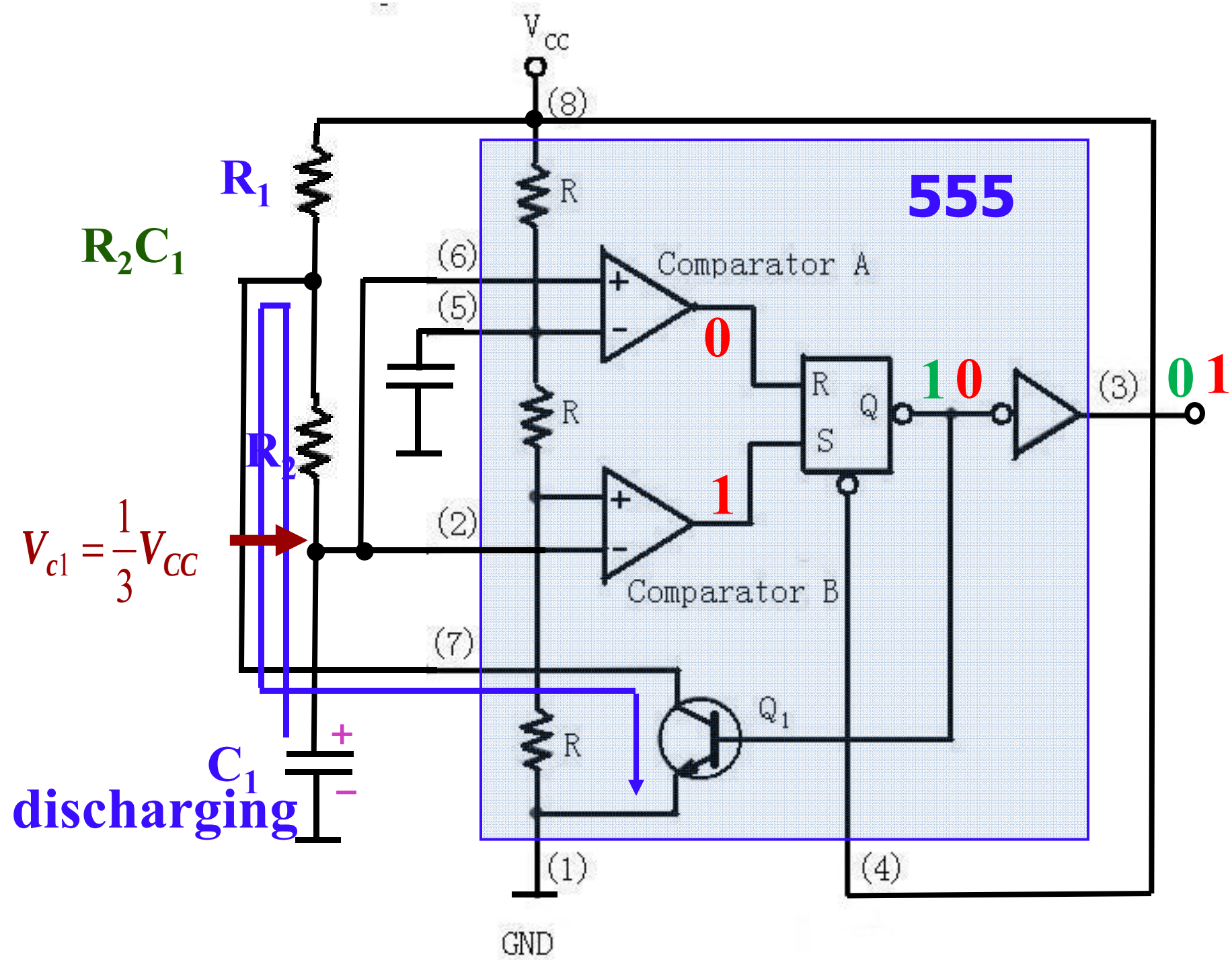
多谐振荡器

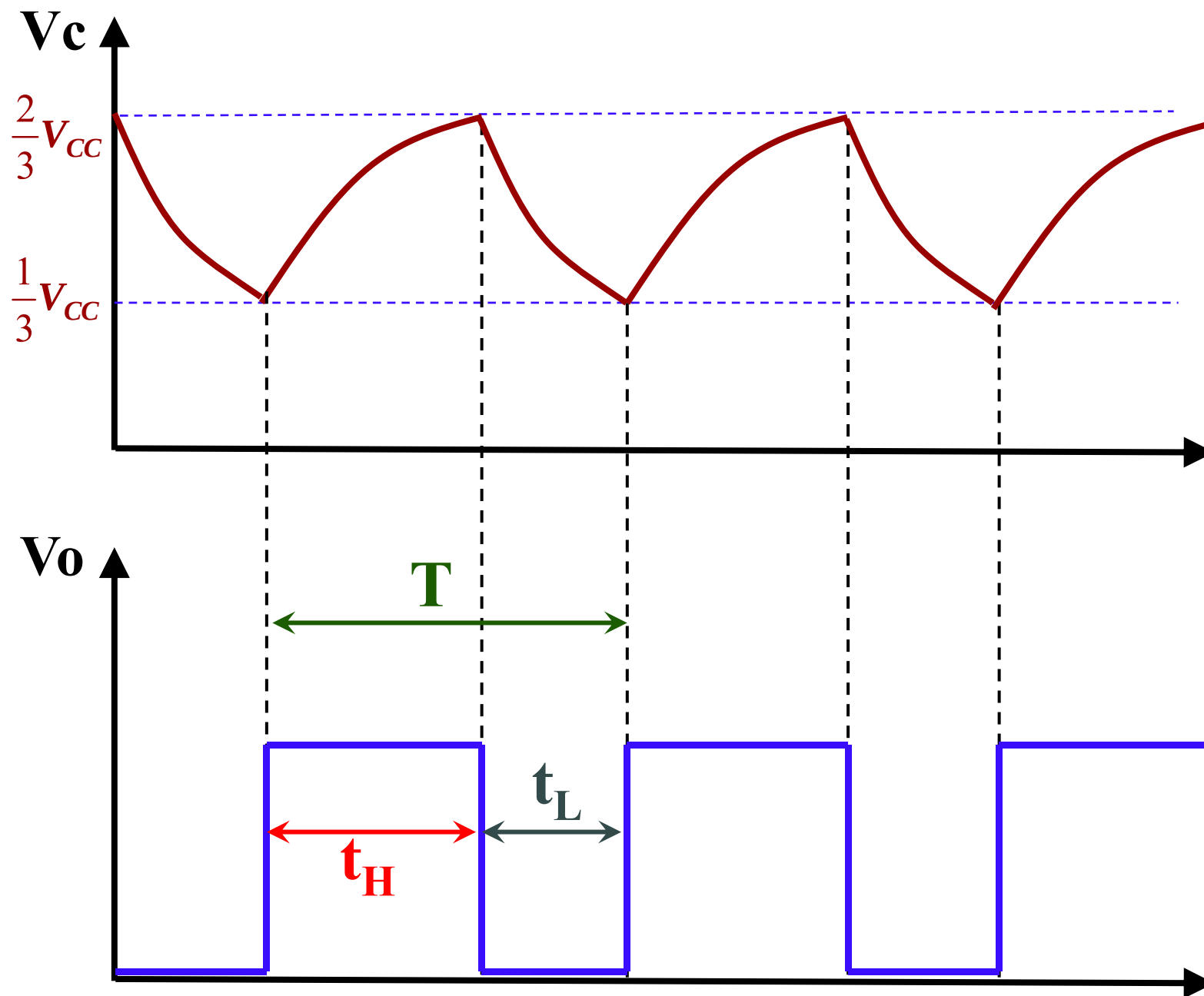
Astable multivibrator devices have no stable states, instead they *continually switch* between two unstable states. Consequently, they are used as *oscillators* to provide clock signals for timing purposes.

555 Astable Operation









Calculation

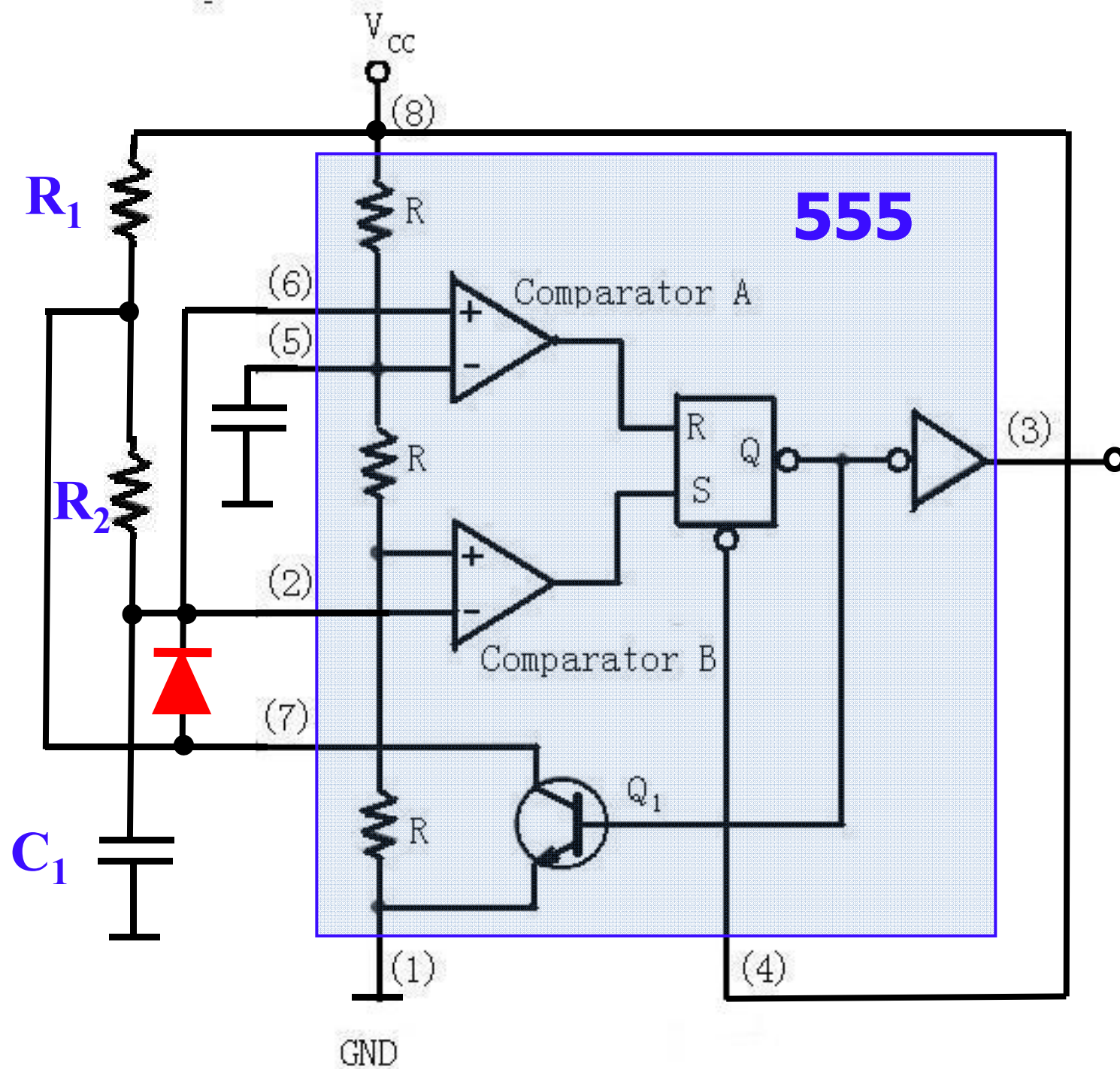
$$T = 0.7(R_1 + 2R_2)C_1$$

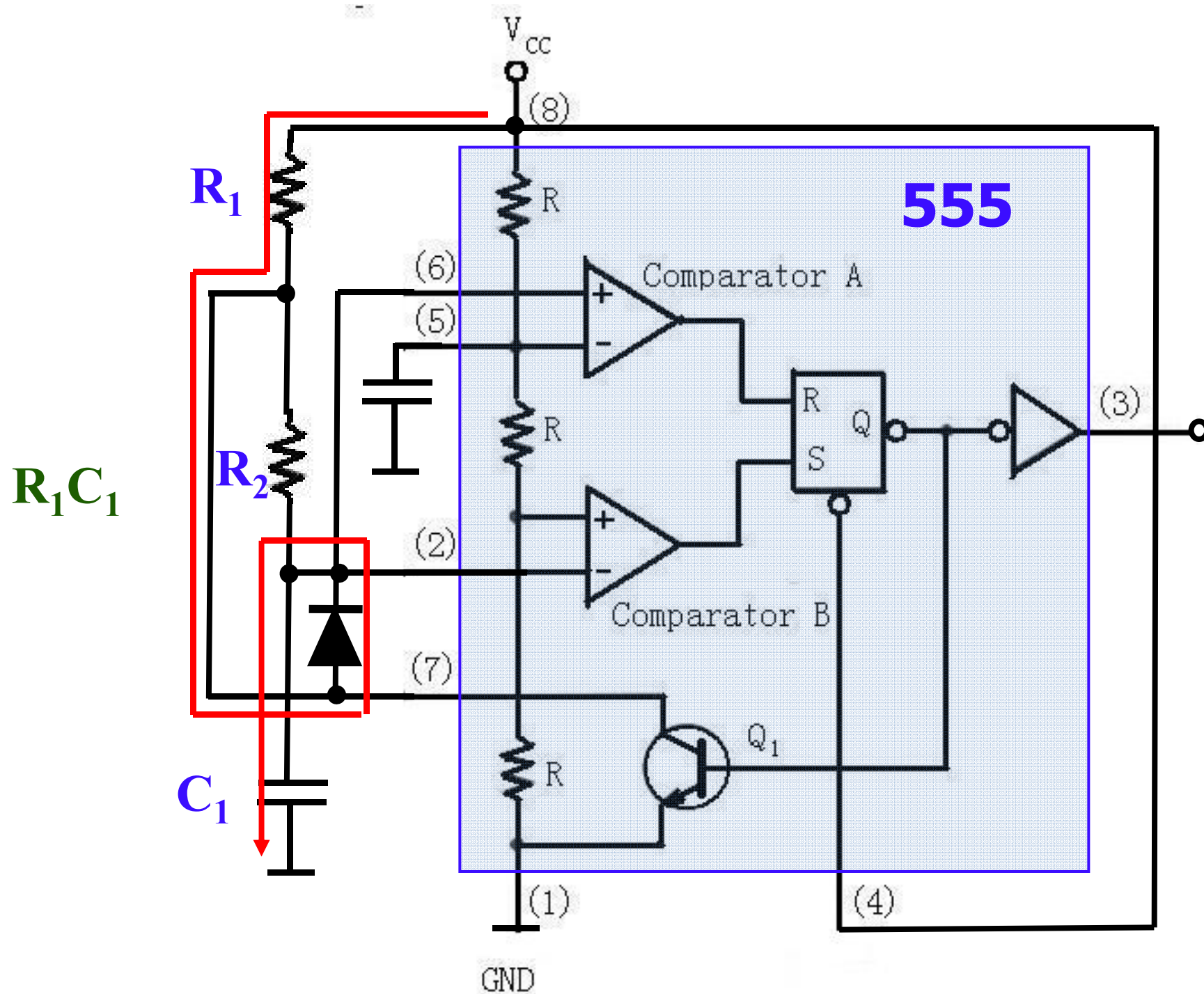
$$t_H = 0.7(R_1 + R_2)C_1$$

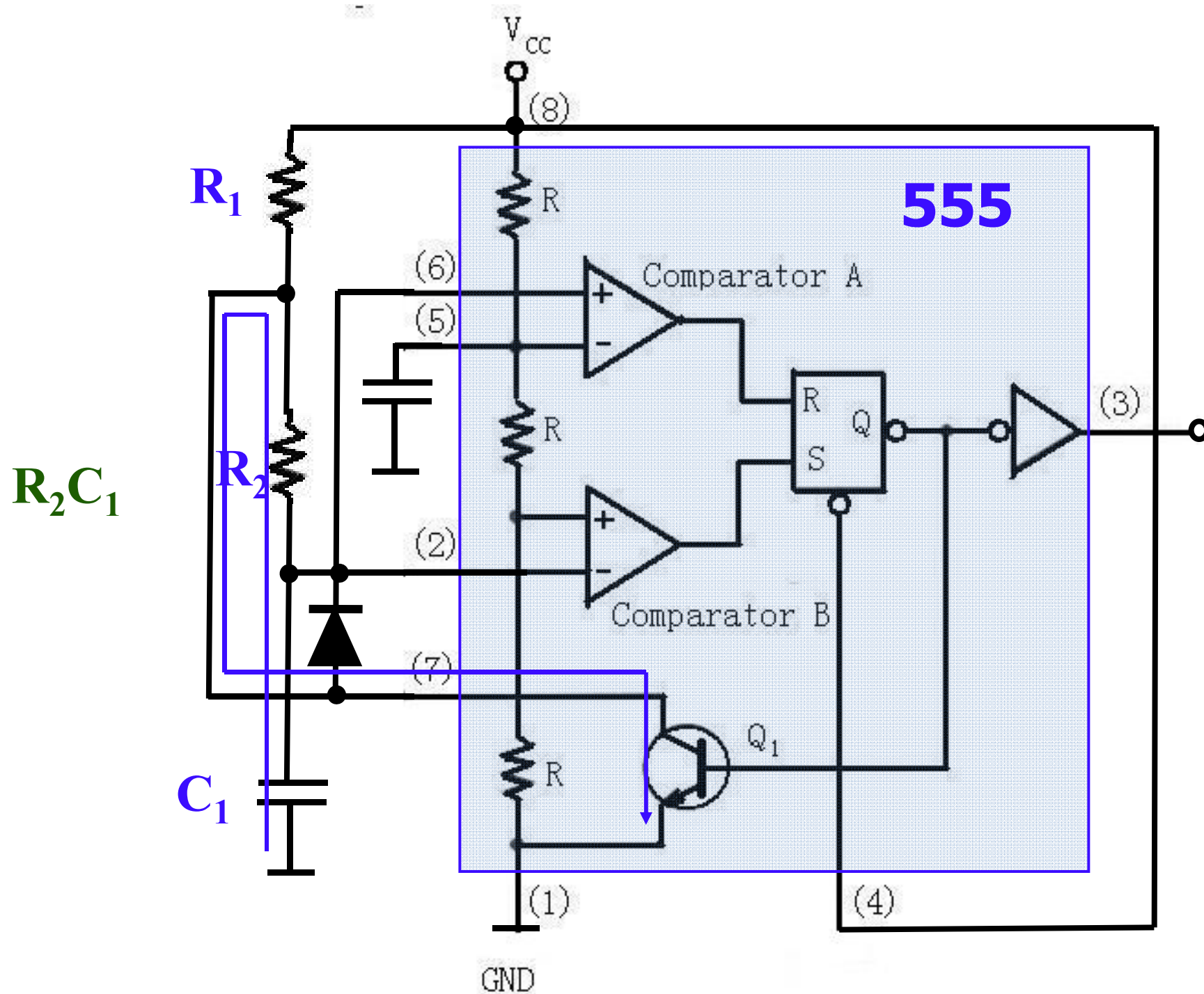
$$t_L = 0.7R_2C_1$$

$$f = \frac{1.44}{(R_1 + 2R_2)C_1}$$

$$\text{Duty cycle} = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\%$$







Calculation

$$T = 0.7(R_1 + R_2)C_1$$

$$t_H = 0.7R_1C_1$$

$$t_L = 0.7R_2C_1$$

$$f = \frac{1.44}{(R_1 + R_2)C_1}$$

$$\text{Duty cycle} = \frac{R_1}{R_1 + R_2} \times 100\%$$

Homework: Exercise (需要画出电路图和设计公式)

A circuit need a 1Hz clock signal with a duty cycle of $2/3$. $C=10\mu\text{F}$. Determine the value of R_1 and R_2 .

- (1) Using the first connection of 555 timer.
- (2) Using the second connection of 555 timer.

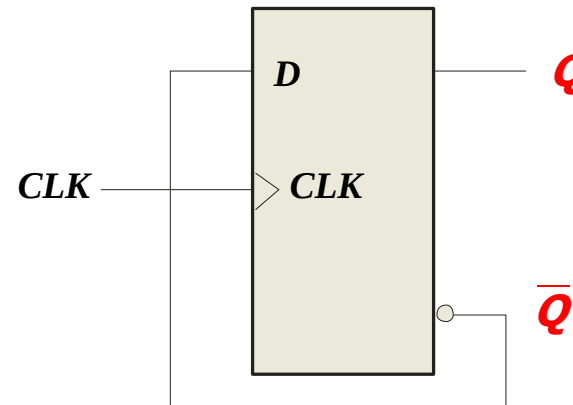
Quiz

1. The output of a D latch will not change if
 - a. the output is LOW
 - b. Enable is not active
 - c. D is LOW
 - d. all of the above

Quiz

2. The D flip-flop shown will

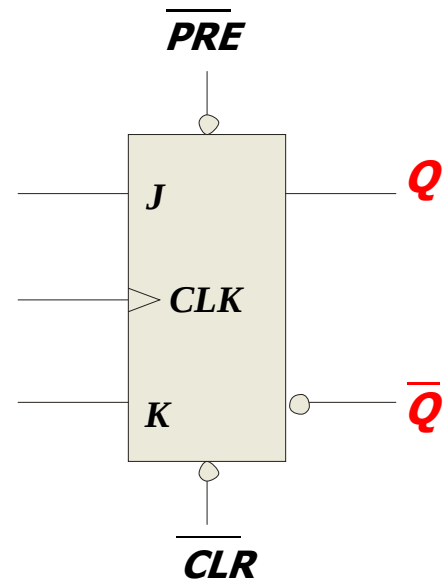
- a. set on the next clock pulse
- b. reset on the next clock pulse
- c. latch on the next clock pulse
- d. toggle on the next clock pulse



Quiz

3. For the J-K flip-flop shown, the number of inputs that are asynchronous is

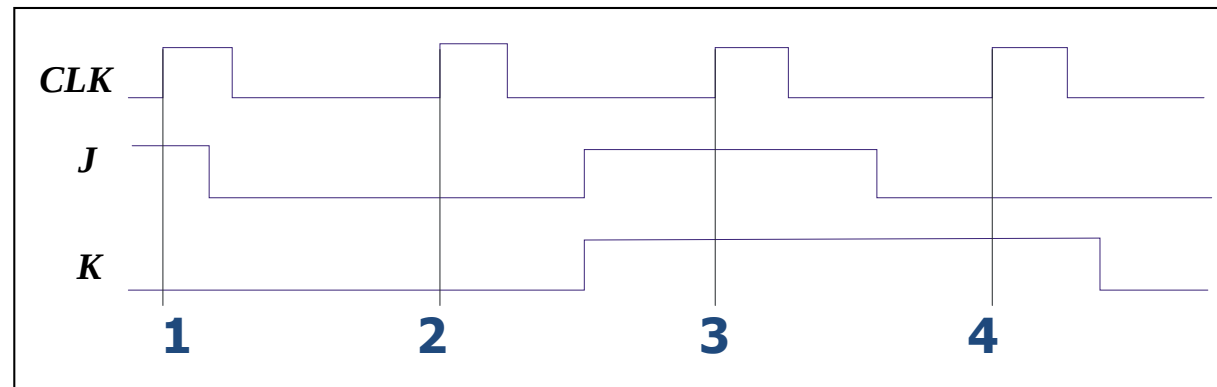
- a. 1
- b. 2
- c. 3
- d. 4



Quiz

4. Assume the output is initially HIGH on a leading edge triggered J-K flip flop. For the inputs shown, the output will go from HIGH to LOW on which clock pulse?

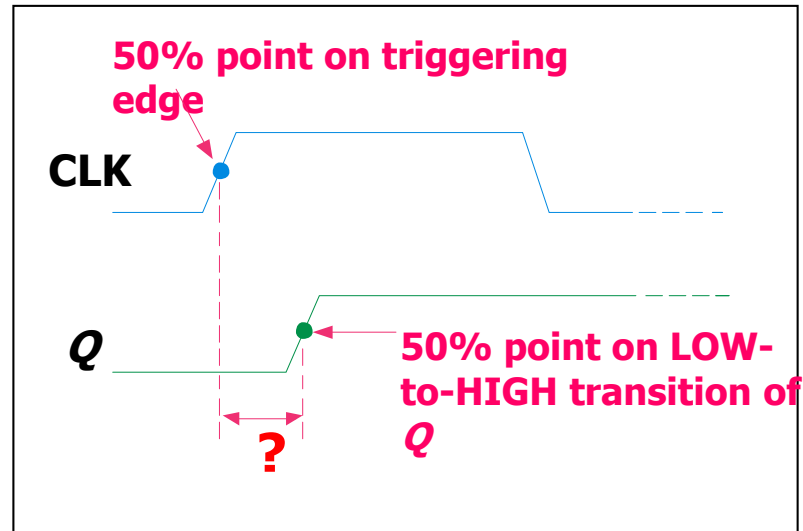
- a. 1
- b. 2
- c. 3
- d. 4



Quiz

5. The time interval illustrated is called

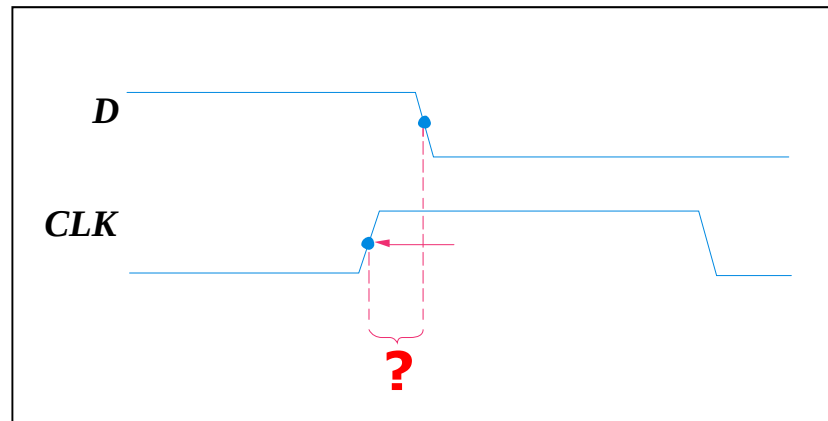
- a. t_{PHL}
- b. t_{PLH}
- c. set-up time
- d. hold time



Quiz

6. The time interval illustrated is called

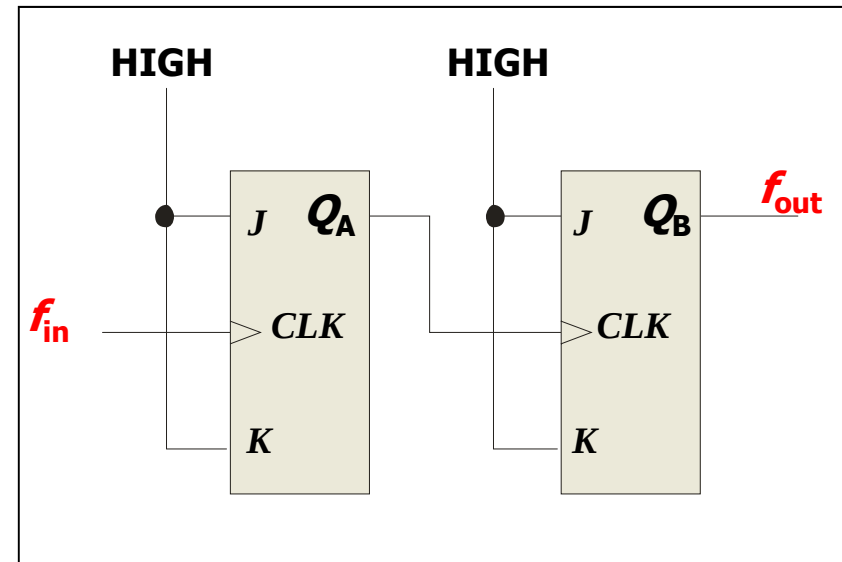
- a. t_{PHL}
- b. t_{PLH}
- c. set-up time
- d. hold time



Quiz

7. The application illustrated is a

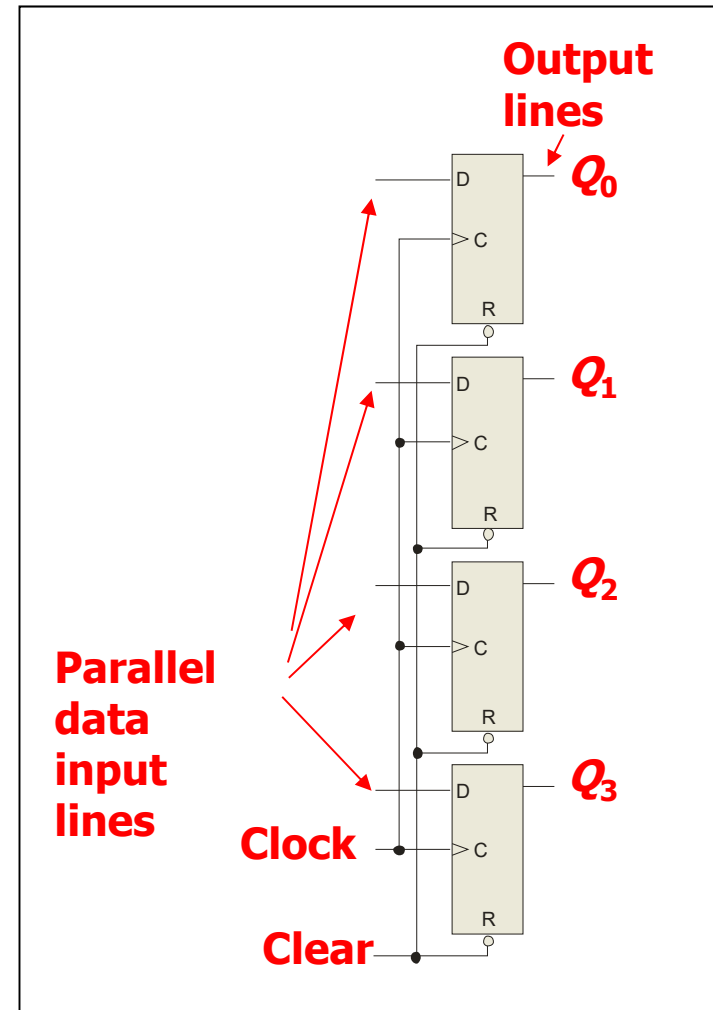
- a. astable multivibrator
- b. data storage device
- c. frequency multiplier
- d. frequency divider



Quiz

8. The application illustrated is a

- a. astable multivibrator
- b. data storage device
- c. frequency multiplier
- d. frequency divider



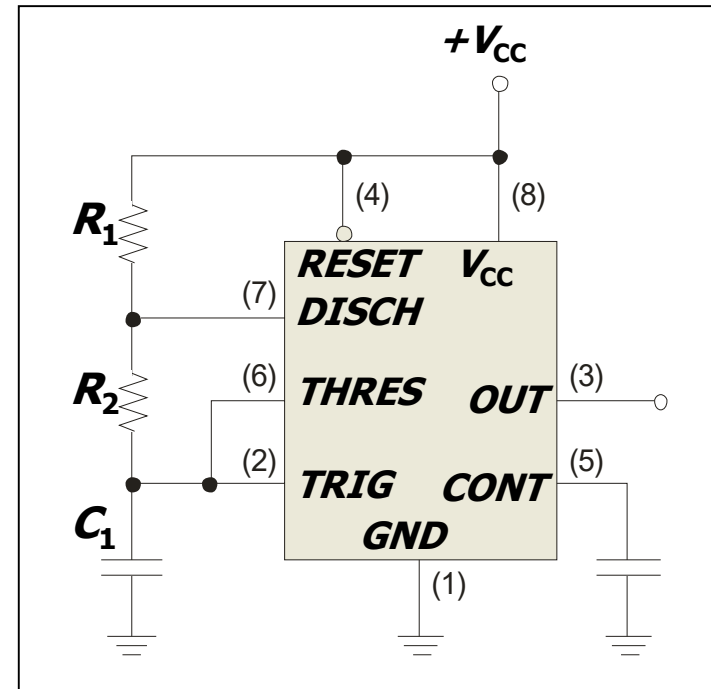
Quiz

9. A retriggerable one-shot with an active HIGH output has a pulse width of 20 ms and is triggered from a 60 Hz line. The output will be a

- a. series of 16.7 ms pulses
- b. series of 20 ms pulses
- c. constant LOW
- d. constant HIGH

Quiz

10. The circuit illustrated is a
- a. astable multivibrator
 - b. monostable multivibrator
 - c. frequency multiplier
 - d. frequency divider



Homework

- Problems: 1, 4, 5, 7, 8, 10, 14, 16, 22, 24, 27, 28, 29, 30, 31
- Exc1. To achieve a one-shot with a pulse width of approximately 10ms, using a 74121. If select $R_{EXT}=20K\Omega$, calculate the capacitance: $C_{EXT}=?$
- Exc2. A circuit need a 1Hz clock signal with a duty cycle of $2/3$. $C=10\mu F$. Determine the value of $R1$ and $R2$.
 - (1) Using the first connection of 555 timer.
 - (2) Using the second connection of 555 timer.(需要画出电路图和设计公式)



Answers

1. b 6. d

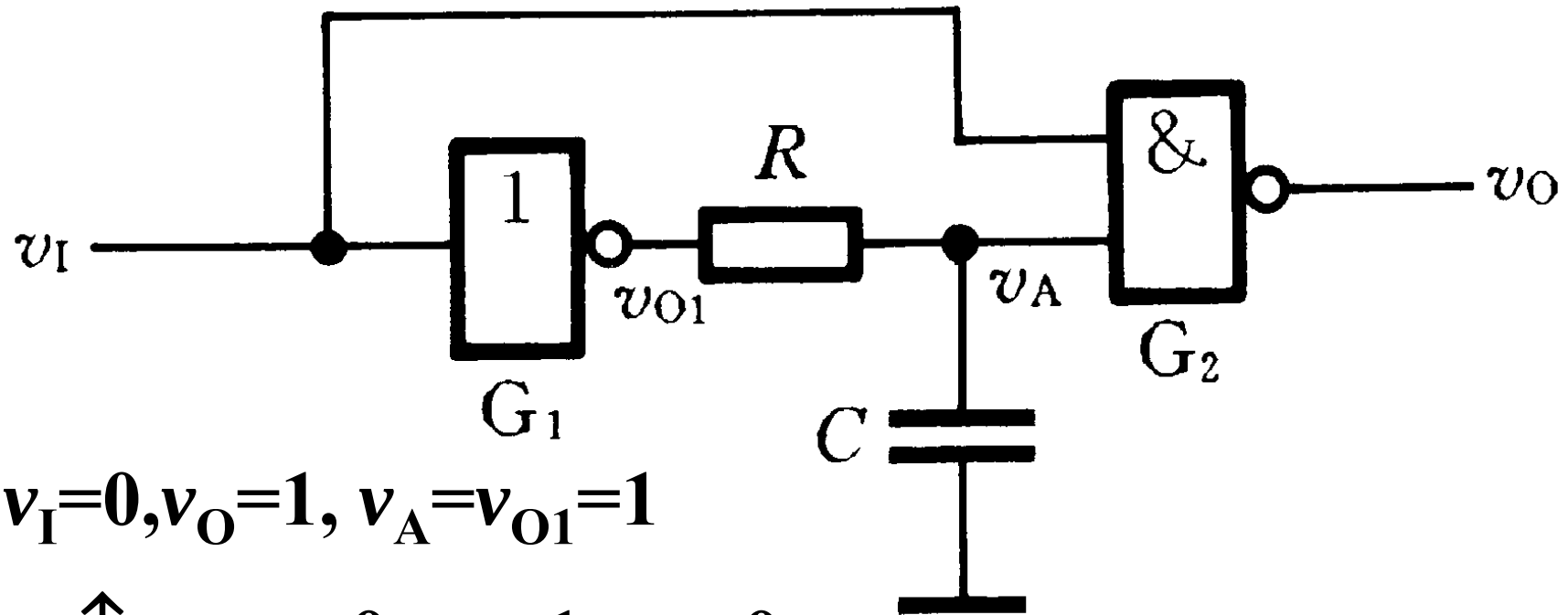
2. d 7. d

3. b 8. b

4. c 9. d

5. b 10. a

RC Integrator One-Shot



$$v_I=0, v_O=1, v_A=v_{O1}=1$$

$$v_I \uparrow \rightarrow v_{O1}=0, v_A=1, v_O=0$$

$$v_C \downarrow \rightarrow v_{I2} \downarrow \rightarrow v_O \uparrow \rightarrow v_{O1} \uparrow$$